CVT System Model Derivation

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March 29, 2026

Abstract

This work develops a first-principles dynamic model for a mechanically actuated dry rubber V-belt continuously variable transmission (CVT) and its coupling to the vehicle drivetrain. The model is motivated by a persistent gap in the available literature: while quasi-static Baja-oriented models are common and detailed dynamic models exist for hydraulically actuated metal-belt CVTs, publicly available formulations for mechanically actuated rubber belt CVTs do not generally combine sheave inertia, flyweight actuation, torque-reactive secondary clamping, belt geometric closure, traction-limited torque transmission, and vehicle longitudinal dynamics within a single unified framework.

Starting from pulley geometry and belt-length closure, the derivation develops the CVT kinematics, the rotational dynamics of the primary and secondary pulleys, the axial equation of motion governing shift, and the torque-transfer closure required to distinguish sticking from traction-limited slip. The primary clutch model includes centrifugal flyweight actuation and spring preload, while the secondary model includes helix-generated torque feedback together with spring forces. A centrifugal belt correction is also derived to account for the reduction in effective normal force at speed. These elements are assembled into a closed nonlinear state-space model suitable for direct numerical time integration.

The resulting formulation is intended both as a simulation model and as a derivation document in which each major modeling step remains physically interpretable. Simulation results are used to examine whether the completed model behaves in mechanically sensible ways under transient launch conditions. Comparisons of launch shift trajectories, pulley axial-force balance, and torque admissibility across multiple tuning cases show a consistent mechanism chain from tuning changes, to clamping-force evolution, to stick–slip behavior, to overall launch outcome. The default tuning produces the most balanced launch response, combining strong early admissible coupling torque, controlled low-ratio retention, and a smoother main shift. More aggressive cases reveal renewed slip and visibly underdamped oscillatory transients during shift onset, highlighting both the usefulness of the model and the limitations introduced by idealized loss assumptions.

Overall, the formulation provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behavior during launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behavior remains traceable back to clear mechanical causes.

Contents

1 Introduction

1.1 Background and Context

The continuously variable transmission (CVT) is a mechanical power transmission device capable of producing a continuously variable speed ratio between an input and output shaft, in contrast to the fixed discrete ratios of a conventional stepped gearbox. In the class of CVTs relevant to this work — dry rubber V-belt CVTs with mechanically actuated conical sheave pairs — the transmission ratio is not commanded by an external actuator or electronic controller. Instead, it is an emergent dynamic response: the equilibrium of axial forces acting on the primary and secondary sheave pairs, arising from centrifugal flyweight mechanisms, torque-reactive helix and ramp assemblies, and return springs, determines the radial position at which the belt seats within each sheave groove, and therefore the effective pitch radii and transmission ratio at every instant.

This self-regulating character is simultaneously the CVT's greatest practical asset and the primary source of difficulty in modeling it rigorously. Because no actuator explicitly commands the ratio, the ratio trajectory must emerge through the evolution of an internal shift degree of freedom whose motion is set by the coupled force balance of the pulleys and belt. In practice, this means that engine speed, transmitted torque, shift position, and traction state evolve together in time, with each affecting the others through the underlying mechanics.

This class of CVT is found throughout small-displacement powersport and utility vehicles. It is the mandated transmission in the SAE Baja competition series, where a 10 hp engine is paired with a CVT that must be designed, selected, and tuned entirely by student engineering teams from first principles. Beyond Baja, mechanically actuated rubber V-belt CVTs are standard equipment in snowmobiles, all-terrain vehicles (ATVs), utility task vehicles (UTVs), and small agricultural and industrial machinery. Despite their prevalence and the practical engineering importance of understanding their behavior, the quality of publicly available engineering models for these systems is strikingly uneven. The present work was motivated directly by this gap.

1.2 Motivation and Literature Review

1.2.1 Overview of the Modeling Landscape

A thorough review of CVT modeling literature reveals a consistent pattern: each body of work captures some aspects of the physics carefully while leaving equally important aspects absent, empiricized, or explicitly assumed away.

A comprehensive survey of CVT dynamics and control research through 2009 is provided by Srivastava and Haque in their review article *"A Review on Belt and Chain Continuously Variable Transmissions: Dynamics and Control"* Srivastava and Haque (2009).

Their review systematically covers the major CVT types — rubber V-belt, metal pushing V-belt, and chain CVT — across static, quasi-static, and fully dynamic formulations, and surveys both the physical models and the control strategies built upon them. Crucially, it emphasizes that rubber belt CVTs are mechanically and dynamically distinct from metal pushing belt CVTs in their actuation mechanisms, belt compliance, and contact mechanics, and that the two should not be treated interchangeably in modeling.

Their central conclusion — that there exists *"considerable disparity in the type of CVT models that have been used"* across the literature, and that fully dynamic, experimentally validated models remain scarce for several CVT classes — indicates that the field has developed in fragments rather than converging toward a unified framework.

The present section builds on that survey, focusing specifically on the subset of literature relevant to the mechanically actuated rubber V-belt CVT considered in this work.

1.2.2 Metal Pushing V-Belt CVT Models

The most analytically complete CVT dynamic models in the literature concern the Van Doorne-type metal pushing V-belt CVT used in automotive applications.

In this class of transmission, clamping force is imposed hydraulically through electronically controlled oil pressure rather than through centrifugal flyweights and spring mechanisms. The belt itself consists of a series of compressive steel elements rather than a flexible rubber belt operating primarily in tension.

Carbone, Mangialardi, and Mantriota Carbone et al. (2005) derive a dynamic model for the shifting behavior of the metal V-belt CVT and identify two distinct operating regimes governing ratio change. In the first regime, micro-slip redistribution within the contact arc governs the shift process, producing relatively slow ratio evolution. In the second regime, macro-slip occurs and rapid ratio change becomes possible.

Srivastava and Haque extend this class of modeling by incorporating dynamic effects associated with belt element inertia and transient force redistribution within the belt system Srivastava and Haque (2006). These studies represent some of the most detailed dynamic analyses of CVT behavior available in the literature.

Experimental characterization of the slip–torque relationship in metal belt CVTs has also been reported by Bonsen et al. Bonsen et al. (2004). Their results show that transmitted torque varies nonlinearly with slip ratio, increasing through a micro-slip region before reaching a maximum near the transition to macro-slip.

While these models achieve a high degree of dynamic fidelity, they describe a fundamentally different system class from the mechanically actuated rubber belt CVT considered here. The actuation mechanisms, belt mechanics, and contact physics differ substantially, limiting the direct applicability of these formulations.

1.2.3 Rubber Belt Mechanical CVT Models

For the mechanically actuated rubber V-belt CVT, the literature is considerably thinner.

The most detailed dynamic model in the open literature appears to be that of Julio and Plante Julio and Plante (2011). Their formulation discretizes the belt into nodal elements and applies Newton's second law to each node individually. Local friction forces are computed from the normal force at each contact point, allowing the torque transmission capacity of the belt to emerge from the cumulative tangential friction forces acting along the wrap arc.

This nodal formulation has several advantages. The belt tension distribution arises naturally from the equilibrium of individual elements rather than being imposed through the classical capstan relationship. The wedging behavior of the belt in the conical groove is captured directly through the element geometry, and centrifugal effects acting on belt elements are included through fictitious forces in the rotating reference frame.

However, two important limitations remain.

First, the inertia of the movable sheaves is neglected. The axial position of the sheaves is determined through instantaneous force balance rather than through an inertial equation of motion, meaning the model cannot capture the finite time required for sheave motion to respond to changing forces.

Second, the authors report difficulty obtaining stable predictions during the transition between clutching and shifting operation. Since mechanically actuated rubber belt CVTs use the belt clamping mechanism itself as the start-up clutch, the clutching regime forms an essential part of the system operating envelope.

Another related effort is presented in SAE technical paper *"Development of CVT Shift Dynamic Simulation Model with Elastic Rubber V-Belt"* Yi et al. (2011). This work develops a dynamic simulation model incorporating belt elasticity and derives a shift motion relationship based on axial force balance between pulleys. Because the full derivation is not publicly accessible, the detailed structure of the dynamic model cannot be evaluated fully from the available abstract and secondary descriptions.

1.2.4 Quasi-Static Baja-Specific Models

Within the SAE Baja community, most CVT modeling work follows a quasi-static force balance approach.

In these models the operating point of the transmission is obtained by solving the condition that the net axial force acting on the sheave assembly is zero. The shift position is therefore treated as an instantaneous function of operating conditions rather than as a dynamic state evolving over time.

A widely cited example is the thesis by Skinner Skinner (2020), which develops a quasi-static model of the flyweight-actuated primary and ramp-actuated secondary and investigates how parameters such as flyweight mass, spring preload, and ramp angle affect engagement speed and operating RPM.

While such models are useful for design tuning and steady-state analysis, they contain no explicit equation of motion for the shift coordinate, no sheave inertia, and no explicit treatment of belt slip. Consequently they cannot simulate transient behavior or predict the dynamic evolution of the transmission ratio.

1.2.5 The Persistent Gap

The literature survey above reveals a consistent pattern. For mechanically actuated dry rubber V-belt CVTs, no publicly available model simultaneously provides

1. an explicit inertial equation of motion for the axial shift coordinate,
2. first-principles modeling of flyweight actuation,
3. a torque-reactive secondary helix force derived from geometry,
4. centrifugal belt mass correction to effective clamping force,
5. an analytical traction-limit slip criterion, and
6. coupling to vehicle longitudinal dynamics.

Metal pushing-belt CVT models such as those of Carbone et al. (2005), Srivastava and Haque (2006), and Bonsen et al. (2004) achieve high levels of dynamic fidelity but describe a different transmission architecture under hydraulic actuation.

Rubber belt models such as Julio and Plante (2011) capture detailed belt mechanics but omit sheave inertia and encounter difficulties during clutching transitions.

Quasi-static Baja models such as Skinner (2020) provide useful design intuition but cannot capture transient behavior.

The absence of a unified dynamic treatment incorporating both mechanical actuation physics and traction-limited torque transmission motivates the development presented in this work.

Reference	Belt	Actuation	Shift ODE	Sheave Inertia	Clutching Mode	Slip Model	Vehicle Dyn.
Carbone et al. (2005)	Metal	Hydraulic	Yes	Yes	No	Yes	No
Srivastava and Haque (2006)	Metal	Hydraulic	Yes	Yes	No	Yes	No
Bonsen et al. (2004)	Metal	Hydraulic	Partial	Yes	No	Yes	No
Julio and Plante (2011)	Rubber	Mechanical	Partial	No	Unstable	Implicit	No
Yi et al. (2011)	Rubber	Mechanical	Partial	Unknown	Unknown	Not explicit	No
Skinner (2020)	Rubber	Mechanical	No	No	No	No	No
Present Work	Rubber	Mechanical	Yes	Yes	Yes	Analytical	Yes

Table 1: Comparison of representative CVT modeling approaches

1.3 Objectives and Scope

This work has two parallel objectives.

The first is technical: to derive a complete, coupled system of nonlinear ordinary differential equations governing the CVT from engine input shaft to vehicle wheel, suitable for direct numerical integration in time.

The second is pedagogical: to make each step of the derivation explicit and physically interpretable, allowing the reader to develop intuition for the mechanisms governing CVT behavior and to trace simulated launch behavior back to clear mechanical causes.

The model includes:

- Explicit shift-coordinate dynamics with sheave inertia
- Primary flyweight actuation and spring loading
- Secondary helix and spring force model
- Belt geometry, ratio kinematics, and belt-length closure
- Traction-limited torque transmission with slip-state logic
- Rotational equations of motion for both pulleys
- Vehicle longitudinal dynamics

The model intentionally excludes thermal effects, belt wear, detailed tribological contact mechanics, and multi-body vehicle dynamics beyond one-dimensional longitudinal motion.

1.4 Contribution Statement

The primary technical contributions of this work are:

1. A unified nonlinear state-space model of the mechanically actuated rubber V-belt CVT coupled to drivetrain and vehicle longitudinal dynamics.
2. First-principles derivation of the primary axial shift equation of motion including sheave inertia and centrifugal flyweight actuation.
3. First-principles derivation of the torque-reactive secondary helix force.
4. Derivation of the centrifugal belt mass correction to effective clamping force.
5. An analytical slip-state switching criterion separating no-slip and traction-limited regimes.
6. Closure of the full drivetrain model into a single time-domain simulation framework in which shift behavior, torque admissibility, slip, and vehicle response evolve consistently together.

1.5 Document Organization

The remainder of this document is organized as follows.

Section 2 introduces the operating principles of a belt-sheave CVT and the specific drivetrain architecture considered in this work, with the goal of establishing mechanical intuition before the mathematical derivation begins.

Section 3 collects the reference material used throughout the document, including coordinate conventions, sign conventions, symbol definitions, units, constants, and parameters.

Section 4 states the modeling assumptions that define the scope and idealizations of the formulation, and Section 5 presents the governing physical principles used in the derivation.

Section 6 defines the dynamic state variables and sign conventions used in the final system formulation.

The core derivation then proceeds through Section 7, Section 8, Section 9, Section 10, and Section 11. These sections develop the CVT geometry and kinematics, the pulley axial force models, the belt transport and contact-state relations, the no-slip coupling torque derivation, and the traction-limited stick–slip closure.

Section 12 assembles these results into the complete coupled dynamic model.

Section 13 presents simulation results organized around launch shift trajectories, axial force balance, torque admissibility, and overall launch performance, with the goal of showing that the completed formulation behaves in mechanically interpretable and physically sensible ways.

Section 14 documents implementation and reproducibility resources, including the code repository, release artifact, and data-access links associated with the reported results.

Section 15 summarizes the main outcomes, limitations, and implications of the present formulation.

The appendices provide supplementary material used by the main text, including belt-length approximation validation (Appendix A), common ramp profile designs (Appendix B), and numerical methods (Appendix C).

2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the fundamental operating principles of a continuously variable transmission (CVT) and then presents the specific CVT and drivetrain architecture considered in this work.

The goal is to establish clear mechanical intuition before developing the mathematical model in later sections.

2.1 Geometric Principle of a Belt-Sheave CVT

A belt-type CVT consists of two pulleys connected by a flexible belt. Each pulley is formed by two opposing conical faces (sheaves). The axial separation between these sheaves determines the effective radius at which the belt rides.

When the sheaves move closer together, the belt is forced outward to a larger radius. When the sheaves move apart, the belt settles at a smaller radius. Because the belt length remains approximately constant, an increase in effective radius on one pulley necessarily causes a corresponding decrease on the other.

As a result, the transmission ratio changes continuously as the pulley geometries change. Unlike a traditional gearbox, there are no discrete gear steps; the ratio evolves smoothly between its minimum and maximum values.

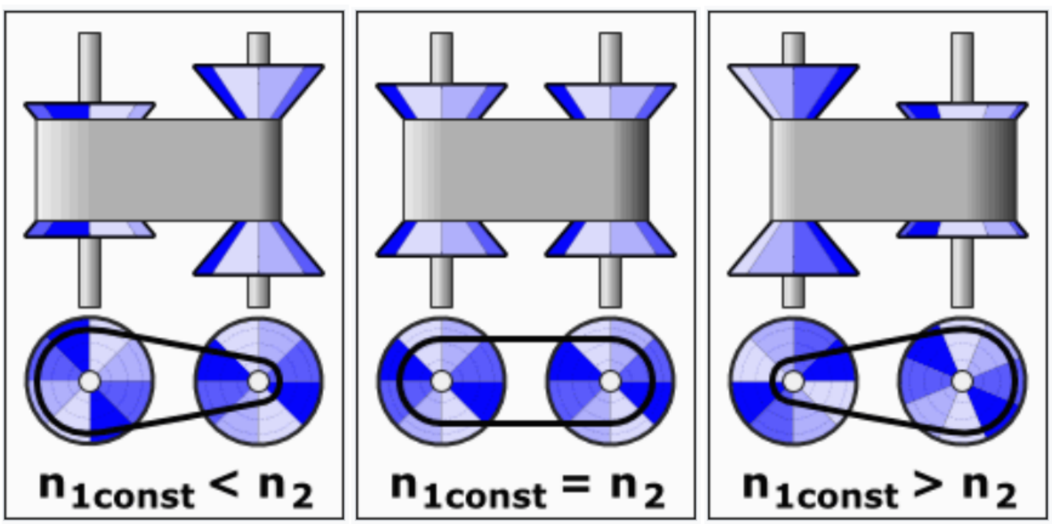


Figure 1: Conceptual illustration of a belt-sheave CVT at three different ratios. Changing the axial separation of the conical sheaves alters the effective belt radius, thereby modifying the transmission ratio continuously.

2.2 Continuous Ratio Versus Discrete Gearing

The defining feature of a CVT is that its transmission ratio can vary smoothly rather than in discrete steps. To understand why this matters, it is useful to first consider how a conventional transmission behaves.

In a stepped gearbox, each gear pair imposes a fixed relationship between engine speed and wheel speed. When vehicle speed increases beyond the useful range of one gear, the transmission must shift to another. This produces an abrupt change in engine speed for the same vehicle speed. As a result, the engine repeatedly accelerates and decelerates as the vehicle progresses through the gears.

A belt-sheave CVT operates differently. Because the effective pulley radii change continuously, the speed relationship between the engine and drivetrain can be adjusted smoothly at any instant. There are no fixed gear pairs that constrain the system to specific ratios. Instead, the ratio can evolve gradually as operating conditions change.

This continuous adjustability allows the engine to operate near a preferred speed while the vehicle accelerates. For example, if an engine produces its peak power within a narrow speed band, a CVT can maintain the engine near that region while vehicle speed increases steadily. Rather than stepping through gears, the transmission adapts its ratio to match the engine's most effective operating condition.

The result is smoother acceleration, improved power utilization, and greater flexibility in matching engine behavior to load demands.

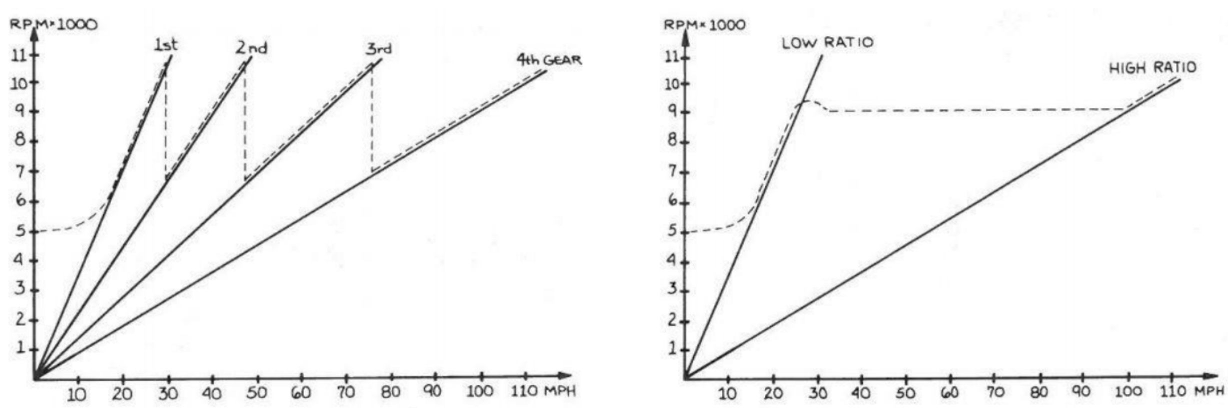


Figure 2: Illustrative comparison of engine speed versus vehicle speed. Left: stepped transmission showing discrete shifts. Right: continuously variable transmission showing smooth ratio evolution.

2.3 Traction-Based Torque Transmission

While the geometric adjustment of pulley radii determines the speed ratio, torque is transmitted through frictional interaction between the belt and the sheave faces.

When a pulley attempts to rotate, it applies a tangential force to the belt. For that force to be transmitted without relative motion, sufficient normal clamping force must press the belt against the conical surfaces. The greater the normal force, the greater the tangential force that can be supported before slip occurs.

This requirement arises directly from the nature of friction. The belt does not contain rigid gear teeth that mechanically lock the pulleys together. Instead, torque transfer depends on traction generated by contact pressure and surface friction. If the transmitted torque exceeds the available traction capacity, the belt will begin to slip relative to the sheave faces.

Slip reduces efficiency, generates heat, and can accelerate component wear. For this reason, CVT designs incorporate mechanisms that actively regulate clamping force in response to operating conditions.

It is therefore useful to distinguish between two interacting aspects of CVT behavior:

- The **geometric ratio**, which determines relative speeds,
- The **traction capacity**, which determines how much torque can be transmitted without slip.

Both must be considered to fully understand performance.

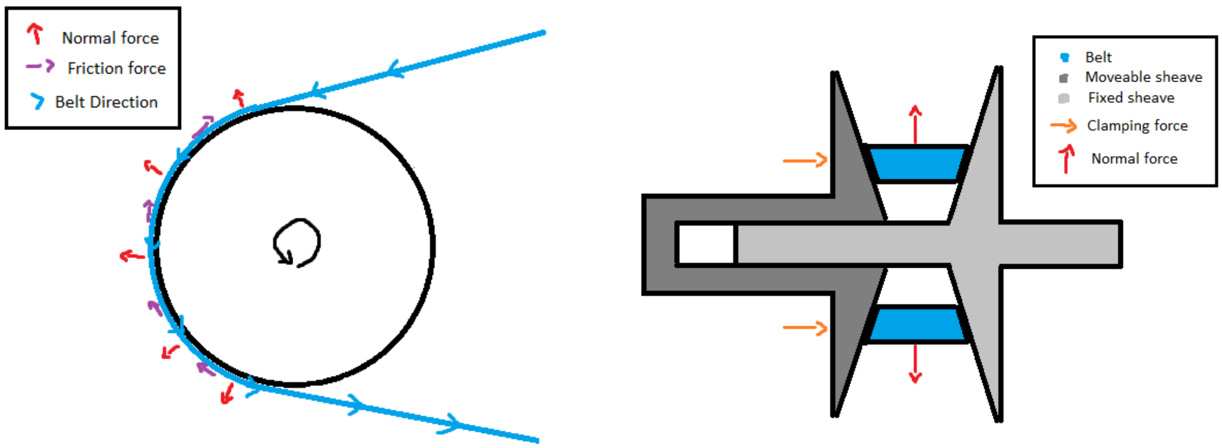


Figure 3: Belt-pulley contact mechanics and force transmission. Left: belt wrapped around a pulley showing normal forces (pressing belt against sheave surface) and tangential friction forces (transmitting torque through traction). Right: cross-sectional view of a CVT pulley assembly showing how axial clamping force between opposing sheaves generates the normal force required for friction and torque transmission.

2.4 Actuation Mechanisms in the Present CVT

While the geometric principles described above apply to all belt-type CVTs, the manner in which axial motion and clamping forces are generated depends on the specific hardware implementation.

The CVT considered in this work employs:

- A **centrifugal flyweight and ramp mechanism** on the primary pulley,
- A **torque-feedback helix mechanism** on the secondary pulley,
- Compression and torsional springs that regulate axial clamping forces.

On the primary side, rotating flyweights move outward under centrifugal effects. Through a ramped interface, this outward motion generates axial force, which adjusts the sheave separation and alters the effective radius.

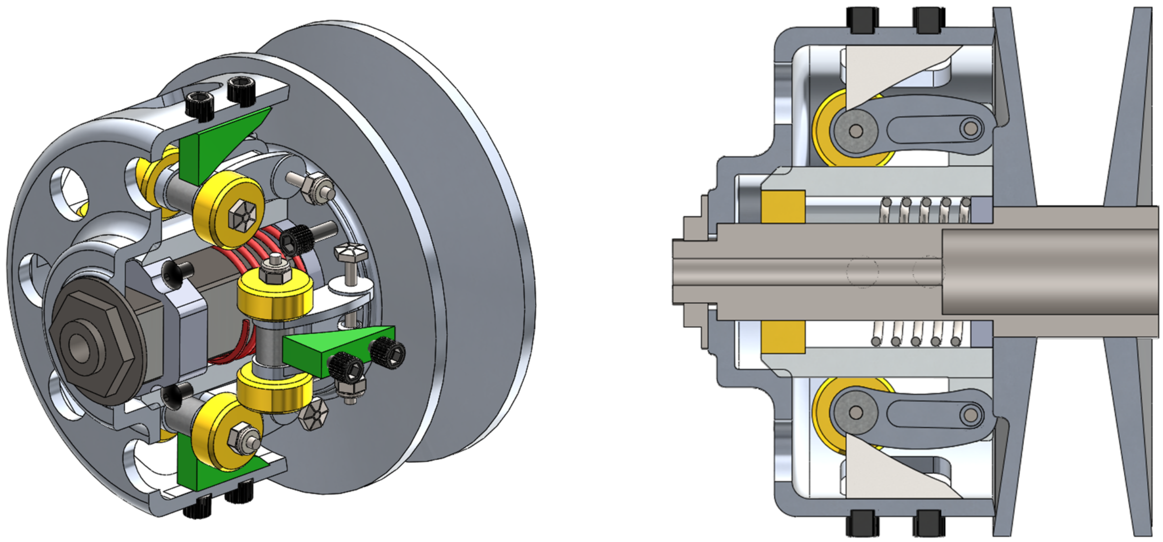


Figure 4: Primary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing centrifugal flyweights (yellow), ramps (green), and compression spring (red). As engine speed increases, centrifugal force drives the flyweights outward along the ramps, generating axial clamping force.

On the secondary side, transmitted torque produces a rotational reaction within a helical interface. This interaction generates axial force that increases clamping in response to torque demand. Springs supplement this mechanism to regulate baseline clamping and system stability.

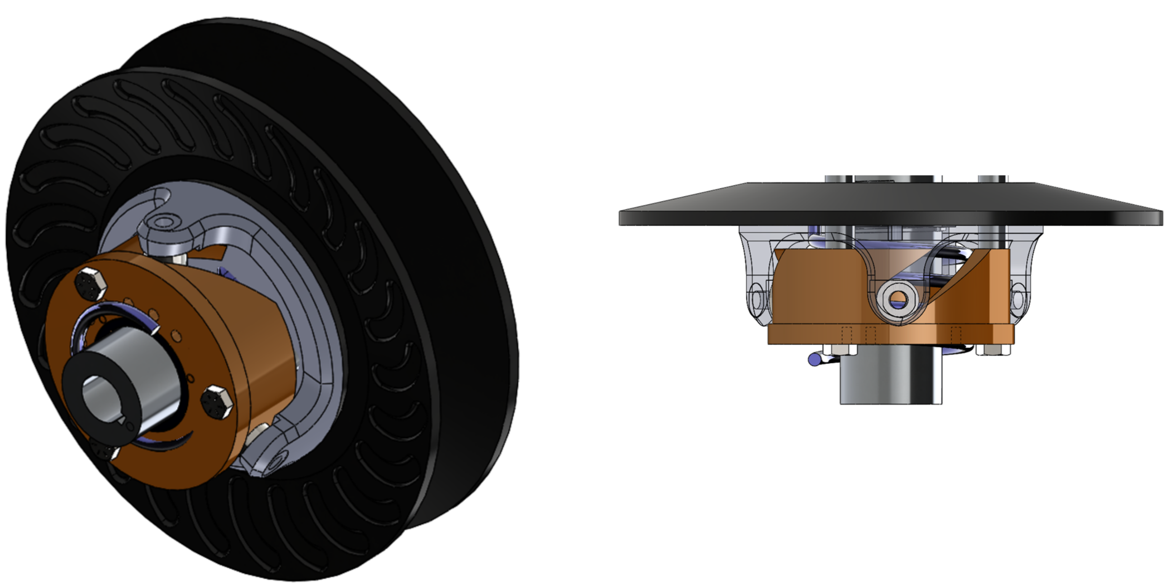


Figure 5: Secondary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing torque-feedback helix mechanism (bronze) and torsional spring (black). Transmitted torque generates a rotational reaction in the helix, which produces axial clamping force proportional to load.

Although these mechanisms are specific to the hardware used in this system, the underlying principles of variable effective radius and traction-based torque transmission remain common to belt-driven CVTs.

2.5 Full Drivetrain Architecture

The CVT does not operate in isolation. In the system considered here, the drivetrain consists of:

- An internal combustion engine,
- The primary CVT pulley,
- A belt connecting to the secondary CVT pulley,
- A fixed-ratio gearbox,
- Final drive components and driven wheels,
- Tire-ground interaction generating tractive forces.

Engine torque enters the primary pulley, is transmitted through the belt to the secondary pulley, passes through a constant gear reduction, and ultimately produces tractive force at the ground.

The interaction between engine dynamics, CVT ratio evolution, drivetrain inertia, and external load forces governs overall vehicle performance.

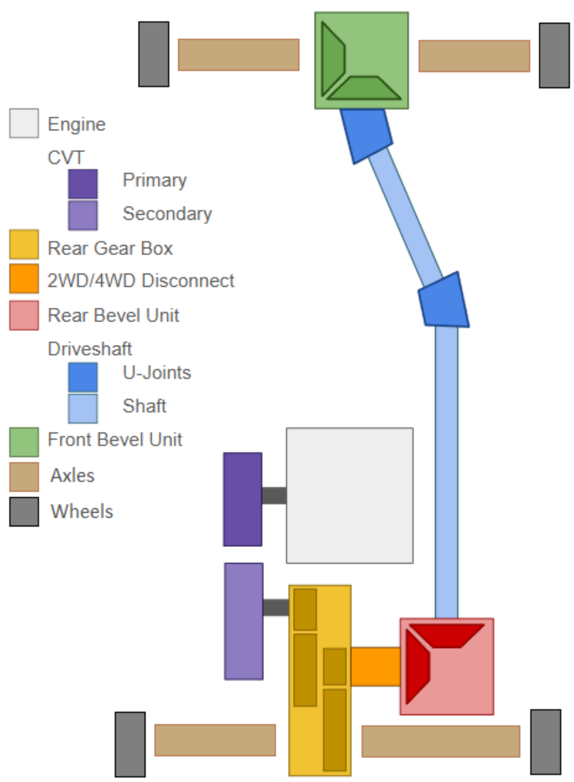


Figure 6: Overview of the complete drivetrain architecture considered in this work.

3 Reference Material

This section records any information for easy reference.

3.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

3.1.1 Vehicle Longitudinal Coordinate and Incline Angle

Vehicle motion is modeled along a single longitudinal coordinate that follows the surface of the road.

The coordinate x is defined locally along the direction of travel, tangent to the road surface. The positive direction, $+x$, corresponds to the forward direction of vehicle motion.

Because the road may be inclined relative to horizontal, the local direction of the x axis depends on the road slope. The incline angle is denoted by α and is measured between the road surface and a horizontal reference line.

$$\alpha > 0 \quad : \text{uphill in the } +x \text{ direction}$$

$$\alpha < 0 \quad : \text{downhill in the } +x \text{ direction}$$

As illustrated in Figure 7, the x direction always lies tangent to the road surface. Consequently, the orientation of the longitudinal axis varies with the local road incline α.

Vehicle velocity is defined along this coordinate. A velocity $v > 0$ corresponds to motion in the forward $+x$ direction along the road, while $v < 0$ corresponds to motion in the opposite direction.

These definitions establish the geometric orientation of vehicle motion and the sign convention for road incline used throughout the model.

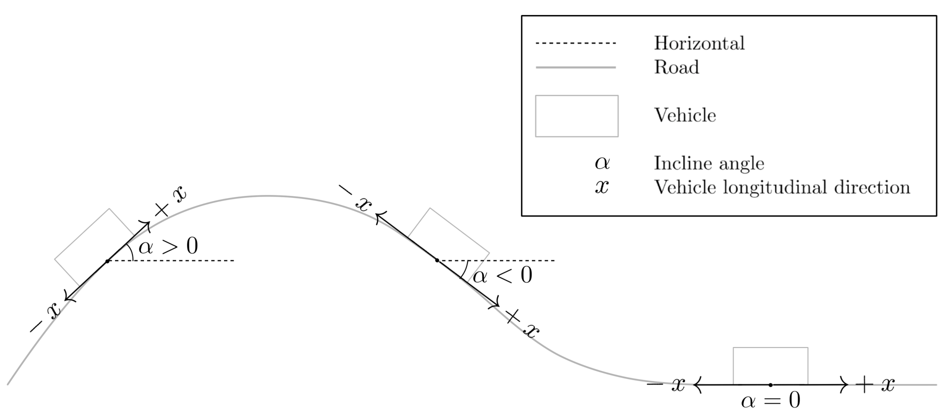


Figure 7: Definition of the longitudinal vehicle coordinate and incline angle. The coordinate x is defined tangent to the road surface and points in the forward direction of travel. The road incline α is measured relative to the horizontal reference line. Positive α corresponds to uphill motion in the $+x$ direction, while negative α corresponds to downhill slope.

3.1.2 Rotational Sign Convention

All rotating components in the drivetrain follow a consistent angular sign convention defined relative to forward vehicle motion.

The positive longitudinal direction $+x$ was defined previously as the forward direction of travel along the road surface. The rotational directions of all drivetrain components are chosen such that positive rotation produces vehicle motion in this $+x$ direction.

As illustrated in Figure 8, positive wheel angular velocity $\omega_w > 0$ corresponds to wheel rotation that drives the vehicle forward along $+x$.

The same convention is propagated upstream through the drivetrain. Positive rotation of the secondary pulley, $\omega_s > 0$, corresponds to the direction of rotation that produces this forward wheel motion through the final reduction gearbox and axle.

Likewise, positive rotation of the primary pulley, $\omega_p > 0$, corresponds to the direction that drives the secondary pulley in this same sense through the CVT belt.

Engine angular velocity follows the same convention. The engine rotates positively when it drives the primary pulley in the defined positive direction.

Torque follows the same sign convention. A torque is considered positive when it acts in the defined positive rotational direction and therefore tends to increase the corresponding angular velocity.

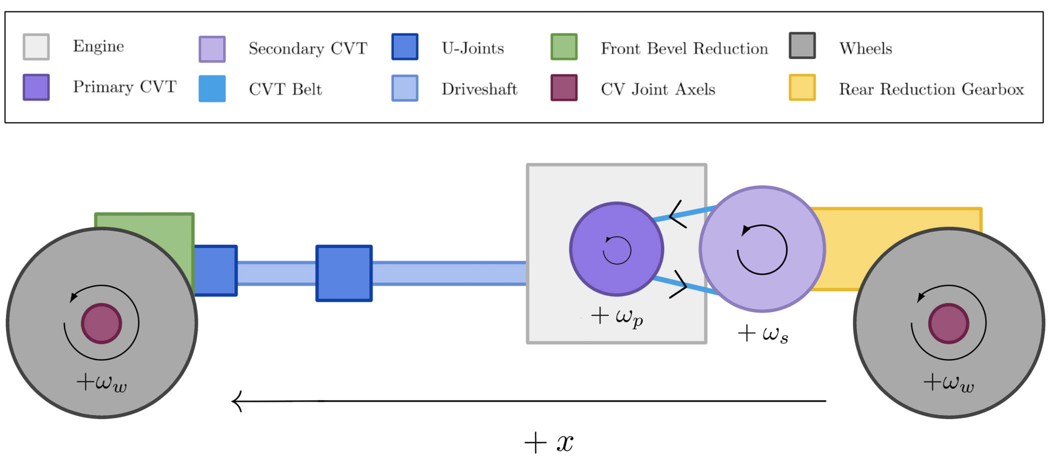


Figure 8: Rotational sign convention for the drivetrain. Positive angular velocities for the wheels (ω_w), secondary pulley (ω_s), and primary pulley (ω_p) are defined such that drivetrain rotation produces forward vehicle motion along the $+x$ direction.

3.1.3 CVT Axial Direction

The axial coordinate of the primary pulley is defined along the axis of the pulley shaft, parallel to the pulley centerline and perpendicular to the plane of belt rotation.

In the reference configuration shown in Figure 9 (left), the primary pulley is fully open. The movable sheave is separated from the fixed sheave by the initial gap, and this configuration defines the origin of the axial coordinate.

The axial variable s measures the translation of the movable sheave relative to this open configuration.

$$s = 0 \quad : \text{fully open configuration}$$

A positive value of s corresponds to the movable sheave translating toward the fixed sheave along the pulley axis.

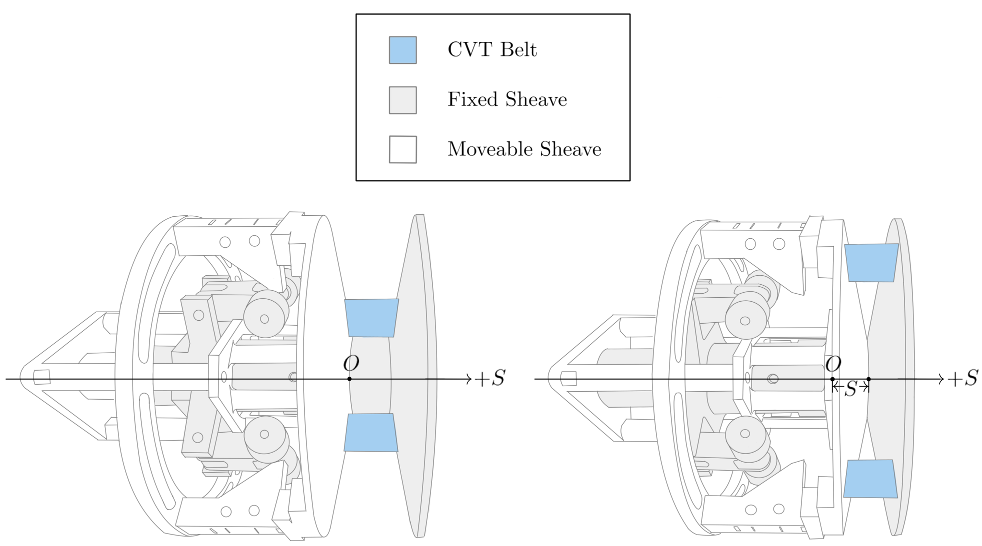
$$s > 0 \quad : \text{sheave closing motion}$$

As the movable sheave closes, the belt is squeezed between the conical faces and is forced outward along the pulley ramps. This increases the effective belt radius on the primary pulley and corresponds to an upshift of the transmission.

The maximum displacement $s = S_{\max}$ corresponds to the fully closed configuration shown in Figure 9 (right).

Although s is defined using the primary pulley, the motion cannot occur independently. Because the belt length remains constrained, closure of the primary pulley forces the secondary pulley to open. The precise geometric relationship between these motions will be derived later.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the movable sheave translating toward the fixed sheave.



$S = O$ (Fully Open)

$S = S_{max}$ (Fully Closed)

Figure 9: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

3.1.4 Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar coordinate convention commonly used in rotational mechanics.

At any point in the plane, two orthogonal directions are defined: the radial direction $\hat{e}_r$ and the tangential direction $\hat{e}_\theta$. These directions form a local coordinate basis centered on the pulley axis.

The radial direction lies in the plane of rotation and points directly outward from the center of the pulley. A radial force is defined as positive when it acts away from the center of rotation. Thus, $F_r > 0$ corresponds to a force directed outward along $\hat{e}_r$.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction. The positive tangential direction $\hat{e}_\theta$ is defined to be consistent with the positive rotational convention introduced in Section 3.1.2. Tangential motion in this direction corresponds to motion produced by positive angular velocity.

Figure 10 illustrates these directions using physical examples from the primary pulley. The centrifugal flyweights generate outward radial forces F_r, while the belt moves along the tangential direction with velocity v_t. These quantities serve as representative examples of forces and motion that act along the radial and tangential directions.

The previously defined axial direction is perpendicular to this plane of rotation and points along the pulley shaft, into the page.

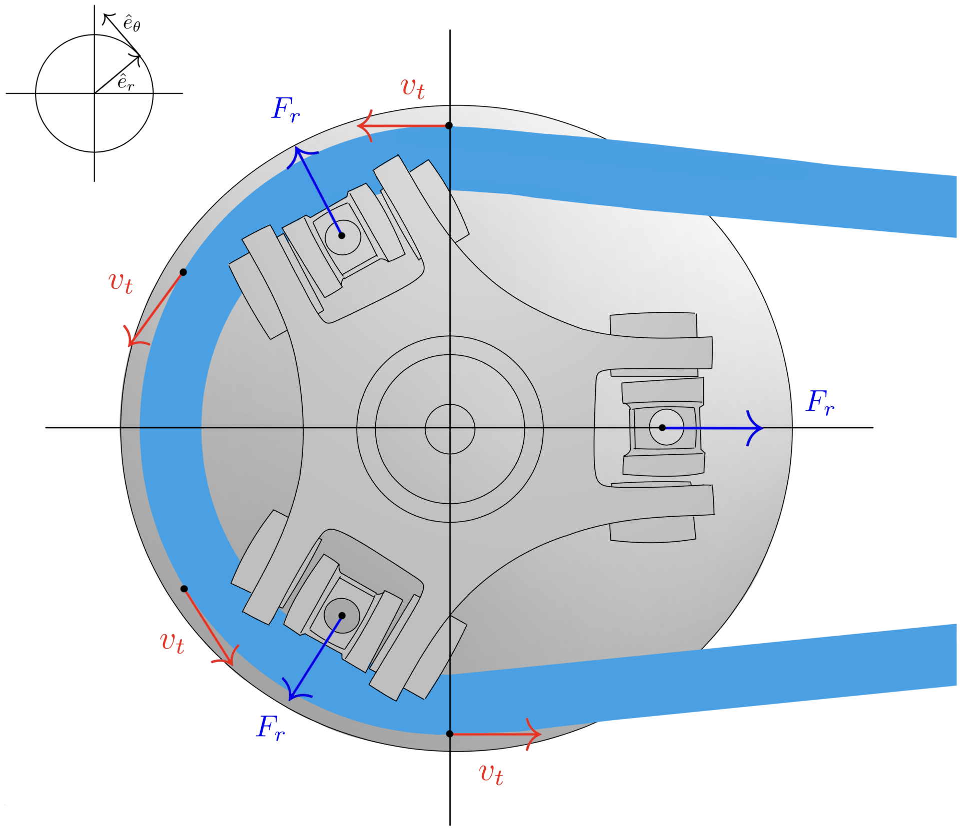


Figure 10: Radial and tangential directions in the plane of pulley rotation. The radial direction $\hat{e}_r$ points outward from the center of the pulley, while the tangential direction $\hat{e}_\theta$ is perpendicular to $\hat{e}_r$ and aligned with positive rotation. Example quantities illustrated include outward radial forces F_r generated by flyweights and tangential belt velocity v_t. For reference, the previously defined axial coordinate s is perpendicular to the page in this view and points along the pulley shaft, into the page.

3.2 Symbol Summary

Symbol	Meaning
State and kinematics	
$\boldsymbol{x}(t)$	state vector
t	time
$\omega_p,\ \omega_s$	primary and secondary angular speed
$s,\ \dot{s},\ \ddot{s}$	shift position, velocity, and acceleration
$R,\ \dot{R}$	CVT ratio and ratio rate
$\hat{v}_b,\ \dot{\hat{v}}_b$	modeled belt transport speed and its time derivative
$v_b^*,\ T_b$	slip-branch target belt speed and belt-speed relaxation time constant
v_Δ	slip metric: primary-imposed minus secondary-imposed belt-line speed
Geometry	
$r_{p,\text{outer}}(s),\ r_{s,\text{outer}}(s)$	primary and secondary outer radius

Symbol	Meaning
$r_{p,\text{eff}}(s),\ r_{s,\text{eff}}(s)$	primary and secondary effective torque radius
$r_{p,\text{outer,min}}$	minimum primary outer radius
$r_{s,\text{outer,max}}$	maximum secondary outer radius
$s_{\text{dz}},\ s_{\max}$	deadzone threshold and maximum shift
β	sheave face half-angle (radial reference)
C	pulley center distance
L_b	total belt length
ϕ	belt wrap angle used in distributed contact model
$r,\ \dot{r}$	local belt path radius and its time derivative
r_{cm}	belt element center-of-mass radius
Constraint and mapping functions	
$f(r_p, r_s, C)$	belt-length compatibility root function ($f = 0$)
$\text{root}_{r_s}(\cdot)$	numerical root solve for secondary radius
$r_f(s),\ \dfrac{dr_f}{ds}$	primary flyweight radius map and gradient
$\theta_s(s),\ \dfrac{d\theta_s}{ds}$	secondary helix angle map and gradient
Forces and torques	
$F_{p,\text{ax}},\ F_{s,\text{ax}}$	primary and secondary axial mechanism force
$F_{c,p},\ F_{c,s}$	primary and secondary belt centrifugal axial contribution
F_{ax}	generic axial clamp force at one pulley contact
$\tau_{\text{eng}},\ \tau_{\text{load}}$	engine input torque and reflected load torque
$\tau_p,\ \tau_s$	coupling torque seen at primary and secondary
$\tau,\ \tau_c$	generic interface torque and applied coupling/friction torque
τ_{ns}	no-slip requested coupling torque
$\tau_-^{\text{stick}},\ \tau_+^{\text{stick}}$	aggregate lower and upper coupling-torque bounds for stick admissibility
$\tau_-^{\text{slip}},\ \tau_+^{\text{slip}}$	aggregate lower and upper coupling-torque bounds on slip branches
$\tau_{c,p,\pm}^{\text{stick}},\ \tau_{c,s,\pm}^{\text{stick}}$	primary/secondary pulley-wise stick-branch lower/upper bounds
$\tau_{c,p,\pm}^{\text{slip}},\ \tau_{c,s,\pm}^{\text{slip}}$	primary/secondary pulley-wise slip-branch lower/upper bounds
Mass/inertia and material parameters	
$I_p,\ I_s$	primary and secondary rotational inertia
m_f	effective flyweight mass
$m_{p,m},\ m_{s,m}$	equivalent primary and secondary translating mechanism mass
ρ_b	belt material density
A_b	belt cross-sectional area
μ	belt–pulley traction coefficient
$k_p,\ x_{p,0}$	primary spring stiffness and preload displacement
$k_{s,\theta},\ \theta_{s,0}$	secondary torsional stiffness and preload angle
$k_{s,x},\ x_{s,0}$	secondary axial stiffness and preload displacement

Table 2: Core symbols used throughout the derivation.

3.3 Base Units

All quantities in this document are expressed in SI units unless otherwise stated.

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 3: Base SI units used by the model variables and parameters.

3.4 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	$\mathrm{m/s^2}$
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	$\mathrm{rad/s^2}$
Area	square meter	$\mathrm{m^2}$
Volume	cubic meter	$\mathrm{m^3}$
Density	kilogram per cubic meter	$\mathrm{kg/m^3}$
Linear spring stiffness	newton per meter	N/m
Torsional spring stiffness	newton meter per radian	N m/rad
Force	newton ($\mathrm{kg\,m/s^2}$)	N
Torque	newton meter ($\mathrm{kg\,m^2/s^2}$)	N m
Moment of inertia	kilogram meter squared	$\mathrm{kg\,m^2}$
Power	watt ($\mathrm{kg\,m^2/s^3}$)	W
Dimensionless ratios/coefs.	unitless	1

Table 4: Derived units and unitless groups used throughout the CVT model.

3.5 Constants and Parameters

All non-varying model parameters are centralized below as the canonical reference. Unless explicitly treated as inputs or state-dependent maps, symbols listed here are held constant during simulation.

Symbol	Name	Nominal value	Unit	Description
Geometry and belt				
L_b	Belt length	0.953262	m	Fixed total belt length
β	Sheave half-angle	0.2007129	rad	Cone half-angle used for radial/axial conversion
h_b	Belt thickness	0.0155702	m	Radial belt thickness
b_{out}	Belt outer width	0.021336	m	Belt width at outer face
b_{in}	Belt inner width	0.0168148	m	Belt width at inner face
$r_{p,\text{outer,min}}$	Primary minimum outer radius	0.0206375	m	Radius at fully open primary
$r_{s,\text{outer,max}}$	Secondary maximum outer radius	0.1016	m	Radius at fully closed secondary
s_{dz}	Shift deadzone threshold	0.0024892	m	Onset of effective radial motion

Symbol	Name	Nominal value	Unit	Description
$s_{\max}$	Maximum shift	0.01905	m	Physical upper travel bound
Primary/secondary actuation				
m_f	Flyweight mass	0.5	kg	Effective flyweight mass
$r_{f,0}$	Initial flyweight radius	0.05	m	Flyweight radius offset at $s = 0$
k_p	Primary spring stiffness	12784	N/m	Primary compression spring rate
$x_{p,0}$	Primary spring preload	0.1	m	Initial compression of primary spring
$k_{s,\theta}$	Secondary torsional stiffness	3.476	N m/rad	Torsional spring rate in helix assembly
$\theta_{s,0}$	Secondary torsional preload	3.49066	rad	Initial torsional deflection preload
r_h	Helix reference radius	0.04445	m	Effective cylindrical radius in helix kinematics
$k_{s,x}$	Secondary axial stiffness	7000	N/m	Secondary axial compression spring rate
$x_{s,0}$	Secondary axial preload	0.1	m	Initial compression of secondary axial spring
Inertia and drivetrain				
I_{eng}	Engine inertia	0.1	kg m^2	Engine rotational inertia
I_{prim}	Primary pulley inertia	–	kg m^2	Primary CVT pulley inertia
I_p	Effective primary inertia	0.1	kg m^2	Implemented primary-side inertia in current setup
I_{sec}	Secondary pulley inertia	–	kg m^2	Secondary CVT pulley inertia
I_{gb}	Gearbox inertia	–	kg m^2	Gearbox/intermediate shaft inertia
I_{wheel}	Wheel inertia	–	kg m^2	Wheel/hub inertia about wheel axis
I_s	Effective secondary inertia	0.5	kg m^2	Lumped secondary-side inertia about secondary axis
G	Final drive ratio	7.556	1	Fixed ratio defined by $\omega_s = G\,\omega_w$
m	Vehicle mass	300	kg	Total vehicle mass
r_w	Wheel rolling radius	0.2794	m	Effective tire rolling radius
Contact and environment				
ρ_b	Belt density	1100	kg/m^3	Belt material density
μ_s	Belt–pulley static friction coefficient	0.65	1	Static traction coefficient used at impending slip
μ_k	Belt–pulley kinetic friction coefficient	0.51	1	Kinetic traction coefficient used on slip branches
C_{rr}	Rolling resistance coefficient	0.015	1	Tire-road rolling resistance coefficient
C_d	Drag coefficient	0.6	1	Aerodynamic drag coefficient
A_f	Frontal area	1.11484	m^2	Vehicle aerodynamic reference area
ρ	Air density	1.225	kg/m^3	Ambient air density used in drag model
g	Gravitational acceleration	9.80665	m/s^2	Gravity constant

Table 5: Canonical non-varying parameters used by the CVT model equations.

4 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

4.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

4.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

4.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Longitudinal elasticity, transient stretch, permanent creep, and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

Assumption 9 (Uniform Contact Radius Around Wrap)

At any given instant, the belt contacts each pulley at a single, uniform effective radius throughout the entire wrap angle. The contact radius is constant around the circumference of contact, with no spiral or progressive radial variation along the wrap. While the effective radius may evolve with axial shift position, it remains spatially uniform at each time instant.

4.4 Dynamic and Inertial Modeling

Assumption 10 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 11 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 12 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 13 (Ideal CVT Power Transmission)

Power transmission through the CVT is modeled as ideal (i.e., no internal dissipation). Although belt slip is explicitly modeled and affects torque capacity and shift dynamics, the associated power loss P_{loss} is neglected.

Assumption 14 (Single Active Slip Interface)

At any instant, the belt is assumed to either adhere to both pulleys or slip at exactly one pulley contact. Simultaneous slip at both pulley contacts is not modeled.
If operating conditions would otherwise cause both contacts to exceed their traction limits, the model assumes an instantaneous transition between single-slip states.

4.5 Environmental and Operational Scope

Assumption 15 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 16 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

5 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s^2)

Source

Standard mechanics identity.

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m^2)
- α is angular acceleration (rad/s^2)

Source

Standard rigid-body dynamics identity.

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Standard mechanics identity.

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Standard mechanics identity.

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v.

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

Standard circular-motion identity.

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

Standard linear-elastic constitutive identity.

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

Standard torsional spring identity.

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

Standard rotational power identity.

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

Standard dry-friction model identity.

Derivation

Coulomb friction model for impending slip.

Theory 10: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

ScienceDirect Topics (n.d.).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{\mathrm{d}T}{T} = \mu\ \mathrm{d}\theta$, then integrate over wrap.

Theory 11: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m^3)
- m is mass (kg)
- V is volume (m^3)

Source

Standard definition.

Derivation

Definition of density.

Theory 12: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m^3)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m^2)

Source

Standard aerodynamic drag model identity.

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 13: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

Standard rolling-resistance model identity.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 14: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

Standard near-Earth mechanics identity.

Derivation

Near Earth's surface, $g \approx 9.81$ m/s^2. On an incline with angle θ, the component parallel to the surface is $F_g \sin\theta$.

6 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables.* A state variable is a quantity whose future evolution is determined by its current value and the governing differential equations of the system.

In numerical simulation, state variables are the quantities that are explicitly integrated forward in time. All other model quantities — such as torque transfer, transmission ratio, belt forces, and contact conditions — are computed algebraically from these states.

6.1 State Vector

State Vector **(6.1)**

$$\boldsymbol{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \\ \hat{v}_b \end{bmatrix}$$

where:

- ω_p is the angular speed of the primary pulley (rad/s),
- ω_s is the angular speed of the secondary pulley (rad/s),
- s is the axial shift position of the primary sheave (m),
- $\dot{s}$ is the axial shift velocity (m/s).
- $\hat{v}_b$ is the modeled belt transport speed (m/s), represented as a closure-oriented dynamic state.

6.2 Independent Dynamic Degrees of Freedom

The five state variables above are the dynamic degrees of freedom of the present state-space model.

The primary pulley can rotate independently under engine input and belt interaction. Its angular speed ω_p therefore represents one rotational degree of freedom.

The secondary pulley also rotates and interacts with both the belt and the external drivetrain load. Its angular speed ω_s represents a second independent rotational degree of freedom.

In addition to rotation, the CVT includes axial motion of the primary sheave. This axial displacement s determines the effective pulley geometry and therefore the transmission ratio. The axial motion is mechanically independent of pure rotation and must be modeled explicitly.

Because axial motion follows second-order dynamics, both the axial position s and its rate $\dot{s}$ are required to fully describe its evolution.

The quantity $\hat{v}_b$ can be viewed as an additional belt-related degree of freedom in a fully resolved belt model. In the present reduced-order formulation, however, detailed belt internal mechanics are not represented explicitly. Instead, $\hat{v}_b$ is introduced as an effective dynamic closure variable governed by compatibility and closure relations introduced in later sections. Thus, $\hat{v}_b$ acts as an approximated surrogate for belt transport dynamics, rather than a fully resolved independent mechanical coordinate.

6.3 Time Integration Perspective

During numerical simulation, the state vector is advanced forward in time.

At each timestep Δt:

- The angular speeds ω_p and ω_s are updated according to their angular accelerations.
- The axial position s is updated using its current velocity $\dot{s}$.
- The axial velocity $\dot{s}$ is updated according to the axial acceleration.
- The modeled belt transport speed $\hat{v}_b$ is updated according to its closure law.

The angular and axial accelerations are determined from the governing equations derived in subsequent sections. Once the state vector has been updated, all other quantities in the model are computed directly from the current values of ω_p, ω_s, s, $\dot{s}$, and $\hat{v}_b$.

7 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies the effective pulley radii and therefore the transmission ratio.

The analysis presented here is purely kinematic. No force balances or dynamic effects are considered. The goal is to describe how pulley geometry alone determines belt position and ratio evolution.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

Radius Definitions Used in This Section Because the belt has finite radial height, two geometric radii are used here: the *outer radius* (belt outer surface/contact geometry) and the *inner radius* (belt inner surface). Let h_b denote the belt height (belt radial thickness). Then,

$$r_{\text{inner}} = r_{\text{outer}} - h_b,$$

The effective torque radius used in torque and ratio relations is introduced later in Section 7.8.

7.1 Conceptual Overview of Variable Geometry

As introduced in Section 2, each pulley consists of two opposing conical sheaves. One sheave is fixed axially, while the other translates along the shaft.

The key controllable motion in the system is this *axial separation* between the sheaves. When the movable sheave translates toward the fixed sheave, the gap narrows and the belt is forced outward along the conical faces. When the sheaves separate, the belt settles deeper between them.

This axial motion does not directly change the pulley radius. Instead, the effective radius arises geometrically from where the belt sits within the sheave gap. In other words, axial displacement determines the belt's radial position, and therefore the effective pulley radius. The precise relationship between these quantities will be derived in the following subsection.

Because the belt length remains constrained, the effective radii of the primary and secondary pulleys cannot vary independently. An increase in the primary effective radius must therefore be accompanied by a corresponding decrease in the secondary effective radius. This geometric coupling is fundamental to CVT operation and forms the basis of the ratio model derived below.

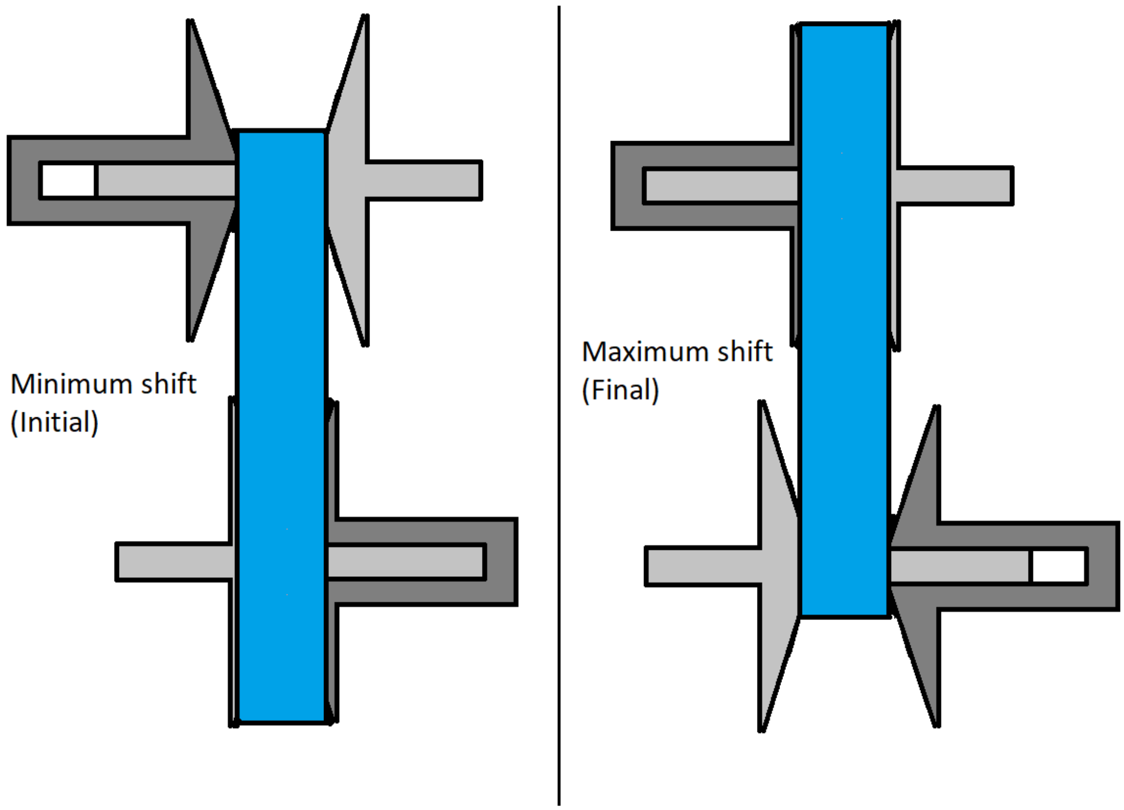


Figure 11: Conceptual illustration of variable pulley geometry. Changes in axial separation alter effective radii. Because belt length is constrained, an increase in primary effective radius corresponds to a decrease in secondary effective radius.

7.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the movable primary sheave, defined according to the sign convention in Section 3.1.3.

At $s = 0$, the primary pulley is in its fully open configuration. The sheaves are at their maximum axial separation, and the belt rests at its minimum effective radius.

The shift coordinate s measures the amount of closure relative to this initial configuration. Increasing s corresponds to the movable sheave translating toward the fixed sheave, reducing the axial gap.

Deadzone and Travel Limits In practice, the initial sheave gap at $s = 0$ is larger than the belt width. As a result, small axial motion does not immediately alter the belt's contact position on the conical faces.

The interval

$$0 \leq s < s_{\mathrm{dz}}$$

defines this *deadzone*. Within this region, the movable sheave translates axially but does not yet sufficiently compress the belt to change its effective contact radius. The belt remains at the same radial position despite the reduction in gap.

Once the sheave gap becomes small enough to fully engage the belt, further axial motion alters the belt position. For

$$s_{\mathrm{dz}} \leq s \leq s_{\mathrm{max}},$$

additional closure forces the belt outward along the conical faces, producing a proportional change in primary effective radius.

The shift coordinate is therefore bounded as

$$0 \leq s \leq s_{\mathrm{max}}.$$

The upper limit s_{max} corresponds to the physical travel limit of the sheave assembly. At this position, the sheaves are fully closed and mechanically contact one another, forming a hard geometric boundary beyond which further axial motion is not possible.

7.3 Primary Outer Radius as a Function of Shift

When the primary sheaves close beyond the deadzone, the belt is forced outward along the conical faces. The relationship between axial closure and radial expansion follows directly from the cone geometry.

As shown in Figure 12, the sheave face forms a straight line in cross-section. The cone half-angle β is defined as the angle between the sheave face and the *radial direction* (As defined in Section 3.1.4).

For a small axial displacement Δs beyond the deadzone, the total closure is shared symmetrically between the two opposing sheaves, so each sheave contributes $\Delta s/2$.

From the right triangle formed by the sheave face,

$$\tan\beta = \frac{\Delta s/2}{\Delta r},$$

which gives

$$\Delta r = \frac{\Delta s}{2\tan\beta}. \tag{7.1}$$

Thus, the cone angle determines how axial motion is converted into radial growth: smaller β produces greater radial change for the same axial displacement.

Taking the infinitesimal limit of Equation (7.1) yields the local geometric relationship between axial shift and effective radius,

$$\frac{dr}{ds} = \frac{1}{2\tan\beta}. \tag{7.2}$$

Because the sheave face is straight, this derivative is constant, meaning that radial position varies linearly with axial displacement.

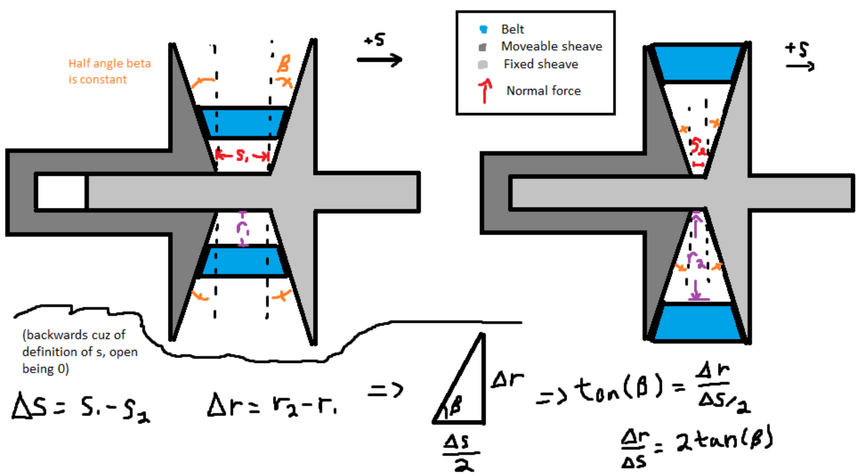


Figure 12: Axial–radial relationship for opposing conical sheaves. The angle β is defined between the sheave face and the radial direction.

Piecewise Radius Model Let $r_{p,\text{outer},\text{min}}$ denote the minimum primary outer radius, corresponding to the fully open configuration ($s = 0$).

Within the deadzone, axial motion does not alter belt position. Once $s \geq s_{\text{dz}}$, only the displacement *beyond the deadzone* contributes to radial growth.

Primary Outer Radius **(7.3)**

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{outer},\text{min}}, & 0 \leq s < s_{\text{dz}}, \\ r_{p,\text{outer},\text{min}} + \dfrac{s - s_{\text{dz}}}{2\tan\beta}, & s_{\text{dz}} \leq s \leq s_{\max}. \end{cases}$$

7.4 Belt Length Constraint

In the previous subsection, the primary effective radius was expressed as a function of axial shift. The secondary radius, however, cannot be chosen independently. The belt has a fixed total length, and therefore any change in primary radius must be accompanied by a corresponding geometric adjustment of the secondary radius.

This section derives the geometric compatibility constraint imposed by constant belt length. By expressing the total belt length in terms of the primary radius r_p, secondary radius r_s, and center-to-center distance C, we obtain an implicit relationship that couples the two pulley radii.

The derivation follows directly from the belt path geometry shown in Figure 13. The geometric quantities used in the derivation are

- r_p — radius of the primary pulley
- r_s — radius of the secondary pulley
- C — center-to-center distance between pulley shafts
- ϕ_p — belt wrap angle on the primary pulley
- ϕ_s — belt wrap angle on the secondary pulley

- ℓ — length of a single straight tangent span

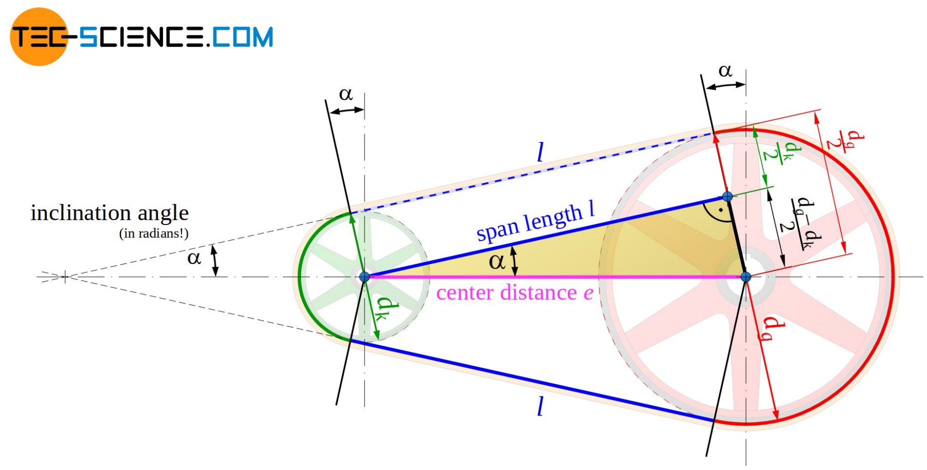


Figure 13: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C. The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans ℓ.

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p\phi_p + r_s\phi_s + 2\ell, \tag{7.4}$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \tag{7.5}$$

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \tag{7.6}$$
$$\phi_s = \pi + 2\lambda. \tag{7.7}$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in Figure 13). We define

$$\sin\lambda = \frac{r_s - r_p}{C}, \tag{7.8}$$

and equivalently,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.9}$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Closed Form Belt-Length Expression Substituting Equation (7.5) and Equation (7.6)–Equation (7.7) into Equation (7.4) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \tag{7.10}$$

Finally, substituting Equation (7.9) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{7.11}$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (7.11) yields

Belt Length Constraint Root Function **(7.12)**

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\,\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b.$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \tag{7.13}$$

The form of Equation (7.13) is unchanged if radii are defined at the belt outer surface, inner surface, or effective mid-height radius, provided all terms are referenced consistently. In particular, L_b must correspond to the same reference as r_p and r_s (e.g., outer-length with outer radii, inner-length with inner radii, or effective-length with effective radii).

Remark (Transcendental Nature and Approximation)

Equation (7.12) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.
In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.
It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.
However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix A quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

7.5 Center-to-Center Distance Determination

In-order to make use of the belt-length constraint derived in Section 7.4, the center-to-center distance C must be determined.

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C, we evaluate the belt-length constraint Equation (7.13) at the *known* initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\text{outer,min}}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\text{outer,max}}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (7.12) yields

$$f(r_{p,\text{outer,min}}, r_{s,\text{outer,max}}, C) = 0. \tag{7.14}$$

This equation contains a single unknown, C, and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b.

$$C = \operatorname*{root}_{C} \; f(r_{p,\text{outer,min}}, r_{s,\text{outer,max}}, C)\,. \tag{7.15}$$

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

7.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 7.5, and the primary radius $r_{p,\text{outer}}(s)$ defined by Equation (7.3), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s, the primary radius $r_{p,\text{outer}}(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_{p,\text{outer}}(s), r_s, C) = 0, \tag{7.16}$$

where $f(\cdot)$ is defined in Equation (7.12).

Numerical Determination of $r_s(s)$ Because Equation (7.16) is transcendental in r_s, the secondary radius is obtained numerically for each value of s:

Secondary Outer Radius **(7.17)**

$$r_{s,\text{outer}}(s) = \operatorname*{root}_{r_s} \; f(r_{p,\text{outer}}(s), r_s, C)\,.$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in Section 7.5, which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

With both $r_{p,\text{outer}}(s)$ and $r_{s,\text{outer}}(s)$ determined, all geometric quantities describing the belt path can now be expressed explicitly as functions of the shift coordinate s.

7.7 Wrap Angles as Functions of Shift

The wrap angles on the primary and secondary pulleys vary as the pulley radii change. Using the geometric relations from Section 7.4, the tangency angle becomes

$$\lambda(s) = \sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{7.18}$$

Substitution into the wrap-angle definitions yields

$$\phi_p(s) = \pi - 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right), \tag{7.19}$$

$$\phi_s(s) = \pi + 2\sin^{-1}\left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C}\right). \tag{7.20}$$

7.8 Effective Torque Radius

All geometric relations derived thus far use the outer surface of the belt as the reference radius. This simplifies the belt-length formulation and pulley geometry.

Torque transmission, however, occurs through the belt cross-section. The appropriate moment arm for torque is therefore located approximately at the mid-height of the belt, as illustrated in Figure 14.

The effective torque radius for each pulley is defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \tag{7.21}$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \tag{7.22}$$

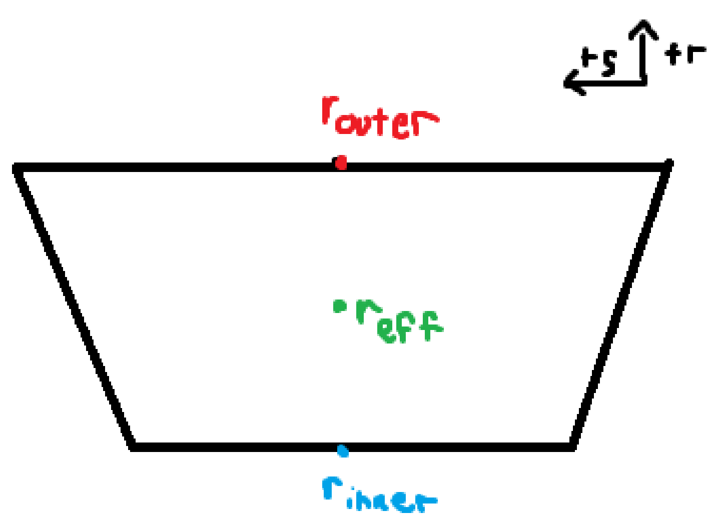


Figure 14: Belt cross-section showing the outer radius r_{outer}, inner radius r_{inner}, and effective torque radius r_{eff} located at the belt mid-height. The effective radius is used for all torque and power transmission calculations.

7.9 CVT Ratio Definition

The instantaneous transmission ratio is determined purely by pulley geometry. It is defined as the ratio of secondary to primary effective radii:

CVT Ratio **(7.23)**

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}.$$

Because both effective radii are functions of the shift coordinate s, the transmission ratio is likewise a function of s alone.

As the primary sheaves close, $r_{p,\text{eff}}(s)$ increases. By belt-length compatibility, the secondary radius decreases. Consequently, the ratio $R(s)$ decreases with increasing s.

Throughout this document, *low ratio* and *high ratio* are named by speed-reduction behavior, not by the numerical magnitude of R. Using the no-slip relation $\omega_p = R\,\omega_s$, larger R means stronger reduction (primary faster than secondary), which we call *low ratio*; smaller R is the opposite, which we call *high ratio*.

Thus:

- Small s (minimum primary radius) corresponds to low ratio.
- Large s (maximum primary radius) corresponds to high ratio.

The CVT ratio is therefore a purely geometric quantity determined by the axial shift position.

7.10 Ratio Rate of Change

The CVT ratio is not only a function of shift position, but also changes in time whenever the sheaves are moving. This time variation matters because a changing ratio introduces additional kinematic coupling between the primary and secondary rotations. In other words, it is not sufficient to know the instantaneous ratio $R(s)$; the model also requires how quickly the ratio is changing.

This section derives an explicit expression for $\dot{R}$ in terms of the shift velocity $\dot{s}$ and geometric derivatives.

7.10.1 Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{7.24}$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds}\dot{s}. \tag{7.25}$$

Thus, the objective is to compute dR/ds.

7.10.2 Differentiate R with Respect to s

Differentiating Equation (7.24) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds}\left(\frac{r_{s,\mathrm{eff}}}{r_{p,\mathrm{eff}}}\right) = \frac{\left(\frac{dr_{s,\mathrm{eff}}}{ds}\right) r_{p,\mathrm{eff}} - r_{s,\mathrm{eff}}\left(\frac{dr_{p,\mathrm{eff}}}{ds}\right)}{r_{p,\mathrm{eff}}^2}. \tag{7.26}$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\mathrm{eff}}}\frac{dr_{s,\mathrm{eff}}}{ds} - \frac{r_{s,\mathrm{eff}}}{r_{p,\mathrm{eff}}^2}\frac{dr_{p,\mathrm{eff}}}{ds}. \tag{7.27}$$

At this point, the remaining task is to express the two derivatives $dr_{p,\mathrm{eff}}/ds$ and $dr_{s,\mathrm{eff}}/ds$ in computable form.

7.10.3 Primary Radius Derivative

The primary radius derivative follows directly from the sheave geometry, since the primary radius is prescribed explicitly as a function of shift. From the piecewise primary radius model Equation (7.3), the primary outer radius satisfies

$$r_{p,\mathrm{outer}}(s) = \begin{cases} r_{p,\mathrm{outer,min}}, & 0 \le s < s_{\mathrm{dz}}, \\ r_{p,\mathrm{outer,min}} + \dfrac{s - s_{\mathrm{dz}}}{2\tan\beta}, & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases}$$

Differentiating with respect to s gives

Primary Radius Derivative **(7.28)**

$$\frac{dr_p}{ds} = \frac{dr_{p,\mathrm{outer}}}{ds} = \begin{cases} 0, & 0 \le s < s_{\mathrm{dz}}, \\ \dfrac{1}{2\tan\beta}, & s_{\mathrm{dz}} \le s \le s_{\max}. \end{cases}$$

Because $r_{p,\mathrm{eff}} = r_{p,\mathrm{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\mathrm{eff}}}{ds} = \frac{dr_p}{ds}. \tag{7.29}$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit $s_{\max}$, then $r_{p,\mathrm{outer}}$ (and thus $r_{p,\mathrm{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\max}$. In this document we restrict to the admissible domain $0 \le s \le s_{\max}$.

7.10.4 Secondary Radius Derivative

Unlike the primary radius, the secondary radius is not prescribed explicitly as a function of shift. Instead, it is determined implicitly by the belt-length compatibility condition derived in Section 7.4. Thus, the secondary radius derivative must be obtained by differentiating the constraint itself.

The belt-length constraint is written as

$$f(r_p, r_s, C) = 0, \tag{7.30}$$

where $f(\cdot)$ is defined in Equation (7.12) and C is constant.

Since both $r_p = r_p(s)$ and $r_s = r_s(s)$ depend on the shift coordinate s, differentiating Equation (7.30) with respect to s gives, by the multivariable chain rule,

$$\frac{\partial f}{\partial r_p}\frac{dr_p}{ds} + \frac{\partial f}{\partial r_s}\frac{dr_s}{ds} = 0. \tag{7.31}$$

Solving Equation (7.31) for the secondary radius derivative yields

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}}\frac{dr_p}{ds}. \tag{7.32}$$

Because the effective secondary radius differs from the outer radius only by the constant belt offset $h_b/2$, the same derivative applies to the effective radius:

$$\frac{dr_{s,\text{eff}}}{ds} = \frac{dr_s}{ds}. \tag{7.33}$$

Equation Equation (7.32) shows that the secondary radial motion is not independent, but is instead determined by the belt-length constraint and the primary radial motion.

At this point, the remaining task is to evaluate the two belt-constraint sensitivities $\partial f/\partial r_p$ and $\partial f/\partial r_s$, which appear in Equation (7.32). The following subsections derive these quantities explicitly for the belt-length function defined in Equation (7.12).

7.10.5 Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \tag{7.34}$$

We now compute the two partial derivatives required in Equation (7.32), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \underbrace{\frac{\partial}{\partial r_p}(r_s - r_p)}_{(1)}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\underbrace{\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right)}_{(2)}. \tag{7.35}$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p}(r_s - r_p) = -1. \tag{7.36}$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx}\sin^{-1}(x) = \frac{1}{\sqrt{1 - x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{1 - u^2}}\left(-\frac{1}{C}\right). \tag{7.37}$$

Now simplify the denominator:

$$\sqrt{1 - u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into Equation (7.37) yields

$$\frac{\partial}{\partial r_p}\sin^{-1}\left(\frac{r_s - r_p}{C}\right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.38}$$

Substitute into the product rule Substituting Equation (7.36) and Equation (7.38) into Equation (7.35) gives

$$\begin{aligned}\frac{\partial}{\partial r_p}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] &= (-1)\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p)\left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}\right) \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \tag{7.39}$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.40}$$

Assemble $\partial f/\partial r_p$ Differentiating Equation (7.34) term-by-term gives

$$\begin{aligned}\frac{\partial f}{\partial r_p} &= \pi + 2\left(-\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \end{aligned} \tag{7.41}$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s}\left[(r_s - r_p)\sin^{-1}\left(\frac{r_s - r_p}{C}\right)\right] = \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}, \tag{7.42}$$

and

$$\frac{\partial}{\partial r_s}\left[2\sqrt{C^2 - (r_s - r_p)^2}\right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \tag{7.43}$$

Assembling terms gives

$$\begin{aligned}\frac{\partial f}{\partial r_s} &= \pi + 2\left(\sin^{-1}\left(\frac{r_s - r_p}{C}\right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}\right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \end{aligned} \tag{7.44}$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (7.34) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{7.45}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.46}$$

These expressions are globally valid under the geometric definition of λ given in Equation (7.9) and require no piecewise case distinction.

7.10.6 Final Expression for the Secondary Radius Derivative

Equation Equation (7.32) expresses the secondary radius derivative in terms of the belt-constraint partial derivatives and the primary radius derivative:

$$\frac{dr_s}{ds} = -\frac{\dfrac{\partial f}{\partial r_p}}{\dfrac{\partial f}{\partial r_s}} \frac{dr_p}{ds}. \tag{7.47}$$

From Section 7.10.5, the partial derivatives of the belt-length constraint Equation (7.12) are

$$\frac{\partial f}{\partial r_p} = \pi - 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right), \tag{7.48}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2\sin^{-1}\left(\frac{r_s - r_p}{C}\right). \tag{7.49}$$

Substituting Equation (7.48) and Equation (7.49) into Equation (7.47) gives

$$\frac{dr_s}{ds} = -\frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} \frac{dr_p}{ds}. \tag{7.50}$$

Finally, substituting the piecewise primary radius derivative Equation (7.28) yields

Secondary Radius Derivative **(7.51)**

$$\frac{dr_s}{ds} = \begin{cases} 0, & 0 \leq s < s_{\mathrm{dz}}, \\ -\dfrac{1}{2\tan\beta} \dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}, & s_{\mathrm{dz}} \leq s \leq s_{\max}. \end{cases}$$

Because $r_{s,\mathrm{eff}} = r_s - h_b/2$ with h_b constant, the same expression applies to the effective secondary radius derivative:

$$\frac{dr_{s,\mathrm{eff}}}{ds} = \frac{dr_s}{ds}. \tag{7.52}$$

7.10.7 Final Expression for dR/ds and $\dot{R}$

From Equation (7.27), the ratio derivative is

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_{p,\text{eff}}}{ds}. \tag{7.53}$$

Using Equation (7.29) and Equation (7.52), this becomes

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}}\frac{dr_s}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\frac{dr_p}{ds}. \tag{7.54}$$

Substituting Equation (7.50) into Equation (7.54) gives

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}}\frac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\right]\frac{dr_p}{ds}. \tag{7.55}$$

Finally, substituting the piecewise primary derivative Equation (7.28) yields the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ \dfrac{1}{2\tan\beta}\left[-\dfrac{1}{r_{p,\text{eff}}}\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} - \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2}\right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases} \tag{7.56}$$

Using the chain rule Equation (7.25), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (7.56) and rearranging gives

Rate of Change of Transmission Ratio **(7.57)**

$$\dot{R} = \begin{cases} 0, & 0 \le s < s_{\text{dz}}, \\ -\dfrac{\dot{s}}{2\tan\beta\, r_{p,\text{eff}}}\left[\dfrac{\pi - 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)}{\pi + 2\sin^{-1}\left(\dfrac{r_s - r_p}{C}\right)} + \dfrac{r_{s,\text{eff}}}{r_{p,\text{eff}}}\right], & s_{\text{dz}} \le s \le s_{\max}. \end{cases}$$

7.11 Section Summary and Forward Connection

This section established the complete *geometric foundation* of the CVT model. The development was intentionally kinematic: the objective was to derive a closed, computable mapping from axial shift to pulley geometry and transmission ratio, without introducing force balance, slip behavior, or torque relations.

At a high level, the geometry defines the chain

$$s(t) \longrightarrow r_{p,\mathrm{eff}}(s),\ r_{s,\mathrm{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with each intermediate relationship derived explicitly from pulley and belt geometry.

Key Results Established The following geometric components are now defined and ready for use in subsequent modeling:

1. **Primary radius law.** Equation (7.3) provides the piecewise relationship between axial shift s and primary outer radius, including deadzone behavior and cone-angle dependence.

2. **Belt-length compatibility constraint.** Equation (7.13) enforces constant belt length and defines the implicit, nonlinear coupling between r_p, r_s, and the center distance C.

3. **Center distance determination.** Section 7.5 fixes the assembled center-to-center distance C by evaluating the belt constraint at a known geometric configuration. This solve is performed once during initialization.

4. **Secondary radius as a function of shift.** Section 7.6 defines $r_s(s)$ implicitly by solving the belt constraint at each shift position, producing the required redistribution of belt radius between pulleys.

5. **Wrap angles as functions of shift.** Section 7.7 defines the wrap angles $\phi_p(s)$ and $\phi_s(s)$ follow directly from the tangency geometry and are available for later contact, work, and belt-distribution calculations.

6. **Effective torque radii.** Section 7.8 corrects outer radii to effective radii using belt thickness, ensuring subsequent torque relations use an appropriate moment arm.

7. **Ratio and ratio rate.** Equation (7.23) defines the geometric ratio $R(s)$, and Section 7.10 provides $\dfrac{dR}{ds}$ and $\dot{R}$ via Equation (7.56) and Equation (7.57), including the deadzone condition where R is constant.

Structural Role in the Full Model These results form the geometric backbone of the CVT simulation. All subsequent sections treat the relationships derived here as fixed mappings. In particular, later torque and force models will use $r_{p,\mathrm{eff}}(s)$ and $r_{s,\mathrm{eff}}(s)$ as inputs, and shifting kinematics will use $R(s)$ and $\dot{R}(s, \dot{s})$ to relate primary and secondary motion during ratio change.

Several key properties are now clear: (i) the geometric coupling is inherently nonlinear (through the belt-length constraint), (ii) the deadzone is explicitly represented, and (iii) within the admissible travel range the mappings are differentiable and suitable for continuous-time simulation.

Transition to Dynamics With the geometry fully specified, the next objective is to determine how the shift coordinate evolves in time. Subsequent sections introduce axial force generation (from pulley mechanisms and belt interaction) to determine $\dot{s}$ and $\ddot{s}$. Once $\dot{s}$ is known, the ratio-rate expression Equation (7.57) provides the corresponding ratio evolution and the required kinematic coupling between rotating components during shifting.

8 Pulley Axial Force Models

The geometric relations developed in the previous section define how the transmission ratio varies with axial shift. What remains is to determine *why* the shift coordinate $s(t)$ changes in time.

Axial motion of the sheaves is driven by clamping forces generated within the primary and secondary assemblies. On the primary side, centrifugal flyweights and springs produce an inward axial force that tends to close the sheaves and increase radius. On the secondary side, torque reaction and spring preload generate an opposing axial force.

The evolution of the shift coordinate is governed by the net axial force acting on the movable sheave assembly. Applying Newton's second law in the axial direction (Theory 1 (Newton's Second Law (Translation))) gives

$$m\,\ddot{s} = F_{p,\mathrm{ax}} - F_{s,\mathrm{ax}}, \tag{8.1}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ denote the axial forces generated by the primary and secondary mechanisms, respectively, and m is the effective axial inertia of the moving components.

The purpose of this section is to derive expressions for $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ as functions of the current system state. Once these force models are established, the shift acceleration $\ddot{s}$ follows directly, completing the dynamic description of axial motion.

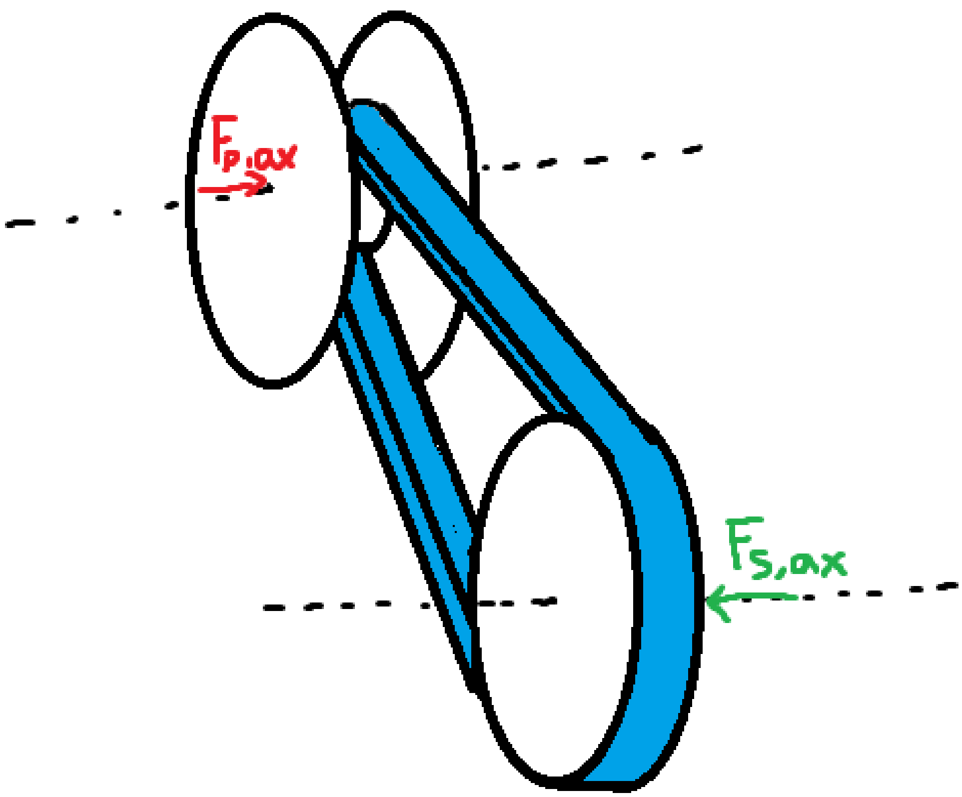


Figure 15: Axial force balance governing sheave motion. Primary and secondary mechanisms generate opposing axial forces. The net force determines the shift acceleration $\ddot{s}$.

Although the specific actuator mechanisms modeled here correspond to a flyweight-driven primary and a torque-reactive helix secondary, the formulation is modular: any actuator model may be substituted provided it returns the net axial force as a function of state.

8.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined contact surfaces. As the shift coordinate s evolves, mechanical elements slide along these surfaces, converting radial or circumferential forces into axial clamping force.

To keep the formulation general and modular, the ramp geometries are parameterized directly as functions of the axial shift coordinate s. Any actuator profile may therefore be substituted provided it supplies the appropriate geometric mapping and derivative.

Primary Ramp Geometry (Axial-to-Radial Mapping) On the primary pulley, centrifugal force acting on the flyweights is redirected into axial closing force through a ramp surface (see Figure 16). As the movable sheave translates axially, the flyweight slides along this ramp.

We represent the ramp by an axial-to-radial profile

$$r_f = r_f(s), \tag{8.2}$$

where $r_f(s)$ is the effective flyweight radius.

An incremental axial displacement ds produces a radial displacement dr_f. The local slope of the ramp is therefore dr_f/ds. The instantaneous primary ramp angle is denoted by $\alpha_p(s)$ and is defined with respect to the axial direction as

$$\tan\alpha_p(s) = \frac{dr_f}{ds}, \qquad \alpha_p(s) = \tan^{-1}\left(\frac{dr_f}{ds}\right). \tag{8.3}$$

This definition follows directly from the geometry of the sliding motion: a steeper ramp (larger dr_f/ds) produces greater radial change per unit axial travel and therefore greater axial gain when radial force is redirected through the surface.

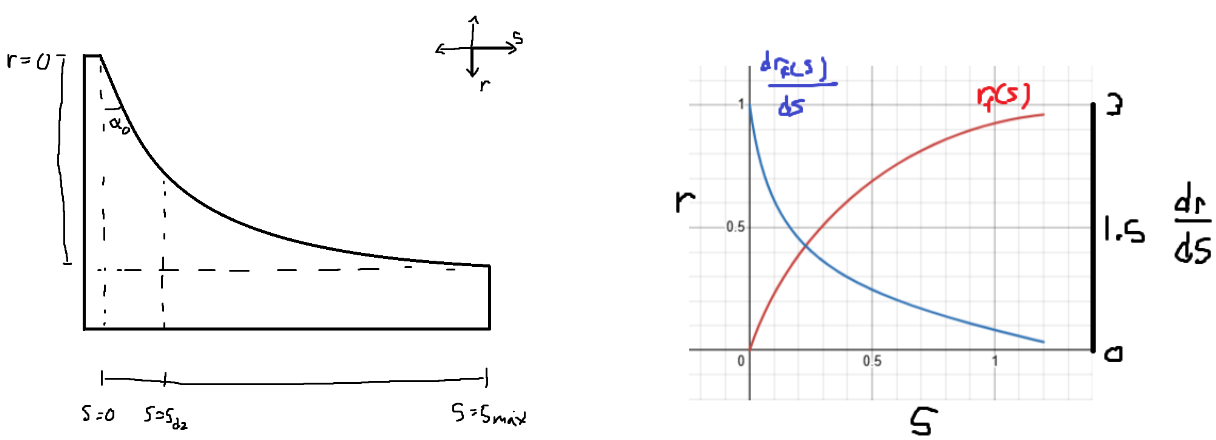


Figure 16: Primary ramp geometry. **Left:** As axial displacement s increases, the flyweight slides along the ramp, producing radial change dr_f. The instantaneous ramp angle satisfies $\tan\alpha_p = dr_f/ds$. **Right:** Example profiles of $r_f(s)$ and its slope dr_f/ds.

Primary Admissibility Conditions In order to be a valid primary ramp profile $r_f(s)$, it must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan\alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces relative rotation of a helical ramp mechanism, loading a torsional spring and generating torque-reactive clamping (see Figure 17).

We parameterize this kinematically as

$$\theta_s = \theta_s(s), \tag{8.4}$$

where $\theta_s(s)$ is the helix-induced rotation.

An incremental rotation $d\theta_s$ corresponds to a circumferential displacement $r_h d\theta_s$ along a cylindrical surface of effective radius r_h. The instantaneous secondary helix angle is denoted by $\alpha_s(s)$ and is defined from the local slope of the ramp surface:

$$\tan\alpha_s = \frac{\text{axial rise}}{\text{circumferential run}} = \frac{ds}{r_h d\theta_s}.$$

Rewriting in differential form gives

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}. \tag{8.5}$$

Thus, a shallower helix (smaller α_s) produces greater rotation per unit axial travel, while a steeper helix produces less rotation. This inverse relationship arises naturally from the geometry of motion on a cylindrical surface.

Remark (Contrast with Primary Ramp)

The primary ramp behaves like a simple wedge: radial displacement is directly proportional to axial travel, so the ramp angle is naturally defined by $\tan\alpha_p = dr_f/ds$.
The secondary mechanism instead behaves like a power screw. Axial motion is produced through rotation around a cylindrical surface, so the relevant geometric ratio is how much rotation occurs per unit axial travel. This leads to the slope definition $ds/(r_h d\theta_s)$ and the inverse relationship in Equation (8.5).
In short, the primary redirects force through a wedge, while the secondary converts torque through a screw-like helix. The difference is purely geometric and reflects the distinct mechanisms by which axial motion is generated.

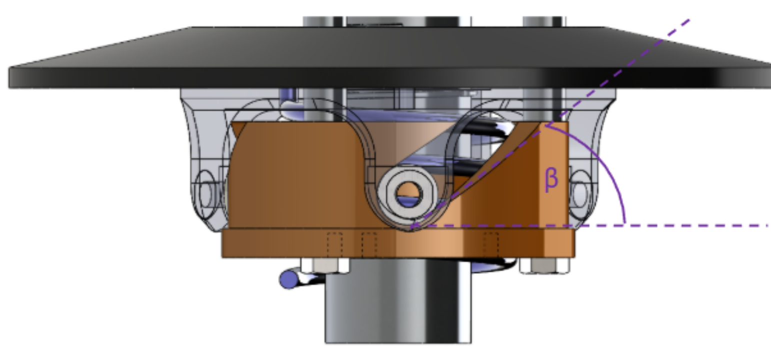


Figure 17: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ around a cylindrical surface of radius r_h. The instantaneous helix angle satisfies $\tan\alpha_s = ds/(r_h d\theta_s)$.

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\dfrac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\dfrac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases rotation. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix B.

8.2 Primary Axial Force Model

The primary axial clamping force is generated by the interaction between centrifugal loading of the flyweights and the restoring force of the primary compression spring.

As shown in Figure 18, each flyweight of mass m_f rotates with angular speed ω_p and experiences an outward radial centrifugal force. This radial force acts along the ramp profile $r_f(s)$ and is redirected into an axial closing force through the ramp geometry described in Section 8.1.

Opposing this motion is the primary compression spring, characterized by stiffness k_p and preload $x_{p,0}$. As the sheaves close (s increases), the spring compresses and produces a restoring axial force resisting further closure.

The net axial force on the movable primary sheave is therefore the axial component of the ramp-redirected centrifugal force minus the spring force. This net force determines the axial acceleration $\ddot{s}$ of the primary assembly.

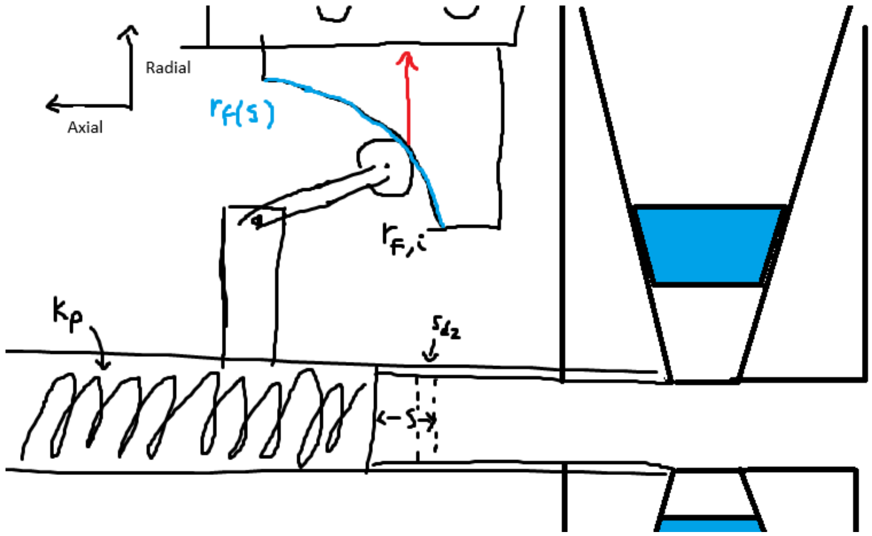


Figure 18: Primary CVT assembly showing flyweight mass m_f, ramp profile $r_f(s)$, compression spring with stiffness k_p and preload $x_{p,0}$, and axial shift coordinate s.

The state variable governing primary rotation is ω_p.

Centrifugal Loading of Flyweights From Theory 5 (Centrifugal Force),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_{f,0}$ denote the initial flyweight radius at $s = 0$,
- $r_f(s)$ denote the incremental radial change produced by the ramp.

The total effective flyweight radius is therefore

$$r_{f,\text{tot}}(s) = r_{f,0} + r_f(s).$$

The radial centrifugal force is

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big). \tag{8.6}$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in Figure 19, the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \tag{8.7}$$

From Equation (8.3), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 \big(r_{f,0} + r_f(s)\big) \frac{dr_f}{ds}. \tag{8.8}$$

Thus the axial gain of centrifugal force is governed by the local ramp slope, while the centrifugal magnitude depends on the total flyweight radius.

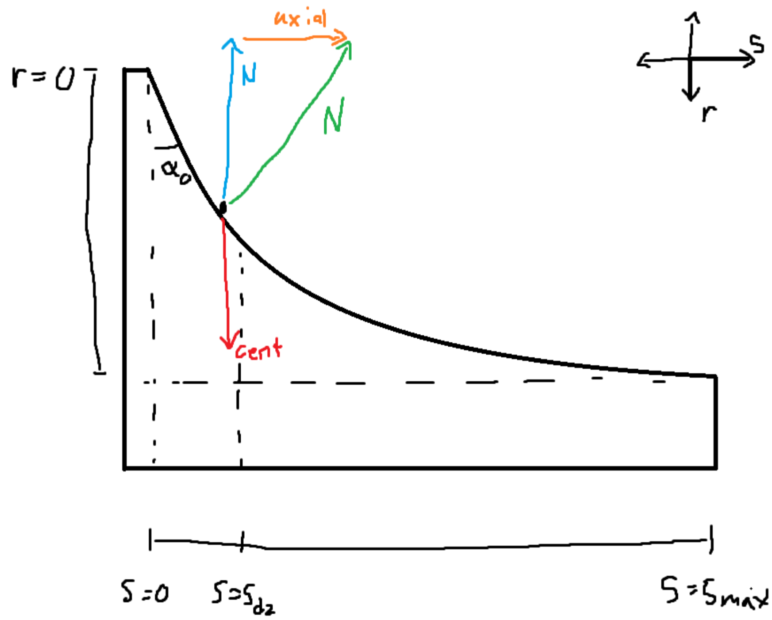


Figure 19: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \tag{8.9}$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

Primary Axial Force **(8.10)**

$$F_{p,\mathrm{ax}}(s,\omega_p) = m_f\omega_p^2\big(r_{f,0}+r_f(s)\big)\frac{dr_f}{ds} - k_p(x_{p,0}+s).$$

8.3 Secondary Axial Force Model

The secondary pulley is *torque-reactive*: its axial clamping force is generated in response to torque transmitted through the CVT. Unlike the primary, which produces clamping based on rotational speed, the secondary produces clamping primarily when torque flows through it.

The mechanism consists of a helical ramp between the two sheaves and a single spring located between them. This spring serves two roles:

- It acts axially in compression as the sheaves separate.
- It resists relative rotation torsionally as the helix turns.

Thus, axial motion and rotational motion of the helix both load the same spring.

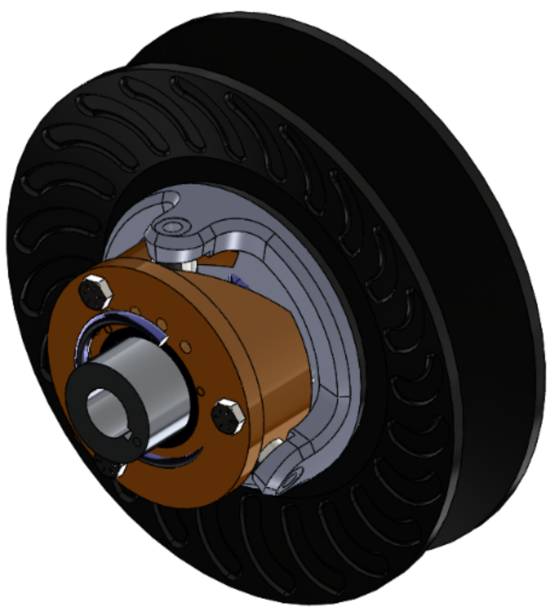


Figure 20: Secondary CVT assembly showing the helix mechanism and spring. Torque τ_s transmitted through the belt produces tangential force at the helix ramps, which is redirected into axial clamping force.

Physical Mechanism (Screw Analogy) The secondary helix behaves mechanically like a screw.

If you try to spin a screw in free space, nothing happens axially. If you spin a screw into a block that is not restrained, the block simply rotates with the screw—again producing no axial force. Axial force only develops when there is resistance to rotation on one side while torque is applied on the other.

In other words, a screw converts *reacted torque* into axial force.

The secondary helix operates on the same principle. The belt applies a driving torque to the secondary pulley. If the driven shaft were free, the entire assembly would rotate without developing tangential reaction force at the helix. However, when the driven load resists rotation, torque is transmitted through the helix ramps.

Those ramps are inclined surfaces—just like screw threads. As torque attempts to rotate one helix face relative to the other, contact forces develop along the inclined surfaces. Because the surfaces are angled, this tangential reaction force has an axial component.

The result is axial clamping force.

The key distinction from a conventional screw is direction: in a normal screw, torque pulls two components together. In the CVT helix, the orientation of the ramp is such that torque causes the movable sheave to be pushed outward, increasing belt clamping.

Thus, the secondary does not clamp because it spins. It clamps because torque is being transmitted and resisted.

Definition of Transmitted Torque Let τ_s denote the torque transmitted through the secondary shaft. This is the torque carried from the belt, through the helix mechanism, and into the driven load.

For the present derivation, τ_s is treated as a known input.

Torsional Spring Torque As the helix rotates relative to the opposing sheave, the spring is twisted torsionally.

From Theory 7 (Hooke's Law (Torsional)),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big). \tag{8.11}$$

This torsional component of the same spring contributes to helix loading even when no torque is transmitted ($\tau_s = 0$), providing baseline clamping.

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \tag{8.12}$$

Both contributions act in the same rotational direction at the helix. The transmitted torque loads the helix through belt force, and the torsional spring resists relative rotation. Both therefore generate axial clamping through the same geometry.

Torque-to-Axial Conversion Through the Helix From Theory 3 (Torque Definition),

$$\tau = F_t r_h \quad \Rightarrow \quad F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The tangential force F_t acts along two symmetric helix ramp faces. Each ramp carries half of the total tangential load, introducing the factor of $1/2$ in the axial projection below.

From helix geometry (see Figure 21),

$$F_{s,\text{helix}} = \frac{F_t}{2\tan\alpha_s(s)}. \tag{8.13}$$

Substituting F_t and using

$$\tan\alpha_s(s) = \frac{1}{r_h \dfrac{d\theta_s}{ds}}$$

gives

$$F_{s,\text{helix}}(s, \tau_s) = \frac{\tau_{s,\text{total}}}{2}\frac{d\theta_s}{ds}. \tag{8.14}$$

Notably, the helix radius r_h cancels: torque is converted into axial force according to the geometric rotation-per-travel relationship $d\theta_s/ds$, consistent with screw mechanics.

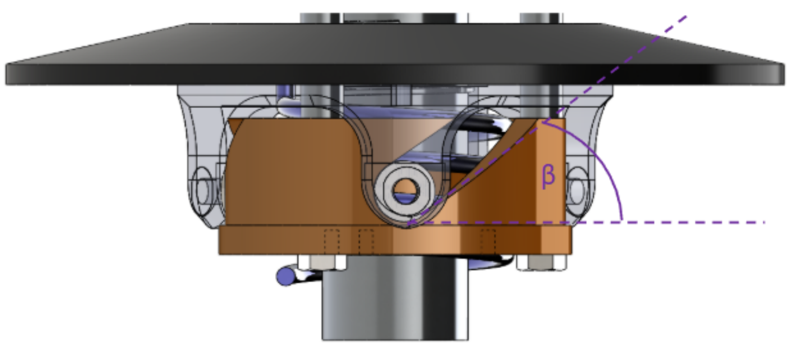


Figure 21: Helix free-body concept. Tangential force from shaft torque is redirected into axial force along the inclined ramps. Two symmetric ramp faces share the tangential load, producing the 1/2 factor.

Secondary Compression Spring The same spring also acts axially between the sheaves. At $s = 0$, the spring is pre-compressed by an amount $x_{s,0}$, producing a baseline separating force that assists upshift.

As the sheaves move apart (increasing s), its compression increases further.

From Theory 6 (Hooke's Law (Compressional)),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote axial stiffness. Because increasing s increases compression, the effective compression is

$$\Delta x = x_{s,0} + s.$$

Thus,

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \tag{8.15}$$

This axial spring force acts in the same direction as the helix force, providing additional clamping as the sheaves separate.

Net Secondary Axial Force The total secondary axial force is the sum of:

- Helix-generated torque-reactive clamping,
- Torsional spring contribution (through the helix),
- Direct axial compression spring force.

Secondary Axial Force **(8.16)**

$$F_{s,\text{ax}}(s, \tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2}\frac{d\theta_s}{ds}}_{\text{torque-reactive helix (including torsional spring)}} + \underbrace{k_{s,x}(x_{s,0} + s)}_{\text{axial spring}}.$$

8.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Throughout this section, let

$$j \in \{p, s\}$$

denote either pulley, where $j = p$ corresponds to the primary and $j = s$ corresponds to the secondary.

From Theory 5 (Centrifugal Force), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around pulley j, three quantities must be defined:

1. the radius at which the belt mass rotates,
2. the mass of belt material carried around the wrap,

3. the angular velocity of that belt mass.

Each is developed below.

1. Radius at Which the Belt Mass Rotates Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner belt surface. The location of this center of mass therefore determines the radius that must be used in the centrifugal force calculation.

The belt cross-section is approximated as a trapezoid (see Figure 22) with:

- radial thickness h_b,
- outer width b_{out},
- inner width b_{in}.

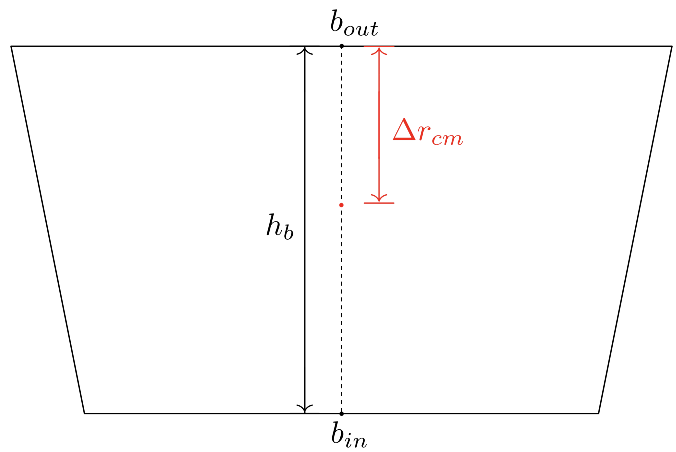


Figure 22: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b and the centroid is located at Δr_{cm} from the outer surface.

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2}\left(b_{\text{out}} + b_{\text{in}}\right). \tag{8.17}$$

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y:

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \tag{8.18}$$

The centroid location Δr_{cm}, measured inward from the outer surface, follows from the area-centroid definition:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y\, b(y)\; \mathrm{d}y. \tag{8.19}$$

Substituting Equation (8.18) into Equation (8.19) gives

$$\begin{aligned}
\int_0^{h_b} y\, b(y)\; \mathrm{d}y &= \int_0^{h_b} y \left(b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right) \mathrm{d}y \\
&= \int_0^{h_b} b_{\text{out}} y\; \mathrm{d}y + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2\; \mathrm{d}y \\
&= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} \left(b_{\text{in}} - b_{\text{out}} \right) \\
&= \frac{h_b^2}{6} \left(b_{\text{out}} + 2 b_{\text{in}} \right).
\end{aligned} \tag{8.20}$$

Dividing by the area from Equation (8.17) yields

$$\Delta r_{\text{cm}} = h_b \frac{b_{\text{out}} + 2 b_{\text{in}}}{3 \left(b_{\text{out}} + b_{\text{in}} \right)}. \tag{8.21}$$

For pulley j, let $r_{j,\text{out}}(s)$ denote the outer belt radius. The radius at which the belt mass rotates is therefore

$$r_{j,\text{cm}}(s) = r_{j,\text{out}}(s) - \Delta r_{\text{cm}}. \tag{8.22}$$

2. Mass of Belt Material on the Wrap To determine the centrifugal loading on pulley j, we next determine the amount of belt mass associated with an infinitesimal portion of its wrap.

Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped region on pulley j spans

$$\theta \in \left[-\frac{\phi_j}{2}, \frac{\phi_j}{2} \right],$$

where $\phi_j(s)$ is the total wrap angle on pulley j (see Figure 23).

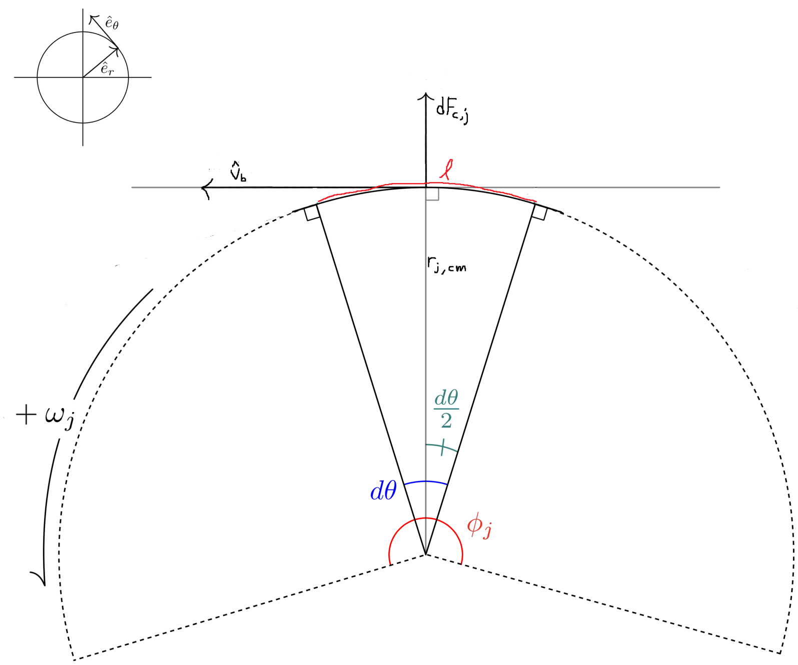


Figure 23: Belt wrapped around pulley j. A differential element at angle θ has arc length $\mathrm{d}\ell = r_{j,\mathrm{cm}}\ \mathrm{d}\theta$ and experiences radially outward centrifugal force.

A differential belt element subtending angle $\mathrm{d}\theta$ at the mass-centroid radius $r_{j,\mathrm{cm}}$ has arc length

$$\mathrm{d}\ell = r_{j,\mathrm{cm}}\ \mathrm{d}\theta.$$

Since the belt has cross-sectional area A_b, the differential belt element has volume

$$\mathrm{d}V_j = A_b\ \mathrm{d}\ell. \tag{8.23}$$

Substituting $\mathrm{d}\ell = r_{j,\mathrm{cm}}\ \mathrm{d}\theta$ gives

$$\mathrm{d}V_j = A_b\, r_{j,\mathrm{cm}}\ \mathrm{d}\theta. \tag{8.24}$$

Let ρ_b denote the belt material density (kg/m^3). The corresponding differential mass is then

$$\begin{aligned} \mathrm{d}m_j &= \rho_b\ \mathrm{d}V_j \\ &= \rho_b A_b\, r_{j,\mathrm{cm}}\ \mathrm{d}\theta. \end{aligned} \tag{8.25}$$

This gives the amount of belt material associated with an infinitesimal portion of the wrap on pulley j.

3. Angular Velocity of the Belt Mass The rotating belt mass on pulley j must also be assigned an angular speed. From Section 6, $\hat{v}_b$ denotes the modeled belt transport speed. The corresponding angular velocity of the belt mass centroid on pulley j is then

$$\omega_{b,j} = \frac{\hat{v}_b}{r_{j,\mathrm{cm}}(s)}. \tag{8.26}$$

This relation expresses the fact that the same belt transport speed corresponds to different local angular velocities depending on the radius of motion.

Distributed Centrifugal Load Along the Wrap With the radius, differential mass, and angular velocity now defined, the centrifugal force on the differential belt element follows from Theory 5 (Centrifugal Force):

$$\begin{aligned} \mathrm{d}F_{c,j} &= \mathrm{d}m_j\, \omega_{b,j}^2\, r_{j,\mathrm{cm}} \\ &= (\rho_b A_b\, r_{j,\mathrm{cm}}\ \mathrm{d}\theta) \left(\frac{\hat{v}_b}{r_{j,\mathrm{cm}}} \right)^2 r_{j,\mathrm{cm}} \\ &= \rho_b A_b\, \hat{v}_b^2\ \mathrm{d}\theta. \end{aligned} \tag{8.27}$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating Equation (8.27) over the full wrap angle gives the total radially-directed centrifugal force on pulley j:

$$\begin{aligned} F_{c,j,\mathrm{rad}} &= \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b\, \hat{v}_b^2\ \mathrm{d}\theta \\ &= \rho_b A_b\, \hat{v}_b^2 \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}\theta \\ &= \rho_b A_b\, \hat{v}_b^2\, \phi_j(s). \end{aligned} \tag{8.28}$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β, radial loading cannot occur without simultaneous axial motion of the sheaves. Thus, a purely radial force applied to the belt generates an axial reaction component that contributes to clamping.

From the pulley geometry derived previously (see Equation (7.2)), the relationship between radial and axial displacement in the active shift region is

$$\frac{\mathrm{d}r}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

For an infinitesimal displacement,

$$\delta W = F_{\mathrm{rad}}\, \mathrm{d}r = F_{\mathrm{ax}}\, \mathrm{d}s.$$

Substituting $\mathrm{d}r = \dfrac{1}{2\tan\beta}\, \mathrm{d}s$ gives

$$F_{\mathrm{rad}}\, \frac{1}{2\tan\beta}\, \mathrm{d}s = F_{\mathrm{ax}}\, \mathrm{d}s.$$

Canceling $\mathrm{d}s$ yields

$$F_{\mathrm{ax}} = \frac{F_{\mathrm{rad}}}{2\tan\beta}.$$

Thus, the axial clamping contribution from belt centrifugal force on pulley j is

$$F_{c,j} = \frac{\rho_b A_b\, \hat{v}_b^2\, \phi_j(s)}{2\tan\beta}. \tag{8.29}$$

8.5 Axial Shift Dynamics

Earlier (see Equation (8.1)), the evolution of the shift coordinate was established as

$$m\, \ddot{s} = F_{p,\mathrm{ax}} - F_{s,\mathrm{ax}}, \tag{8.30}$$

where $F_{p,\mathrm{ax}}$ and $F_{s,\mathrm{ax}}$ are the axial forces generated by the primary and secondary mechanisms, respectively.

We now complete this model by:

1. Expanding the force terms to include centrifugal belt loading,
2. Replacing the lumped mass m with the equivalent inertia associated with coordinated shift motion.

Equivalent Inertia Associated with s Newton's second law applies to physical masses in translation. Since s parameterizes coordinated motion of several bodies, we construct an equivalent mass m_{eq} such that the total kinetic energy associated with shift can be written in the form of Theory 4 (Kinetic Energy (Translational)):

$$T_{\mathrm{shift}} = \frac{1}{2}\, m_{\mathrm{eq}}\, \dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$ in the positive coordinate direction. The movable secondary sheave translates with equal magnitude speed (but opposite physical direction), so its kinetic energy also contributes proportionally to $\dot{s}^2$.

Let $m_{p,m}$ and $m_{s,m}$ denote the masses of the movable primary and secondary sheaves.

Belt Mass Contribution The total belt mass is

$$m_b = \rho_b A_b L_b, \tag{8.31}$$

where A_b is the trapezoidal cross-sectional area from Equation (8.17) and L_b is the constant belt length.

The belt moves continuously with a single belt speed, but the axial component of that motion associated with a change in s is determined by geometry. From the axial-radial coupling in Equation (7.28), a change in sheave separation s is shared between the two conical faces supporting the belt wedge. Thus, the belt's axial velocity relative to the pulley axis is

$$\dot{s}_b = \frac{\dot{s}}{2}. \tag{8.32}$$

The kinetic energy associated with this axial motion is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2}\right)^2 = \frac{1}{2}\left(\frac{m_b}{4}\right)\dot{s}^2.$$

Expressed in the generalized coordinate s, the belt contributes an effective mass of $m_b/4$.

Total Equivalent Mass The total inertia associated with shift becomes

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \tag{8.33}$$

Net Axial Force in the s Coordinate Collecting the axial force contributions derived previously:

- $F_{p,\text{ax}}(s, \omega_p)$ from Equation (8.10),
- $F_{s,\text{ax}}(s, \tau_s)$ from Equation (8.16),
- $F_{c,p}(s, \hat{v}_b)$ from Equation (8.29) with $j = p$,
- $F_{c,s}(s, \hat{v}_b)$ from Equation (8.29) with $j = s$.

With the sign convention defined above, the generalized force in the positive s direction is

$$\sum F_s = \Big(F_{p,\text{ax}} + F_{c,p}\Big) - \Big(F_{s,\text{ax}} + F_{c,s}\Big). \tag{8.34}$$

Shift Equation of Motion Applying Newton's second law in the generalized coordinate,

$$\sum F_s = m_{\text{eq}}\, \ddot{s},$$

and substituting Equation (8.33) and Equation (8.34), we obtain

Axial Shift Acceleration **(8.35)**

$$\ddot{s} = \frac{\Big(F_{p,\text{ax}}(s,\omega_p) + F_{c,p}(s,v_b)\Big) - \Big(F_{s,\text{ax}}(s,\tau_s) + F_{c,s}(s,v_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}.$$

Equation (8.35) provides the axial shift acceleration as a function of the current state variables $(s, \dot{s}, \omega_p, \omega_s)$ and the reacted secondary torque τ_s. The remaining coupling between axial and rotational dynamics enters through τ_s, which will be determined later using Theory 2 (Newton's Second Law (Rotation)).

8.6 Section Summary and Forward Outlook

In this section we established a complete axial force model for the CVT mechanism and derived a closed-form equation of motion for the generalized shift coordinate $s(t)$.

At this stage, the axial subsystem is formally closed with respect to the state variables of the model, but it still depends on one remaining coupling quantity that is not determined internally by the axial force balance: the reacted secondary torque τ_s. Although τ_s appears explicitly in the secondary axial force model, its value is determined by the rotational dynamics of the drivetrain and by torque transmission through the belt. Thus, aside from the shared state variables, the axial and rotational subsystems are coupled through this torque term.

The next step is to determine how torque is transmitted across the CVT as a function of pulley radii and speed ratio. This will establish the relationship between τ_s, input torque at the primary, and the evolving ratio $R = r_s/r_p$. Once this transmission relationship is obtained, the axial and rotational equations can be combined into a unified, state-space representation of the full CVT dynamics.

With the axial force structure now complete, attention turns to the torque transmission mechanism that closes the remaining dynamical loop.

9 Belt Motion, Torque Transmission, and Contact States

The axial force model derived in Section 8 depends on quantities that are not determined by axial motion alone. These remaining quantities enter through the belt, which couples the primary and secondary pulleys in two distinct ways. First, the belt moves through the pulley system with a transport speed that enters the belt centrifugal loading terms. Second, through frictional contact with the pulley faces, it transmits tangential force and therefore torque between the pulleys. The purpose of this section is to introduce the belt as the coupling medium of the CVT, define the belt transport motion and torque path through the system, and then distinguish the stick and slip states that govern transmission behavior.

9.1 The Belt as the Coupling Medium

The primary and secondary pulleys do not interact directly. All mechanical interaction between them occurs through the belt. The belt is therefore the coupling medium of the CVT.

This coupling occurs in two distinct ways.

First, the belt moves through the pulley system with a transport speed. Because the belt has mass, this motion contributes to the dynamics through belt-speed-dependent effects such as centrifugal loading.

Second, the belt transmits tangential traction through frictional contact with the pulley faces. This tangential interaction is what allows torque to pass from the primary to the secondary.

These two roles are illustrated in Figure 24. The primary pulley drives the belt, the belt moves through the transmission path, and the belt in turn drives the secondary pulley.

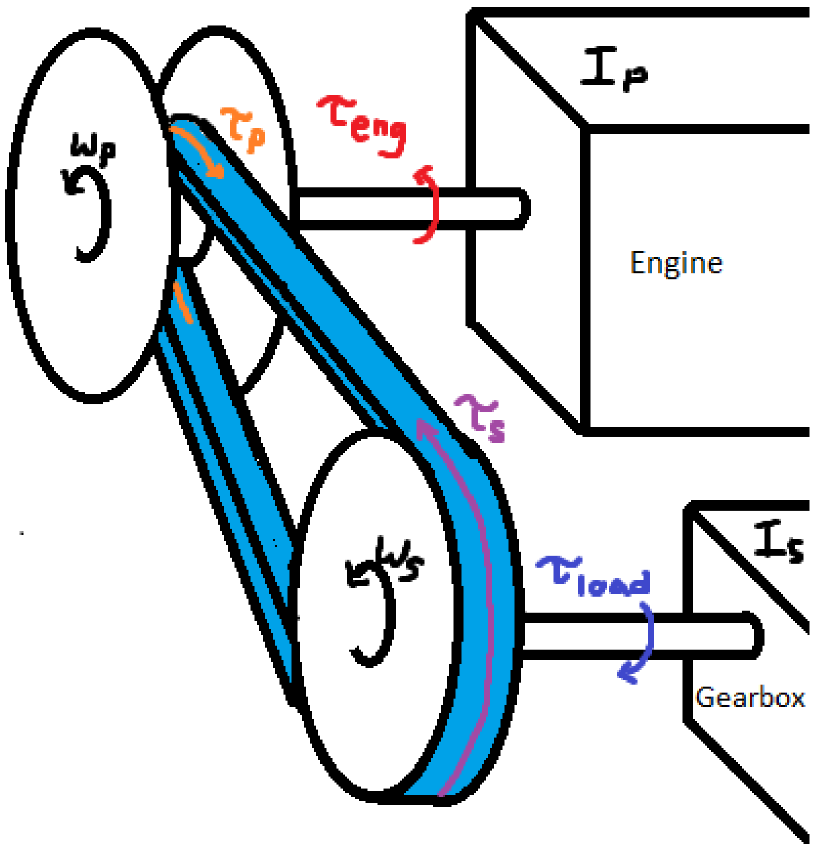


Figure 24: Belt-mediated coupling in the CVT. The primary pulley applies torque to the belt, the belt moves through the transmission path, and the belt applies torque to the secondary pulley. Through frictional contact, the belt also transmits tangential force between the pulley faces.

To describe this interaction, the following belt-mediated quantities will be used:

$$\tau_p : \text{ torque applied by the primary pulley onto the belt,} \tag{9.1}$$

$$\tau_s : \text{ torque applied by the belt onto the secondary pulley,} \tag{9.2}$$

$$F_t : \text{ effective tangential force transmitted through the belt,} \tag{9.3}$$

$$\hat{v}_b : \text{ modeled belt transport speed.} \tag{9.4}$$

The quantities τ_p and τ_s describe how torque enters and leaves the belt transmission path. The quantity F_t describes the tangential traction carried through the belt. The quantity $\hat{v}_b$ describes the motion of the belt itself through the pulley system.

The remainder of this section develops these belt-mediated quantities in order. We first examine the transport motion of the belt, then the tangential force and torque path carried through it, and finally the contact states that govern whether the transmission is in stick or slip.

9.2 Belt Transport Motion

Before torque transmission can be described, the motion of the belt itself must be established.

Under Assumption 8 (Constant Belt Length), the belt is assumed to remain taut and its total length is fixed. As a result, the belt does not admit independent local stretching or compression along different portions of its path. Instead, it moves through the CVT as a single continuous body with one transport speed at any given instant.

To represent this motion in the dynamic model, let

$$\hat{v}_b(t)$$

denote the modeled belt transport speed. This quantity describes the rate at which belt material moves along the transmission path and is the belt-speed state introduced in Section 6.

If a belt segment is adhering locally to pulley $j \in \{p, s\}$, then the belt speed at that contact must match the circumferential speed of the pulley at the radius where the belt mass is modeled to move. Using the centroid radius from Equation (8.22), the pulley-imposed belt-line speeds are therefore

$$v_{p,\mathrm{cm}} = r_{p,\mathrm{cm}}(s)\,\omega_p, \qquad v_{s,\mathrm{cm}} = r_{s,\mathrm{cm}}(s)\,\omega_s. \tag{9.5}$$

These quantities represent the belt transport speeds that would be imposed by the primary and secondary pulleys, respectively, if the belt were adhering locally at each contact.

If both pulley contacts are adhering simultaneously, then the belt cannot move at two different speeds at once. The transport speed must therefore satisfy the compatibility condition

$$\hat{v}_b = v_{p,\mathrm{cm}} = v_{s,\mathrm{cm}}. \tag{9.6}$$

This is the kinematic condition associated with sticking contact.

If the two pulley-imposed belt-line speeds are unequal, then simultaneous adherence at both contacts is not kinematically possible. In that case, the transmission must depart from the sticking condition and enter a slipping state, which is discussed later in this section.

With the belt transport motion now identified, the next step is to determine how tangential force and power are carried through this moving belt.

9.3 Belt Power Flow and Torque Transmission

With the belt transport motion now identified, the next step is to determine how force and power are carried through the moving belt.

Since the belt is the only mechanical connection between the pulleys, it must carry the power transmitted through the CVT.

Torque transmission arises from tangential forces acting between the belt and the pulley faces. Let F_t denote the effective tangential force transmitted through the belt.

The mechanical power carried by the belt is therefore

$$P = F_t \, \hat{v}_b. \tag{9.7}$$

Under Assumption 13 (Ideal CVT Power Transmission), this power is conserved through the transmission, so that the power entering the belt at the primary equals the power leaving at the secondary.

The same transmitted tangential force F_t acts at the effective contact radius of each pulley. The torques associated with this force are therefore

$$\tau_p = r_{p,\text{eff}} \, F_t, \tag{9.8}$$

$$\tau_s = r_{s,\text{eff}} \, F_t. \tag{9.9}$$

Here, $r_{p,\text{eff}}$ and $r_{s,\text{eff}}$ are the effective contact radii defined by Equation (7.21) and Equation (7.22), respectively. These radii determine the torque produced by the transmitted tangential force. Since both torques arise from the same transmitted force, F_t can be eliminated directly. From the primary expression,

$$F_t = \frac{\tau_p}{r_{p,\text{eff}}}. \tag{9.10}$$

Substituting this into the secondary torque relation gives

$$\tau_s = r_{s,\text{eff}} \left(\frac{\tau_p}{r_{p,\text{eff}}} \right) = \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}} \, \tau_p. \tag{9.11}$$

Recalling the geometric definition of the CVT ratio from Equation (7.23),

$$R = \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}}, \tag{9.12}$$

it follows that

$$\tau_s = R \, \tau_p. \tag{9.13}$$

Thus, the torque carried through the belt scales directly with the current CVT ratio. The primary and secondary torques are not independent quantities, but two descriptions of the same belt-mediated transmission process.

This observation motivates the introduction of a single *coupling torque*.

$$\tau_c \equiv \tau_p. \tag{9.14}$$

The corresponding secondary torque is then

$$\tau_s = R\,\tau_c. \tag{9.15}$$

The transmitted torque is thus not a property of either pulley in isolation, but of the belt–pulley interaction as a whole. The quantities τ_p and τ_s represent the primary-side and secondary-side expressions of this same coupling torque.

9.4 Transmission Regimes

The coupling torque τ_c defined in Equation (9.13) is the torque actually transmitted through the belt. It is not prescribed externally. Instead, its value depends on the contact state of the transmission.

In Section 9.2, the belt was introduced as a moving body with modeled transport speed $\hat{v}_b$. That quantity describes the motion of belt material through the CVT and is used in belt-speed-dependent effects such as centrifugal loading. The present subsection addresses a different issue: whether the belt is locally adhering or slipping at the pulley contacts. For this purpose, the relevant kinematic quantities are the tangential contact speeds imposed by the pulley surfaces.

No-slip torque. If the belt remains adhered at the pulley contacts, then the transmission behaves as a sticking constraint. In that case, the transmitted torque is whatever value is required to preserve that adhered motion.

We denote this required torque by

$$\tau_{\mathrm{ns}}, \tag{9.16}$$

where the subscript "ns" denotes *no slip*. Thus, in a sticking state,

$$\tau_c = \tau_{\mathrm{ns}}. \tag{9.17}$$

Traction limits. Adherence cannot be maintained for arbitrary transmitted torque. The belt–pulley contact can only sustain a finite tangential traction, and therefore only a finite coupling torque.

We therefore introduce directional traction bounds and write the admissible sticking interval as

$$\tau_- \leq \tau_{\mathrm{ns}} \leq \tau_+. \tag{9.18}$$

If Equation (9.18) holds, then the no-slip torque lies within the available traction limits, so sticking is physically admissible.

If instead τ_{ns} lies outside this interval, then the torque required to maintain adherence cannot be supported by the contact, and slip must occur.

Why a slip metric is still needed. The traction condition alone is not enough to determine the regime. Once slipping has begun, the transmission does not return immediately to stick simply because τ_{ns} later re-enters the admissible interval. Stick is only recovered once the existing relative motion has decayed to zero.

For this reason, one further quantity is needed: a measure of the relative contact-speed mismatch.

In analogy with the pulley-imposed belt-line speeds introduced in Section 9.2, define the pulley-imposed tangential contact speeds at the effective radii by

$$u_p = r_{p,\text{eff}}\omega_p, \qquad u_s = r_{s,\text{eff}}\omega_s. \tag{9.19}$$

These are the tangential speeds that the primary and secondary pulley surfaces would impose at their effective contact radii if each were enforcing local adherence.

Under Assumption 14 (Single Active Slip Interface), the model permits slip at only one pulley contact at a time. A single scalar mismatch is therefore sufficient:

$$v_\Delta = u_p - u_s = r_{p,\text{eff}}\omega_p - r_{s,\text{eff}}\omega_s. \tag{9.20}$$

If

$$v_\Delta = 0, \tag{9.21}$$

then the two pulley motions are compatible with a common adhered contact motion, so a sticking state is kinematically possible.

If instead

$$v_\Delta \neq 0, \tag{9.22}$$

then the two pulley contacts are demanding different tangential contact speeds, and simultaneous adherence at both contacts is not possible.

The sign of v_Δ indicates the slip direction: if $v_\Delta > 0$, the primary tends to drive the belt faster than the secondary is accepting it; if $v_\Delta < 0$, the opposite is true.

Stick and slip states. The transmission can therefore be in stick only when two conditions are satisfied simultaneously:

1. the required no-slip torque lies within the admissible traction interval, and
2. the relative slip velocity has vanished.

This gives the regime structure

$$\tau_c = \begin{cases} \tau_{\text{ns}}, & v_\Delta = 0 \text{ and } \tau_- \le \tau_{\text{ns}} \le \tau_+, \\ \text{traction-limited value}, & \text{otherwise}. \end{cases} \tag{9.23}$$

Physical interpretation. If the required no-slip torque leaves the admissible interval, the contact can no longer maintain adherence and slip begins.

Once slip is present, the transmission remains on a slip branch until the existing mismatch has been driven back to zero. This is why the slip metric is needed in addition to the traction limits: the traction limits determine whether sticking is admissible, while v_Δ determines whether slipping is still occurring and, if so, in which direction. Importantly, this does not determine at *which* contact slip occurs, only which is overruning the other.

9.5 Regularized Belt-Speed Evolution

The modeled belt transport speed $\hat{v}_b$ introduced in Section 9.2 enters the axial force model through the belt centrifugal terms derived in Section 8.4. Its evolution must therefore be specified as part of the complete dynamic model.

Exact closure in stick. From Equation (9.5), the pulley-imposed belt-line speeds are

$$v_{p,\mathrm{cm}} = r_{p,\mathrm{cm}}(s)\,\omega_p, \qquad v_{s,\mathrm{cm}} = r_{s,\mathrm{cm}}(s)\,\omega_s.$$

When the transmission is in a sticking state, these two speeds are compatible. In that case, the belt transport speed is determined exactly by the adhered kinematics, so that

$$\hat{v}_b = v_{p,\mathrm{cm}} = v_{s,\mathrm{cm}}. \tag{9.24}$$

In the same way, the time derivative of the belt transport speed is also determined exactly in stick by differentiating the compatibility relation. Using the primary-side expression gives

$$\dot{\hat{v}}_b = \frac{\mathrm{d}}{\mathrm{d}t}(r_{p,\mathrm{cm}}\,\omega_p) = \dot{r}_{p,\mathrm{cm}}\,\omega_p + r_{p,\mathrm{cm}}\,\dot{\omega}_p, \tag{9.25}$$

and, equivalently, using the secondary-side expression gives

$$\dot{\hat{v}}_b = \frac{\mathrm{d}}{\mathrm{d}t}(r_{s,\mathrm{cm}}\,\omega_s) = \dot{r}_{s,\mathrm{cm}}\,\omega_s + r_{s,\mathrm{cm}}\,\dot{\omega}_s. \tag{9.26}$$

Thus, no additional approximation is required while the contacts remain stuck.

Need for a slip closure. When the transmission is slipping, the two pulley-imposed belt-line speeds no longer agree. At that point, the exact belt transport speed is no longer determined by kinematic compatibility alone. A fully resolved treatment would require a simultaneous branch-dependent solution for contact state, transmitted torque, and adhered belt motion.

Such a closure is beyond the level of detail adopted in the present model. Instead, the belt transport speed is represented during slip by a regularized dynamic approximation. This preserves a continuous belt-speed state for use in belt-speed-dependent terms, while avoiding a substantially more complicated contact solve.

Regularized target speed. To construct this approximation, define the target belt speed

$$v_b^* = \frac{1}{2}\left(v_{p,\mathrm{cm}} + v_{s,\mathrm{cm}}\right). \tag{9.27}$$

This quantity represents the midpoint of the two pulley-imposed belt-line speeds and therefore provides a natural reference speed during slip.

Evolution law. Outside the sticking regime, the modeled belt speed is evolved toward v_b^* using the first-order relaxation law

$$\dot{\hat{v}}_b = \frac{v_b^* - \hat{v}_b}{T_b}, \tag{9.28}$$

where $T_b > 0$ is a belt-speed regularization time constant.

A small value of T_b causes $\hat{v}_b$ to track the target speed rapidly, while a larger value produces a more gradual response. In either case, $\hat{v}_b$ remains continuous through transitions between stick and slip.

Resulting closure. The modeled belt transport speed is therefore obtained from the regime-dependent rule

$$\hat{v}_b = \begin{cases} v_{p,\mathrm{cm}} = v_{s,\mathrm{cm}}, & \text{sticking regime,} \\ \text{solution of } \dot{\hat{v}}_b = \dfrac{v_b^* - \hat{v}_b}{T_b}, & \text{slipping regime.} \end{cases} \tag{9.29}$$

Its corresponding time derivative is

$$\dot{\hat{v}}_b = \begin{cases} \dot{r}_{p,\mathrm{cm}}\,\omega_p + r_{p,\mathrm{cm}}\,\dot{\omega}_p = \dot{r}_{s,\mathrm{cm}}\,\omega_s + r_{s,\mathrm{cm}}\,\dot{\omega}_s, & \text{sticking regime,} \\ \dfrac{v_b^* - \hat{v}_b}{T_b}, & \text{slipping regime.} \end{cases} \tag{9.30}$$

This closure is exact in stick and regularized in slip. Its purpose is not to resolve the detailed local adhered contact during slip, but to provide a consistent and continuous modeled belt speed for the belt-speed-dependent terms already present in the dynamic formulation.

9.6 Summary and Forward Outlook

This section established the belt-level quantities required to couple the axial and rotational subsystems of the CVT model.

The belt was introduced as the mechanical connection between the pulleys, with two distinct dynamical roles: it transports belt mass through the system and it transmits tangential force and therefore torque between the pulley faces. On this basis, the modeled belt transport speed $\hat{v}_b$ and the coupling torque τ_c were defined, and the stick and slip regimes governing transmission behavior were identified. A regularized evolution law for $\hat{v}_b$ was then introduced to close the belt-speed-dependent terms of the model during slip while preserving exact kinematic compatibility in stick.

At this stage, the transmission structure has been established, but two ingredients remain to be determined explicitly: the no-slip torque τ_{ns} and the traction limits τ_- and τ_+. The next section derives the no-slip torque required to maintain adhered motion. The section following that then derives the admissible traction bounds that determine when sticking can be sustained and when the transmission must remain on a slip branch.

10 Derivation of the No-Slip Coupling Torque

In Section 9.4, the coupling torque τ_c was introduced as an internal torque whose value depends on the active transmission regime. The purpose of this section is to derive the explicit form of τ_c on the sticking branch, i.e. the no-slip torque τ_{ns}.

We proceed in four steps:

1. Write the rotational equations of motion for the primary and secondary pulleys.
2. Recall the external torque and inertia models on each side.
3. Impose the no-slip kinematic constraint.
4. Eliminate angular accelerations to obtain the no-slip coupling torque.

The result is the sticking-branch torque law used later in the regime model.

10.1 System Definition and Torque Path

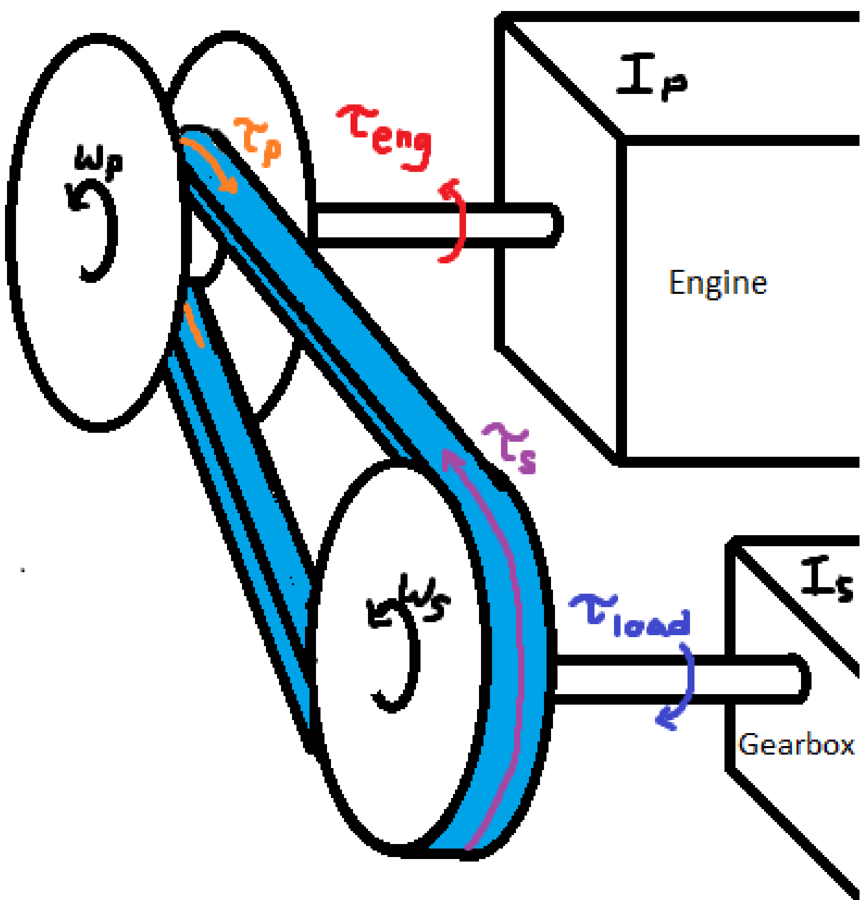


Figure 25: Torque transmission schematic. Engine torque τ_{eng} drives the primary pulley (angular velocity ω_p). Coupling torque τ_c is transmitted through the belt to the secondary pulley (angular velocity ω_s). The secondary delivers torque to the downstream drivetrain, which resists motion with torque τ_{load}.

We model two rotating bodies:

- the primary side, rigidly connected to the engine,
- the secondary side, driving the downstream drivetrain and vehicle.

The external torques and effective inertias used below are

- $\tau_{\text{eng}}(t)$: engine output torque applied to the primary pulley,
- $\tau_{\text{load}}(t)$: resisting torque reflected to the secondary pulley,
- I_p : total rotational inertia of the engine + primary CVT pulley,
- I_s : total effective rotational inertia about the secondary pulley.

Their explicit models are recalled in the following subsections. The remaining internal unknown is the transmitted coupling torque $\tau_c(t)$.

10.2 Rotational Equations of Motion

Applying Theory 2 (Newton's Second Law (Rotation)) to the primary side gives

Primary Pulley Rotational Dynamics **(10.1)**

$$\tau_{\text{eng}}(t) - \tau_c(t) = I_p\,\dot{\omega}_p(t)$$

Using the secondary-side no-slip torque mapping established in Equation (9.13), the secondary rotational equation on the sticking branch is

Secondary Pulley Rotational Dynamics **(10.2)**

$$R(t)\tau_c(t) - \tau_{\text{load}}(t) = I_s\,\dot{\omega}_s(t)$$

At this point, Equation (10.1) and Equation (10.2) provide two equations for the three internal unknowns

$$\dot{\omega}_p(t), \qquad \dot{\omega}_s(t), \qquad \tau_c(t).$$

The remaining quantities appearing in these equations — namely the applied torques $\tau_{\text{eng}}(t)$ and $\tau_{\text{load}}(t)$, the effective inertias I_p and I_s, and the geometric ratio $R(t)$ — are treated as known inputs, parameters, or previously defined geometric quantities. Their explicit forms are recalled in the following subsections.

Thus, the rotational system is not yet closed: two equations are available for three unknowns. A third relation is therefore required to determine the coupling torque. For the no-slip branch, that final relation will come from the sticking kinematic constraint introduced after the torque and inertia models are stated.

10.3 Engine Torque Model

The engine provides the driving torque $\tau_{\text{eng}}(t)$ acting on the primary pulley.

In this model, engine output torque is represented as a user-supplied function of primary angular velocity. Under the full-throttle assumption (Assumption 16 (Full-Throttle Engine Model)),

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}\big(\omega_p(t)\big), \tag{10.3}$$

so torque depends only on instantaneous primary speed.

This representation is consistent with standard engine characterization, where torque is measured as a function of crankshaft speed under steady full-load conditions.

Admissibility Constraints The supplied torque function must satisfy:

- It is defined over a bounded operating domain $\omega_{\text{min}} \leq \omega_p \leq \omega_{\text{max}}$.
- It is finite and real-valued over this domain.
- Outside this domain, torque is either saturated or ω_p is constrained to remain within bounds.

No specific functional form is required. The dynamical model only requires the mapping $\omega_p \mapsto \tau_{\text{eng}}$ to evaluate Equation (10.1).

Example: Kohler CH440 Engine Curve As an illustrative example, Figure 26 shows a representative torque and power curve for the Kohler CH440 engine.

SAE Baja Restricted Kohler CH440							
Engine Speed (rpm)	3600	3400	3200	3000	2800	2600	2400
Torque Net Corrected (ft-lb)	13.5	14.5	15.4	16.6	17.4	18.1	18.5
Power Net Corrected (HP)	9.3	9.4	9.4	9.5	9.3	9.0	8.5

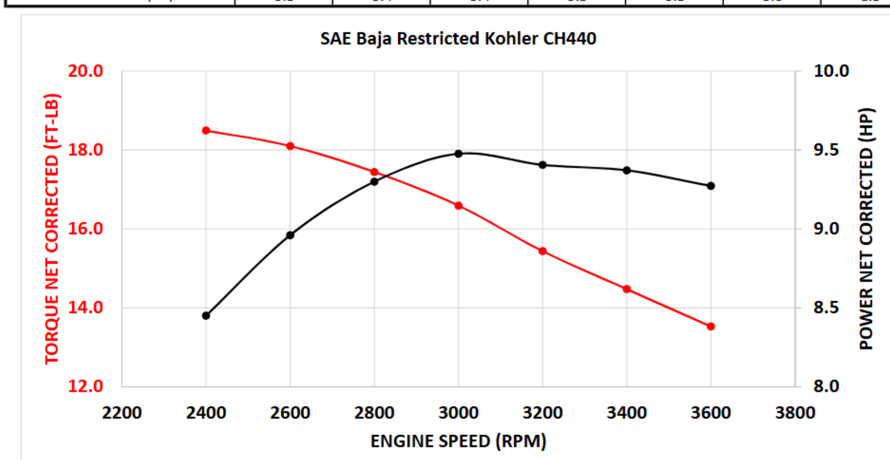


Figure 26: Representative torque and power curves for the Kohler CH440 engine. Torque is supplied as a function of angular velocity and serves as the input mapping in Equation (10.3).

In simulation, the discrete curve is interpolated to provide a continuous admissible torque function over the operating range.

10.4 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ in Equation (10.2) represents the longitudinal road load reflected to the secondary pulley.

We proceed in three stages:

1. Express vehicle speed in terms of secondary rotation.
2. Derive the total longitudinal road force.
3. Reflect that force back to the secondary shaft.

Kinematic Relations Let r_w denote effective wheel rolling radius and $G > 0$ the fixed gear ratio defined by

$$\omega_s = G\,\omega_w.$$

Vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G}\,\omega_s(t). \tag{10.4}$$

Longitudinal Road Force Using Theory 1 (Newton's Second Law (Translation)), the total longitudinal resistive force on the vehicle is modeled as

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \tag{10.5}$$

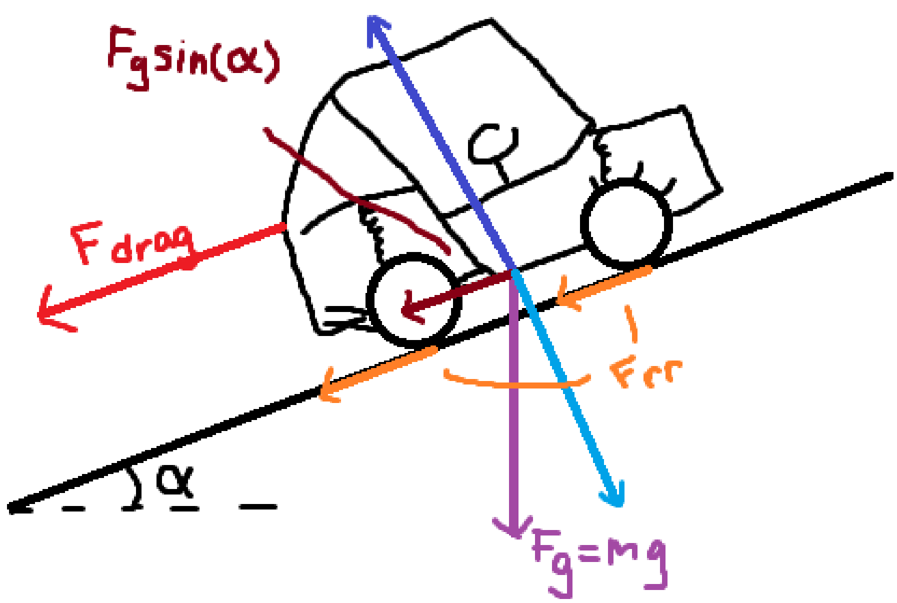


Figure 27: Free-body diagram of a vehicle on an incline. The weight mg resolves into $mg\sin\alpha$ along the slope and $mg\cos\alpha$ normal to the surface. Rolling resistance and aerodynamic drag oppose motion.

Rolling Resistance From Theory 13 (Rolling Resistance),

$$F_{\mathrm{rr}} = C_{\mathrm{rr}}\, N. \tag{10.6}$$

From the incline geometry,

$$N = mg\cos\alpha. \tag{10.7}$$

Thus

$$F_{\mathrm{rr}}(v,\alpha) = C_{\mathrm{rr}} mg\cos\alpha\ \mathrm{sgn}(v). \tag{10.8}$$

Grade Resistance From Theory 14 (Gravitational Force), the gravitational component along the slope is

$$F_{\mathrm{grade}}(\alpha) = mg\sin\alpha. \tag{10.9}$$

Aerodynamic Drag From Theory 12 (Air Resistance (Quadratic Drag)),

$$F_{\mathrm{drag}}(v) = \frac{1}{2}\rho C_d A_f\, v^2\, \mathrm{sgn}(v). \tag{10.10}$$

Total Road Force Substituting the above components into Equation (10.5) yields

$$\begin{aligned} F_{\mathrm{road}}(t) = {} & C_{\mathrm{rr}} mg\cos\alpha\ \mathrm{sgn}(v(t)) \\ & + mg\sin\alpha \\ & + \frac{1}{2}\rho C_d A_f\, v(t)^2\, \mathrm{sgn}(v(t))\,. \end{aligned} \tag{10.11}$$

Using Equation (10.4),

$$v(t) = \frac{r_w}{G}\,\omega_s(t),$$

the road force is now fully expressed in terms of the secondary state variable $\omega_s(t)$.

Reflection to Secondary Torque Wheel torque is

$$\tau_{\mathrm{road},w}(t) = r_w F_{\mathrm{road}}(t). \tag{10.12}$$

Assuming ideal power transmission (Assumption 13 (Ideal CVT Power Transmission)), the resisting torque reflected to the secondary pulley is

$$\tau_{\text{load}}(t) = \frac{1}{G}\,\tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \tag{10.13}$$

Substituting Equation (10.11) gives

$$\boxed{\tau_{\text{load}}(t) = \frac{r_w}{G}\left[C_{\text{rr}} mg\cos\alpha\;\text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big) + mg\sin\alpha + \frac{1}{2}\rho C_d A_f\left(\frac{r_w}{G}\omega_s(t)\right)^2 \text{sgn}\Big(\frac{r_w}{G}\omega_s(t)\Big)\right].} \tag{10.14}$$

10.5 Inertia Definitions

The rotational equations Equation (10.1)–Equation (10.2) require the effective rotational inertia about each pulley axis.

Primary Pulley Inertia The primary pulley is rigidly connected to the engine. All components on this side rotate with angular velocity ω_p, so no reflection is required.

The effective inertia is therefore the direct sum

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \tag{10.15}$$

where

- I_{eng} is the engine rotational inertia,
- I_{prim} is the primary CVT pulley inertia.

Secondary Pulley Inertia On the secondary side, all downstream components must be expressed as an equivalent inertia about the secondary axis.

Let

- I_{sec} : secondary CVT pulley inertia,
- I_{gb} : gearbox + intermediate shaft inertia,
- I_{wheel} : wheel + hub inertia about the wheel axis,
- m : total vehicle mass,
- r_w : effective wheel rolling radius,
- G : fixed gear ratio defined by $\omega_s = G\,\omega_w$.

Vehicle Mass as Equivalent Rotational Inertia The vehicle's translational kinetic energy is

$$T_{\text{trans}} = \frac{1}{2}mv^2,$$

with $v = r_w\omega_w$.

Substituting,

$$T_{\text{trans}} = \frac{1}{2} m (r_w \omega_w)^2 = \frac{1}{2} (m r_w^2)\, \omega_w^2.$$

Thus the vehicle mass behaves as an effective rotational inertia

$$I_{\text{veh,wheel}} = m r_w^2$$

about the wheel axis.

Reflection to the Secondary Axis Because $\omega_s = G\,\omega_w$, we have $\omega_w = \omega_s / G$. Expressing the downstream kinetic energy in terms of ω_s:

$$T = \frac{1}{2} I_{\text{wheel}} \left(\frac{\omega_s}{G}\right)^2 + \frac{1}{2} (m r_w^2) \left(\frac{\omega_s}{G}\right)^2.$$

Factoring,

$$T = \frac{1}{2} \left(\frac{I_{\text{wheel}} + m r_w^2}{G^2} \right) \omega_s^2.$$

Therefore, inertias at the wheel reflect to the secondary pulley as

$$I_{\text{reflected}} = \frac{I_{\text{wheel}} + m r_w^2}{G^2}.$$

Total Secondary Inertia The total effective inertia about the secondary pulley is

$$I_s = I_{\text{sec}} + I_{\text{gb}} + \frac{I_{\text{wheel}} + m r_w^2}{G^2}. \tag{10.16}$$

Equation Equation (10.16) defines the total inertia opposing angular acceleration of the secondary pulley.

10.6 No-Slip Ratio Kinematics

As established in Section 9.4, the sticking branch is characterized by the condition

$$v_\Delta = 0,$$

where v_Δ was defined in Equation (9.20) as

$$v_\Delta = r_{p,\text{eff}}(t)\, \omega_p(t) - r_{s,\text{eff}}(t)\, \omega_s(t).$$

Setting this equal to zero gives

$$r_{p,\text{eff}}(t)\, \omega_p(t) = r_{s,\text{eff}}(t)\, \omega_s(t). \tag{10.17}$$

Using the geometric CVT ratio

$$R(t) = \frac{r_{s,\text{eff}}(t)}{r_{p,\text{eff}}(t)},$$

Equation (10.17) becomes

$$\omega_p(t) = R(t)\,\omega_s(t). \tag{10.18}$$

This is the defining kinematic relation of the sticking branch. If Equation (10.18) does not hold, the present derivation no longer applies and the transmission must instead be described by the slip branch.

Differentiating Equation (10.18) with respect to time gives

$$\dot{\omega}_p(t) = R(t)\,\dot{\omega}_s(t) + \omega_s(t)\,\dot{R}(t), \tag{10.19}$$

which is the corresponding acceleration-level compatibility condition.

10.7 Eliminating Angular Accelerations

We now combine the primary rotational dynamics Equation (10.1), the secondary rotational dynamics Equation (10.2), and the no-slip compatibility relation Equation (10.19) to solve for the coupling torque on the sticking branch.

From Equation (10.1),

$$\tau_c = \tau_{\text{eng}} - I_p\,\dot{\omega}_p.$$

Substituting Equation (10.19) gives

$$\tau_c = \tau_{\text{eng}} - I_p\big(R\dot{\omega}_s + \omega_s\dot{R}\big). \tag{10.20}$$

From Equation (10.2),

$$\dot{\omega}_s = \frac{R\tau_c - \tau_{\text{load}}}{I_s}.$$

Substituting this into Equation (10.20) gives

$$\begin{aligned}\tau_c &= \tau_{\text{eng}} - I_p\left[R\left(\frac{R\tau_c - \tau_{\text{load}}}{I_s}\right) + \omega_s\dot{R}\right] \\ &= \tau_{\text{eng}} - \frac{I_p}{I_s}\big(R^2\tau_c - R\tau_{\text{load}}\big) - I_p\omega_s\dot{R}. \end{aligned} \tag{10.21}$$

Collecting terms yields

$$\tau_c\left(1 + \frac{I_p}{I_s}R^2\right) = \tau_{\text{eng}} + \frac{I_p}{I_s}R\tau_{\text{load}} - I_p\omega_s\dot{R}. \tag{10.22}$$

Because this derivation has imposed the sticking constraint Equation (10.18), the resulting solution is precisely the no-slip coupling torque.

10.8 Closed-Form No-Slip Coupling Torque

The no-slip coupling torque is therefore

No-Slip Coupling Torque **(10.23)**

$$\tau_{\rm ns}(t) = \frac{\tau_{\rm eng}(t) + \dfrac{I_p}{I_s} R(t)\, \tau_{\rm load}(t) - I_p\, \omega_s(t)\, \dot{R}(t)}{1 + \dfrac{I_p}{I_s} R(t)^2}.$$

Equation Equation (10.23) is the torque required to maintain the no-slip constraint Equation (10.18), or equivalently the condition $v_\Delta = 0$.

10.9 Section Summary

This section derived the no-slip coupling torque $\tau_{\rm ns}$ by specializing the rotational dynamics to the sticking branch.

The derivation used three ingredients:

- the primary and secondary rotational equations of motion written in terms of the internal coupling torque τ_c,
- the no-slip kinematic constraint $\omega_p = R\,\omega_s$ and its time derivative $\dot{\omega}_p = R\dot{\omega}_s + \omega_s \dot{R}$,
- the no-slip torque mapping $\tau_{s,\rm ns} = R\,\tau_c$.

Together, these relations close the sticking branch and yield the explicit result Equation (10.23). This quantity is the ideal adhered-branch coupling torque: the value required to maintain sticking kinematics if adherence can be sustained. The next sections determine whether this required torque is admissible under the traction limits and, when it is not, how the simulation transitions to and evolves on the slip branch.

11 Traction Limits and Slip Conditions

In Section 9.4, the sticking branch was defined by the requirement that the no-slip coupling torque remain within an admissible traction interval. In Section 10, the corresponding no-slip torque $\tau_{\rm ns}$ was derived explicitly.

The remaining task is therefore to determine the traction bounds themselves.

Because belt–pulley friction is finite, the contact cannot sustain arbitrary tangential traction. For any instantaneous clamping state, only a bounded range of transmitted coupling torque can be supported before adherence is lost. Once the demanded torque leaves that admissible range, the no-slip branch is no longer physically realizable and the transmission must enter slip.

Accordingly, the objective of this section is to derive bounds of the form

$$\tau_- \le \tau \le \tau_+, \tag{11.1}$$

where τ_- and τ_+ are the negative- and positive-direction traction limits, respectively.

These bounds will later be compared against the no-slip torque τ_{ns} to determine whether the transmission remains in stick or transitions to slip.

11.1 Tight–Slack Tension Difference and Torque Capacity

The same traction-capacity argument applies to both pulleys. Accordingly, in this section we use the subscript

$$j \in \{p, s\},$$

where $j = p$ denotes the primary pulley and $j = s$ denotes the secondary pulley.

Consider the wrapped pulley shown in Figure 28. If torque is transmitted from pulley j to the belt, the belt tension cannot be the same on both spans. One side must carry a higher tension than the other. For a given transmission direction, we therefore define

$$\Delta T_j \equiv T_{j,\mathrm{tight}} - T_{j,\mathrm{slack}}, \tag{11.2}$$

where $T_{j,\mathrm{tight}}$ and $T_{j,\mathrm{slack}}$ are the belt tensions on the two straight spans shown in the figure.

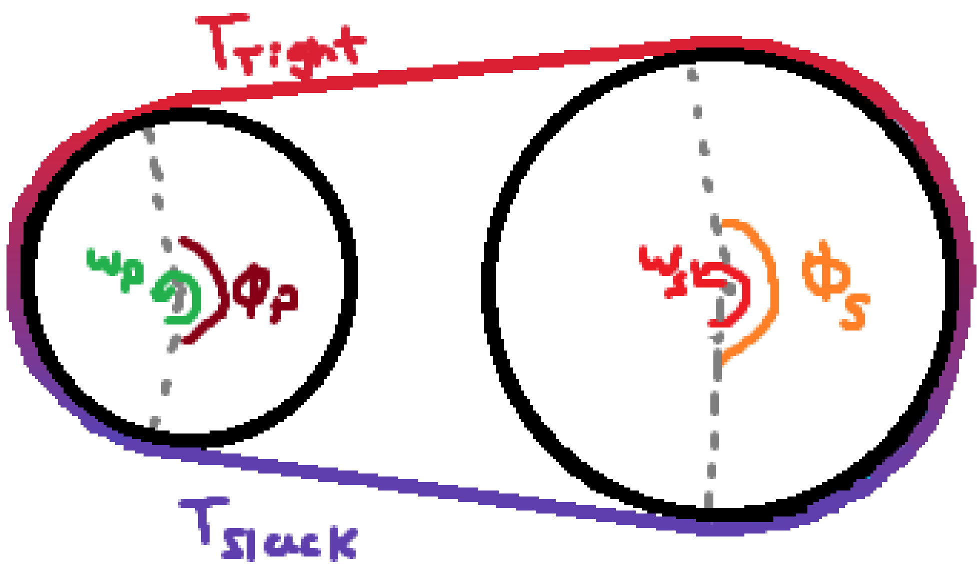


Figure 28: Torque transmission requires a tight–slack tension difference across the wrap.

This tension difference arises because the pulley applies tangential traction to the belt along the wrap angle ϕ_j. The cumulative effect of this traction changes the belt tension progressively from the slack side to the tight side.

At effective contact radius $r_{j,\mathrm{eff}}$, the transmitted torque is

$$\tau_j = r_{j,\mathrm{eff}}\, \Delta T_j, \qquad |\tau_j| = r_{j,\mathrm{eff}}\, |\Delta T_j|. \tag{11.3}$$

Thus, limiting the transmitted torque is equivalent to limiting the admissible tension difference ΔT_j.

Under Assumption 4 (Idealized Belt–Pulley Friction Law) and Assumption 5 (Constant Effective Friction Coefficient), the belt–pulley interface obeys a Coulomb friction law with constant coefficient μ. The tangential traction at any point along the wrap is therefore bounded by the local normal load:

$$|F_f| \le \mu N.$$

The largest admissible tension difference for a given pulley state and transmission direction occurs when friction is fully mobilized everywhere along the wrap, i.e. at impending slip. Beyond this point, additional tension rise cannot be supported and sliding must occur.

We therefore define a maximum supported tension difference $\Delta T_{j,\max}$, corresponding to the friction-saturated state, and the associated torque capacity

$$\tau_{j,\max} = r_{j,\text{eff}}\, \Delta T_{j,\max}. \tag{11.4}$$

The next subsection computes $\Delta T_{j,\max}$ by examining the incremental change in belt tension along the wrap.

Remark (Other Friction Models)

Although the present simulator adopts a Coulomb friction law for simplicity and analytic transparency, richer friction descriptions are available if greater fidelity near the stick–slip transition is desired. A common next step is a Stribeck-type law, in which the effective friction level varies with relative slip speed so that breakaway friction near zero speed can exceed the kinetic friction observed at larger slip speeds. Such a model can better represent engagement and low-slip behaviour than a purely constant Coulomb coefficient, while still remaining relatively simple to implement Armstrong-Hélouvry et al. (1994); de Wit et al. (1995).
For still higher fidelity, dynamic friction models introduce internal state variables that represent the evolving contact condition. The LuGre model is a standard example and can capture presliding displacement, hysteresis, friction lag, and smoother stick–slip transitions, while related multistate models such as the generalized Maxwell–slip model can represent more detailed microslip and memory effects de Wit et al. (1995); Al-Bender et al. (2005). These models are often more physically expressive, but they also require additional parameter identification and increase numerical and modelling complexity. For the present work, Coulomb friction is therefore retained as a deliberate low-complexity approximation, while Stribeck- or state-based friction models remain natural extensions for future refinement Armstrong-Hélouvry et al. (1994).

11.2 Differential Belt Element and Tangential Force Balance

To determine the maximum admissible tension difference $\Delta T_{j,\max}$ introduced in Section 11.1, we now examine how belt tension changes *locally* as the belt passes around a generic pulley $j \in \{p, s\}$.

This derivation uses a differential force balance on a belt element in an inertial frame of reference. Accordingly, the belt element is described through its actual acceleration, so the radial inertia appears through centripetal acceleration rather than through an explicit centrifugal force. This differs from the treatment in Section 8.4, where the same underlying belt inertia on the wrap was represented through an equivalent centrifugal-loading picture in order to derive an axial clamping contribution. That earlier formulation was convenient because the goal there was to obtain a net axial force on the sheaves. Here, by contrast, the goal is to determine how belt tension changes along the wrap, so a local inertial-frame force balance is the more natural description.

Specifically, we seek an expression for the incremental tension change $\mathrm{d}T_j$ over a differential wrap angle $\mathrm{d}\theta$. Once $\mathrm{d}T_j$ is related to the available friction at the interface, integration over the total wrap angle ϕ_j will yield $\Delta T_{j,\max}$.

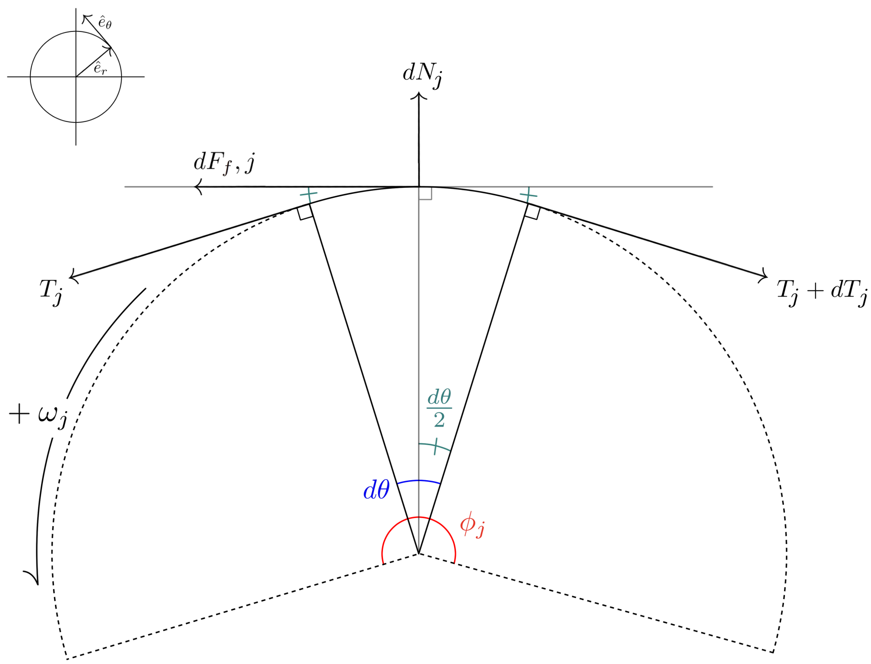


Figure 29: Free-body diagram of an infinitesimal belt segment in contact with pulley j, written in the local radial–tangential coordinate basis $(\hat{e}_r, \hat{e}_\theta)$ shown in the inset. The element subtends an infinitesimal contact angle $\mathrm{d}\theta$ with the total wrap angle ϕ_j. The boundary tensions T_j and $T_j + \mathrm{d}T_j$ act tangentially, while the pulley applies a normal force $\mathrm{d}N_j$ in the radial direction and a friction force $\mathrm{d}F_{f,j}$ in the tangential direction. The positive pulley rotation $+\omega_j$ is shown for reference.

Consider the differential belt element in Figure 29, which spans an infinitesimal wrap angle $\mathrm{d}\theta$ on pulley j. The positive tangential direction is taken along the midpoint tangent of the element, consistent with increasing wrap angle and with increasing belt tension from the slack side toward the tight side.

Tangential Newton's Second Law Applying Theory 1 (Newton's Second Law (Translation)) in the tangential direction gives

$$\sum F_t = \mathrm{d}m_j \, a_{t,j}, \tag{11.5}$$

where $a_{t,j}$ is the tangential acceleration of the belt material along the contact arc on pulley j.

From the element geometry, the tangential components of the boundary tensions are

$$(T_j)_t = T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right), \tag{11.6}$$

$$(T_j + \mathrm{d}T_j)_t = (T_j + \mathrm{d}T_j) \cos\left(\frac{\mathrm{d}\theta}{2}\right). \tag{11.7}$$

The left-end tension acts in the positive tangential direction, while the right-end tension acts in the negative tangential direction. Including the differential friction force $\mathrm{d}F_{f,j}$, the tangential force balance becomes

$$T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) - (T_j + \mathrm{d}T_j) \cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \mathrm{d}m_j \, a_{t,j}. \tag{11.8}$$

Combining the two tension terms gives

$$-\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \mathrm{d}m_j \, a_{t,j}. \tag{11.9}$$

Belt mass element From Equation (8.25), the belt mass associated with the differential contact element on pulley j is

$$\mathrm{d}m_j = \rho_b A_b r_{j,\mathrm{cm}} \, \mathrm{d}\theta. \tag{11.10}$$

Substituting this into Equation (11.9) gives

$$-\mathrm{d}T_j \cos\left(\frac{\mathrm{d}\theta}{2}\right) + \mathrm{d}F_{f,j} = \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \, \mathrm{d}\theta. \tag{11.11}$$

Rearranging for the local tension change yields

$$\mathrm{d}T_j = \frac{\mathrm{d}F_{f,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \, \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{11.12}$$

This expression shows that the local rise in belt tension is produced by the available tangential friction force, reduced by the tangential inertial force required to accelerate the belt element.

Friction limit and impending slip The quantity of interest is the *maximum* admissible tension increase. From Equation (11.12), this occurs when the tangential friction force reaches its largest value, since $\mathrm{d}F_{f,j}$ is the term that directly drives the increase in belt tension.

As discussed in Section 11.1, the maximum tension rise therefore occurs at impending slip, when friction is fully mobilized at the interface. At the differential level, the largest available friction force is

$$\mathrm{d}F_{f,j,\max} = \mu \, \mathrm{d}N_j. \tag{11.13}$$

Substituting Equation (11.13) into Equation (11.12) gives the maximum local tension increase that can be supported without sliding:

$$\mathrm{d}T_{j,\max} = \frac{\mu\ \mathrm{d}N_j - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\ \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{11.14}$$

11.3 Integration Across the Wrap

The tight–slack tension difference introduced in Section 11.1 is obtained by accumulating the incremental tension changes $\mathrm{d}T_j$ along the contact arc of pulley $j \in \{p, s\}$. Since the belt tension evolves continuously around the wrap,

$$\Delta T_j = T_{j,\mathrm{tight}} - T_{j,\mathrm{slack}} = \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}T_j. \tag{11.15}$$

Accordingly, the maximum admissible tension difference is obtained by integrating the local maximum increment $\mathrm{d}T_{j,\max}$ from Equation (11.14):

$$\Delta T_{j,\max} = \int_{-\phi_j/2}^{\phi_j/2} \frac{\mu\ \mathrm{d}N_j - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\ \mathrm{d}\theta}{\cos\left(\frac{\mathrm{d}\theta}{2}\right)}. \tag{11.16}$$

In the differential limit $\mathrm{d}\theta \to 0$, $\cos(\mathrm{d}\theta/2) \to 1$, so this reduces to

$$\Delta T_{j,\max} = \int_{-\phi_j/2}^{\phi_j/2} \mu\ \mathrm{d}N_j - \int_{-\phi_j/2}^{\phi_j/2} \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\ \mathrm{d}\theta. \tag{11.17}$$

The first integral is the total normal load carried over the wrap of pulley j, which we denote by

$$N_{\phi,j} \equiv \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}N_j. \tag{11.18}$$

The explicit relationship between $N_{\phi,j}$ and the axial clamping force is derived in Section 11.5.

From Assumption 9 (Uniform Contact Radius Around Wrap), $r_{j,\mathrm{cm}}$ and $a_{t,j}$ are uniform over the wrap, so the second integral simplifies to $\rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \phi_j$. Hence,

$$\Delta T_{j,\max} = \mu N_{\phi,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j} \phi_j. \tag{11.19}$$

Thus, the maximum admissible tension difference for pulley j depends on two remaining quantities: the total wrap normal load $N_{\phi,j}$ and the tangential belt acceleration $a_{t,j}$.

11.4 Tangential Acceleration of a Belt Element on a Circular Wrap

In Equation (11.19), Newton's second law is applied *along the local direction of belt motion*, so we require an explicit expression for the tangential acceleration $a_{t,j}$ of the belt material on pulley j. Since the inertial term acts on the belt element mass, the relevant kinematics are those of the belt element centroid, located at radius $r_{j,\mathrm{cm}}(t)$.

As discussed in Section 8.4, belt inertia on the wrap may be represented in different but equivalent ways depending on the force balance of interest. There, the goal was to obtain an axial clamping contribution, so the belt inertia was introduced through an equivalent centrifugal-loading picture. Here, by contrast, we require a local force balance on a differential belt element in an inertial frame. In this description, the belt element has an actual acceleration, whose radial component contains the usual centripetal term, while the tangential force balance depends only on the tangential component.

Goal For a belt element of differential mass $\mathrm{d}m_j$ moving along the wrap of pulley j, the tangential force balance uses

$$\sum F_t = \mathrm{d}m_j \, a_{t,j}.$$

Our goal is therefore to compute $a_{t,j}$ from the belt kinematics when both the centroid radius and the modeled belt transport speed may vary:

$$r_{j,\mathrm{cm}}(t), \qquad \dot{r}_{j,\mathrm{cm}}(t), \qquad \hat{v}_b(t), \qquad \dot{\hat{v}}_b(t).$$

Coordinate system and unit vectors Because we care about forces *along the belt*, we choose a basis aligned with the local wrap geometry on pulley j:

- $\mathbf{e}_{r,j}(t)$ points radially outward from the pulley center to the belt point,
- $\mathbf{e}_{\theta,j}(t)$ is the unit tangent, obtained by rotating $\mathbf{e}_{r,j}$ by $+90°$ in the direction of increasing θ_j.

The tangential direction of belt motion is therefore exactly $\mathbf{e}_{\theta,j}$.

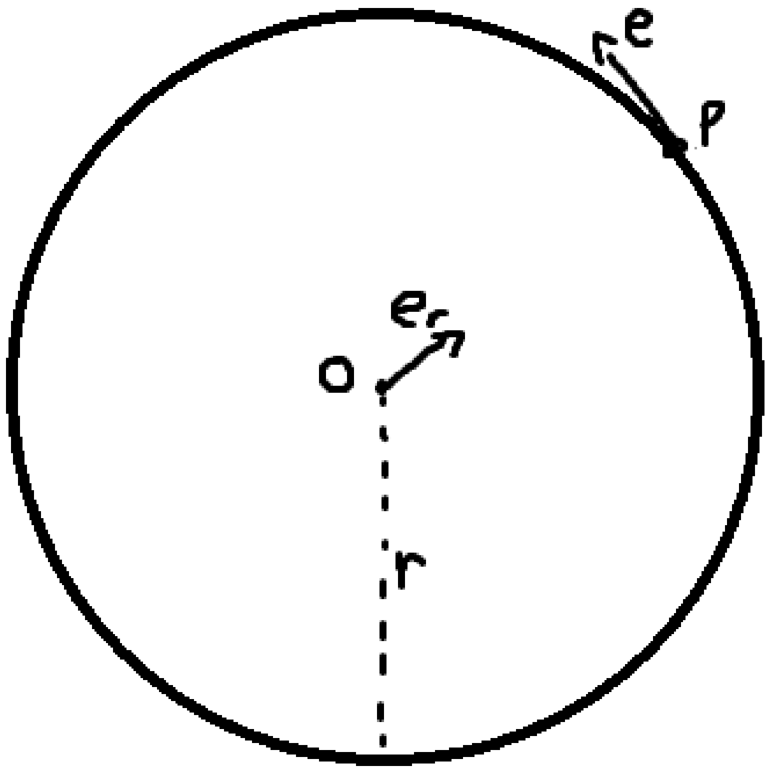


Figure 30: Definition of the radial and tangential unit vectors.

Tangential acceleration as a component of a Tangential acceleration is the component of the acceleration vector along $\mathbf{e}_{\theta,j}$:

$$a_{t,j} \equiv \mathbf{a}_j \cdot \mathbf{e}_{\theta,j}. \tag{11.20}$$

We therefore compute the full acceleration $\mathbf{a}_j = \dot{\mathbf{v}}_j$ and then read off its $\mathbf{e}_{\theta,j}$ component.

Step 1: position and velocity The position of the belt element centroid relative to the pulley center is

$$\mathbf{r}_j(t) = r_{j,\text{cm}}(t)\,\mathbf{e}_{r,j}(t). \tag{11.21}$$

The belt transport speed is the tangential speed of the belt element along the wrap. Since the modeled belt transport speed is $\hat{v}_b$, the velocity of the belt element centroid is

$$\mathbf{v}_j = \dot{r}_{j,\text{cm}}\,\mathbf{e}_{r,j} + \hat{v}_b\,\mathbf{e}_{\theta,j}. \tag{11.22}$$

Thus, $\dot{r}_{j,\text{cm}}$ is the radial speed and $\hat{v}_b$ is the tangential speed.

Step 2: time derivatives of the unit vectors Over a small angular change $\mathrm{d}\theta_j$, the unit vectors rotate rigidly by the same angle. Hence

$$\frac{\mathrm{d}\mathbf{e}_{r,j}}{\mathrm{d}\theta_j} = \mathbf{e}_{\theta,j}, \qquad \frac{\mathrm{d}\mathbf{e}_{\theta,j}}{\mathrm{d}\theta_j} = -\mathbf{e}_{r,j}. \tag{11.23}$$

The local angular speed of the belt element centroid on pulley j is related to the belt transport speed by

$$\dot{\theta}_j = \omega_{b,j} = \frac{\hat{v}_b}{r_{j,\mathrm{cm}}}. \tag{11.24}$$

Using the chain rule, the unit-vector time derivatives therefore become

$$\dot{\mathbf{e}}_{r,j} = \frac{\hat{v}_b}{r_{j,\mathrm{cm}}}\,\mathbf{e}_{\theta,j}, \qquad \dot{\mathbf{e}}_{\theta,j} = -\frac{\hat{v}_b}{r_{j,\mathrm{cm}}}\,\mathbf{e}_{r,j}. \tag{11.25}$$

Step 3: acceleration and tangential component Differentiating Equation (11.22) gives

$$\begin{aligned} \mathbf{a}_j &= \frac{\mathrm{d}}{\mathrm{d}t}\big(\dot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j}\big) + \frac{\mathrm{d}}{\mathrm{d}t}\big(\hat{v}_b\,\mathbf{e}_{\theta,j}\big) \\ &= \ddot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j} + \dot{r}_{j,\mathrm{cm}}\,\dot{\mathbf{e}}_{r,j} + \dot{\hat{v}}_b\,\mathbf{e}_{\theta,j} + \hat{v}_b\,\dot{\mathbf{e}}_{\theta,j}. \end{aligned} \tag{11.26}$$

Substituting Equation (11.25) gives

$$\begin{aligned} \mathbf{a}_j &= \ddot{r}_{j,\mathrm{cm}}\,\mathbf{e}_{r,j} + \dot{r}_{j,\mathrm{cm}}\left(\frac{\hat{v}_b}{r_{j,\mathrm{cm}}}\,\mathbf{e}_{\theta,j}\right) + \dot{\hat{v}}_b\,\mathbf{e}_{\theta,j} - \hat{v}_b\left(\frac{\hat{v}_b}{r_{j,\mathrm{cm}}}\,\mathbf{e}_{r,j}\right) \\ &= \left(\ddot{r}_{j,\mathrm{cm}} - \frac{\hat{v}_b^2}{r_{j,\mathrm{cm}}}\right)\mathbf{e}_{r,j} + \left(\dot{\hat{v}}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}\hat{v}_b\right)\mathbf{e}_{\theta,j}. \end{aligned} \tag{11.27}$$

By Equation (11.20), the tangential acceleration is therefore

$$a_{t,j} = \dot{\hat{v}}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}\hat{v}_b. \tag{11.28}$$

Interpretation The first term, $\dot{\hat{v}}_b$, is the direct time rate of change of the belt transport speed. The second term appears because the belt element is not only moving along the wrap, but also moving to a different wrap radius at the same time. As the element shifts inward or outward while continuing to travel around the pulley, the local radial direction changes, and that changing radial direction contributes a tangential component to the acceleration.

The radial component $\ddot{r}_{j,\mathrm{cm}} - \hat{v}_b^2/r_{j,\mathrm{cm}}$ contains the centripetal term, but it does not enter the tangential force balance used in Section 11.2.

11.5 Normal Force from Axial Clamping

The traction bound Equation (11.19) depends on the total normal load over the wrap of pulley j,

$$N_{\phi,j} \equiv \int_{-\phi_j/2}^{\phi_j/2} \mathrm{d}N_j.$$

We now express $N_{\phi,j}$ in terms of the axial clamping force generated by that pulley.

Axial clamp produces radial normal load The belt is pressed between two opposing conical sheaves. On pulley j, an axial clamping force $F_{\mathrm{ax},j}$ pushes the sheaves together. Because the sheave faces are inclined at cone half-angle β, this axial force generates a radial normal load on the belt. Here, $F_{\mathrm{ax},j}$ is clamping force only; belt inertia (centrifugal effects) is already accounted for by the acceleration term in Equation (11.19).

From the conical geometry relation established in Equation (7.2),

$$\frac{\mathrm{d}r_j}{\mathrm{d}s} = \frac{1}{2\tan\beta}.$$

To relate the axial and radial forces consistently, we use virtual work. The incremental work done by the axial clamp must equal the incremental work associated with radial compression of the belt:

$$F_{\mathrm{ax},j}\ \mathrm{d}s = F_{\mathrm{rad},j}\ \mathrm{d}r_j.$$

Substituting $\mathrm{d}r_j = (\mathrm{d}r_j/\mathrm{d}s)\ \mathrm{d}s$ gives

$$F_{\mathrm{rad},j} = \frac{F_{\mathrm{ax},j}}{\mathrm{d}r_j/\mathrm{d}s} = 2F_{\mathrm{ax},j}\tan\beta. \tag{11.29}$$

Thus, the axial clamp force on pulley j produces a total radial normal load

$$F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j}\tan\beta. \tag{11.30}$$

Identification with the wrap-normal load The radial load $F_{\mathrm{rad},j}$ derived above represents the total inward force applied by the two sheave faces of pulley j onto the belt. That load is carried by the distributed belt–pulley contact forces $\mathrm{d}N_j$ along the wrap.

Therefore, the total normal load available for traction on pulley j is

$$\boxed{N_{\phi,j} = F_{\mathrm{rad},j} = 2F_{\mathrm{ax},j}\tan\beta.} \tag{11.31}$$

This result states that the total wrap normal load, and hence the normal load available for traction, is set directly by the axial clamping force through the cone geometry.

11.6 Torque Capacity

From Section 11.1, the maximum transmissible torque on pulley j is determined by the maximum admissible tension difference across the wrap:

$$\tau_{j,\max} = r_{j,\mathrm{eff}}\,\Delta T_{j,\max}. \tag{11.32}$$

Using the integrated result from Equation (11.19), together with the wrap-normal relation Equation (11.31) and the belt tangential acceleration from Equation (11.28),

$$\Delta T_{j,\max} = \mu N_{\phi,j} - \rho_b A_b r_{j,\mathrm{cm}} a_{t,j}\phi_j, \qquad N_{\phi,j} = 2F_{\mathrm{ax},j}\tan\beta, \qquad a_{t,j} = \dot{\hat{v}}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}}\hat{v}_b,$$

we obtain

$$\Delta T_{j,\max} = 2\mu F_{\mathrm{ax},j} \tan\beta - \rho_b A_b \phi_j r_{j,\mathrm{cm}} \left(\dot{\hat{v}}_b + \frac{\dot{r}_{j,\mathrm{cm}}}{r_{j,\mathrm{cm}}} \hat{v}_b \right). \tag{11.33}$$

Substituting this into Equation (11.32) and re-arranging gives the explicit torque-capacity expression

Pulley Torque Capacity **(11.34)**

$$\tau_{j,\max} = r_{j,\mathrm{eff}} \left[2\mu F_{\mathrm{ax},j} \tan\beta - \rho_b A_b \phi_j \left(r_{j,\mathrm{cm}} \dot{\hat{v}}_b + \dot{r}_{j,\mathrm{cm}}\, \hat{v}_b \right) \right].$$

The remaining task is to substitute the appropriate axial clamp force $F_{\mathrm{ax},j}$ for each pulley to obtain explicit bounds on the coupling torque.

Remark (Why this differs from the classical capstan exponential)

In the classical capstan problem, the normal load is generated *by the belt tension* itself (a larger tension implies a proportionally larger contact normal), which leads to the exponential relation $T_1/T_2 = e^{\mu\phi}$.
Here, the normal load is imposed externally by the clamping mechanism: N_ϕ is determined by F_{ax} through Equation (11.31). Therefore, the available traction scales *linearly* with clamping force, not exponentially with belt tension. Centrifugal loading and ratio change enter additionally through the inertial term in Equation (11.34).

11.7 Primary Pulley Traction Bounds

The generic capacity law Equation (11.34) must now be specialized to the primary pulley. Because the inertial correction contains $\dot{\hat{v}}_b$, and the belt-speed derivative is defined differently in stick and slip, the primary traction bounds must be derived separately for the two transmission regimes.

In the sticking regime, the belt speed is imposed by compatibility. Accordingly, we use the primary-side sticking derivative from Equation (9.25), together with the primary axial-force model Equation (8.10). This causes the inertial term to depend on the primary angular acceleration, and therefore on the coupling torque itself. The resulting sticking bound is thus implicit in τ_c, but it can still be solved in closed form.

In the slipping regime, the belt-speed derivative is instead given directly by the regularized evolution law Equation (9.28). In that case, the primary traction bound no longer depends on τ_c through the inertial term, so the upper and lower limits follow directly.

Stick regime On the primary pulley, the transmitted torque at the belt contact is the coupling torque itself. Applying Equation (11.34) with $j = p$ gives

$$|\tau_c| \le r_{p,\mathrm{eff}} \left[2\mu F_{\mathrm{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\mathrm{cm}} \dot{\hat{v}}_b + \dot{r}_{p,\mathrm{cm}}\, \hat{v}_b \right) \right]. \tag{11.35}$$

In the sticking regime,

$$\hat{v}_b = r_{p,\mathrm{cm}} \omega_p, \qquad \dot{\hat{v}}_b = \dot{r}_{p,\mathrm{cm}}\, \omega_p + r_{p,\mathrm{cm}}\, \dot{\omega}_p,$$

from Equation (9.24) and Equation (9.25). Substituting these into Equation (11.35) gives

$$|\tau_c| \leq r_{p,\text{eff}} \left[2\mu F_{\text{ax},p} \tan\beta - \rho_b A_b \phi_p \left(r_{p,\text{cm}}^2 \dot{\omega}_p + 2 r_{p,\text{cm}} \dot{r}_{p,\text{cm}}\, \omega_p\right)\right]. \tag{11.36}$$

Now substitute the primary axial-force model from Equation (8.10),

$$F_{\text{ax},p}(s, \omega_p) = m_f \omega_p^2 \left(r_{f,0} + r_f(s)\right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\big(x_{p,0} + s\big), \tag{11.37}$$

together with the primary rotational equation of motion from Equation (10.1)

$$\dot{\omega}_p = \frac{\tau_{\text{eng}} - \tau_c}{I_p}. \tag{11.38}$$

This yields

$$\begin{aligned} |\tau_c| \leq r_{p,\text{eff}} \Bigg\{ & 2\mu \tan\beta \left[m_f \omega_p^2 \left(r_{f,0} + r_f(s)\right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\big(x_{p,0} + s\big) \right] \\ & - \rho_b A_b \phi_p \left[\frac{r_{p,\text{cm}}^2}{I_p} \big(\tau_{\text{eng}} - \tau_c\big) + 2 r_{p,\text{cm}} \dot{r}_{p,\text{cm}}\, \omega_p \right] \Bigg\}. \end{aligned} \tag{11.39}$$

The coupling torque now appears on both sides of the inequality, so the positive and negative directions must again be treated separately.

Positive-direction sticking bound $\tau_c \geq 0$ For $\tau_c \geq 0$, $|\tau_c| = \tau_c$, so Equation (11.39) becomes

$$\begin{aligned} \tau_c \leq r_{p,\text{eff}} \Bigg\{ & 2\mu \tan\beta \left[m_f \omega_p^2 \left(r_{f,0} + r_f(s)\right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\big(x_{p,0} + s\big) \right] \\ & - \rho_b A_b \phi_p \left[\frac{r_{p,\text{cm}}^2}{I_p} \tau_{\text{eng}} - \frac{r_{p,\text{cm}}^2}{I_p} \tau_c + 2 r_{p,\text{cm}} \dot{r}_{p,\text{cm}}\, \omega_p \right] \Bigg\}. \end{aligned} \tag{11.40}$$

Collecting the τ_c terms on the left gives

$$\begin{aligned} \left(1 - r_{p,\text{eff}} \rho_b A_b \phi_p \frac{r_{p,\text{cm}}^2}{I_p}\right) \tau_c \leq r_{p,\text{eff}} \Bigg\{ & 2\mu \tan\beta \left[m_f \omega_p^2 \left(r_{f,0} + r_f(s)\right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\big(x_{p,0} + s\big) \right] \\ & - \rho_b A_b \phi_p \left[\frac{r_{p,\text{cm}}^2}{I_p} \tau_{\text{eng}} + 2 r_{p,\text{cm}} \dot{r}_{p,\text{cm}}\, \omega_p \right] \Bigg\}. \end{aligned} \tag{11.41}$$

Hence,

$$\tau_c \leq \tau_{c,p,+}^{\text{stick}}, \tag{11.42}$$

with

Primary Stick-Branch Upper Bound **(11.43)**

$$\tau^{\text{stick}}_{c,p,+} = \frac{r_{p,\text{eff}}\left\{2\mu\tan\beta\left[m_f\omega_p^2\left(r_{f,0}+r_f(s)\right)\dfrac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\left(x_{p,0}+s\right)\right] - \rho_b A_b\phi_p\left[\dfrac{r^2_{p,\text{cm}}}{I_p}\tau_{\text{eng}} + 2r_{p,\text{cm}}\dot{r}_{p,\text{cm}}\,\omega_p\right]\right\}}{1 - r_{p,\text{eff}}\rho_b A_b\phi_p\dfrac{r^2_{p,\text{cm}}}{I_p}}.$$

Negative-direction sticking bound $\tau_c \le 0$ For $\tau_c \le 0$, $|\tau_c| = -\tau_c$, so Equation (11.39) becomes

$$\begin{aligned}\left(1 + r_{p,\text{eff}}\rho_b A_b\phi_p\frac{r^2_{p,\text{cm}}}{I_p}\right)\tau_c \ge\; &-r_{p,\text{eff}}\left\{2\mu\tan\beta\left[m_f\omega_p^2\left(r_{f,0}+r_f(s)\right)\frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\left(x_{p,0}+s\right)\right]\right.\\ &\left.- \rho_b A_b\phi_p\left[\frac{r^2_{p,\text{cm}}}{I_p}\tau_{\text{eng}} + 2r_{p,\text{cm}}\dot{r}_{p,\text{cm}}\,\omega_p\right]\right\}.\end{aligned} \tag{11.44}$$

Collecting the τ_c terms gives

$$\left(1 + r_{p,\text{eff}}\rho_b A_b\phi_p\frac{r^2_{p,\text{cm}}}{I_p}\right)\tau_c \ge -r_{p,\text{eff}}\left\{2\mu\tan\beta\left[m_f\omega_p^2\left(r_{f,0}+r_f(s)\right)\frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\left(x_{p,0}+s\right)\right] - \rho_b A_b\phi_p\left[\frac{r^2_{p,\text{cm}}}{I_p}\tau_{\text{eng}} + 2r_{p,\right.\right. \tag{11.45}$$

Hence,

$$\tau_c \ge \tau^{\text{stick}}_{c,p,-}, \tag{11.46}$$

with

Primary Stick-Branch Lower Bound **(11.47)**

$$\tau^{\text{stick}}_{c,p,-} = -\frac{r_{p,\text{eff}}\left\{2\mu\tan\beta\left[m_f\omega_p^2\left(r_{f,0}+r_f(s)\right)\dfrac{\mathrm{d}r_f}{\mathrm{d}s} - k_p\left(x_{p,0}+s\right)\right] - \rho_b A_b\phi_p\left[\dfrac{r^2_{p,\text{cm}}}{I_p}\tau_{\text{eng}} + 2r_{p,\text{cm}}\dot{r}_{p,\text{cm}}\,\omega_p\right]\right\}}{1 + r_{p,\text{eff}}\rho_b A_b\phi_p\dfrac{r^2_{p,\text{cm}}}{I_p}}.$$

Therefore, in the sticking regime the primary pulley contributes the interval

$$\boxed{\tau^{\text{stick}}_{c,p,-} \le \tau_c \le \tau^{\text{stick}}_{c,p,+}.} \tag{11.48}$$

Slip regime In the slipping regime, the belt-speed derivative is given directly by the regularized closure Equation (9.28):

$$\dot{\hat{v}}_b = \frac{v_b^* - \hat{v}_b}{T_b}.$$

Applying Equation (11.34) with $j = p$, and again substituting the primary axial-force model Equation (8.10), gives

$$|\tau_c| \leq r_{p,\text{eff}} \left\{ 2\mu \tan\beta \left[m_f \omega_p^2 \left(r_{f,0} + r_f(s) \right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p \big(x_{p,0} + s \big) \right] - \rho_b A_b \phi_p \left[r_{p,\text{cm}} \frac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{p,\text{cm}}\, \hat{v}_b \right] \right\}. \tag{11.49}$$

In contrast to the sticking case, the right-hand side no longer depends on τ_c. The positive and negative limits therefore follow directly.

Positive-direction slipping bound $\tau_c \geq 0$ For $\tau_c \geq 0$, $|\tau_c| = \tau_c$, so

$$\tau_c \leq \tau_{c,p,+}^{\text{slip}}, \tag{11.50}$$

with

Primary Slip-Branch Upper Bound (11.51)

$$\tau_{c,p,+}^{\text{slip}} = r_{p,\text{eff}} \left\{ 2\mu \tan\beta \left[m_f \omega_p^2 \left(r_{f,0} + r_f(s) \right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p \big(x_{p,0} + s \big) \right] - \rho_b A_b \phi_p \left[r_{p,\text{cm}} \frac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{p,\text{cm}}\, \hat{v}_b \right] \right\}.$$

Negative-direction slipping bound $\tau_c \leq 0$ For $\tau_c \leq 0$, $|\tau_c| = -\tau_c$, so

$$\tau_c \geq \tau_{c,p,-}^{\text{slip}}, \tag{11.52}$$

with

Primary Slip-Branch Lower Bound (11.53)

$$\tau_{c,p,-}^{\text{slip}} = -r_{p,\text{eff}} \left\{ 2\mu \tan\beta \left[m_f \omega_p^2 \left(r_{f,0} + r_f(s) \right) \frac{\mathrm{d}r_f}{\mathrm{d}s} - k_p \big(x_{p,0} + s \big) \right] - \rho_b A_b \phi_p \left[r_{p,\text{cm}} \frac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{p,\text{cm}}\, \hat{v}_b \right] \right\}.$$

Therefore, in the slipping regime the primary pulley contributes the interval

$$\boxed{\tau_{c,p,-}^{\text{slip}} \leq \tau_c \leq \tau_{c,p,+}^{\text{slip}}.} \tag{11.54}$$

11.8 Secondary Pulley Traction Bounds

The secondary pulley follows the same overall procedure as the primary, but with one important difference. At the secondary, the torque at the belt contact is not treated as an independent quantity. Instead, from the coupling relation introduced in Equation (9.13),

$$\tau_s = R\, \tau_c. \tag{11.55}$$

Accordingly, the secondary traction bounds will be written directly in terms of the coupling torque τ_c.

In addition, the secondary clamp force is torque-reactive. Unlike the primary, whose axial clamping force depends only on pulley speed and shift position, the secondary axial-force model depends explicitly on the transmitted secondary torque. Thus, the substitution $\tau_s = R\tau_c$ must be made not only in the secondary torque capacity relation, but also in the secondary axial clamp-force model itself.

As with the primary, the secondary bounds must be derived separately in the sticking and slipping regimes because the inertial correction contains $\dot{\hat{v}}_b$. In the sticking regime, we use the secondary-side sticking derivative from Equation (9.26), together with the secondary rotational equation of motion from Equation (10.2). In the slipping regime, we instead use the regularized belt-speed evolution law Equation (9.28).

Stick regime Applying Equation (11.34) with $j = s$, and using $\tau_s = R\tau_c$, gives

$$|R\tau_c| \le r_{s,\mathrm{eff}} \left[2\mu F_{\mathrm{ax},s} \tan\beta - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{\hat{v}}_b + \dot{r}_{s,\mathrm{cm}}\, \hat{v}_b\right)\right]. \tag{11.56}$$

Since $R > 0$, dividing by R preserves the inequality direction, so

$$|\tau_c| \le \frac{r_{s,\mathrm{eff}}}{R} \left[2\mu F_{\mathrm{ax},s} \tan\beta - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}} \dot{\hat{v}}_b + \dot{r}_{s,\mathrm{cm}}\, \hat{v}_b\right)\right]. \tag{11.57}$$

In the sticking regime,

$$\hat{v}_b = r_{s,\mathrm{cm}} \omega_s, \qquad \dot{\hat{v}}_b = \dot{r}_{s,\mathrm{cm}}\, \omega_s + r_{s,\mathrm{cm}}\, \dot{\omega}_s,$$

from Equation (9.24) and Equation (9.26). Substituting these into Equation (11.57) gives

$$|\tau_c| \le \frac{r_{s,\mathrm{eff}}}{R} \left[2\mu F_{\mathrm{ax},s} \tan\beta - \rho_b A_b \phi_s \left(r_{s,\mathrm{cm}}^2 \dot{\omega}_s + 2 r_{s,\mathrm{cm}} \dot{r}_{s,\mathrm{cm}}\, \omega_s\right)\right]. \tag{11.58}$$

Now substitute the secondary rotational dynamics from Equation (10.2),

$$\dot{\omega}_s = \frac{R\tau_c - \tau_{\mathrm{load}}}{I_s}, \tag{11.59}$$

and the secondary axial-force model from Equation (8.16), using $\tau_s = R\tau_c$ from Equation (11.55), so that

$$F_{\mathrm{ax},s}(s, R\tau_c) = \frac{R\tau_c + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2} \frac{\mathrm{d}\theta_s}{\mathrm{d}s} + k_{s,x}(x_{s,0} + s). \tag{11.60}$$

This yields

$$\begin{aligned} |\tau_c| \le \frac{r_{s,\mathrm{eff}}}{R} \Bigg[& \mu \tan\beta \left(R\tau_c + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)\right) \frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ & + 2\mu \tan\beta\, k_{s,x}(x_{s,0} + s) - \rho_b A_b \phi_s \left(\frac{r_{s,\mathrm{cm}}^2}{I_s} (R\tau_c - \tau_{\mathrm{load}}) + 2 r_{s,\mathrm{cm}} \dot{r}_{s,\mathrm{cm}}\, \omega_s \right) \Bigg]. \end{aligned} \tag{11.61}$$

The coupling torque now appears through both the torque-reactive clamp force and the secondary angular acceleration, so the positive and negative directions must be treated separately.

Positive-direction sticking bound $\tau_c \geq 0$ For $\tau_c \geq 0$, $|\tau_c| = \tau_c$, so Equation (11.61) becomes

$$\begin{aligned}\tau_c \leq \frac{r_{s,\text{eff}}}{R}\Bigg[&\mu \tan\beta\,\big(R\tau_c + k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big)\big)\,\frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu\tan\beta\,k_{s,x}(x_{s,0}+s) - \rho_b A_b \phi_s\left(\frac{r_{s,\text{cm}}^2}{I_s}(R\tau_c - \tau_{\text{load}}) + 2r_{s,\text{cm}}\dot{r}_{s,\text{cm}}\,\omega_s\right)\Bigg].\end{aligned} \tag{11.62}$$

Collecting the τ_c terms on the left gives

$$\begin{aligned}\left(1 - r_{s,\text{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s} + r_{s,\text{eff}}\rho_b A_b\phi_s\frac{r_{s,\text{cm}}^2}{I_s}\right)\tau_c \;\leq\; &\frac{r_{s,\text{eff}}}{R}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\,k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big) \\ &+ \frac{r_{s,\text{eff}}}{R}2\mu\tan\beta\,k_{s,x}(x_{s,0}+s) \\ &+ \frac{r_{s,\text{eff}}}{R}\rho_b A_b\phi_s\left(\frac{r_{s,\text{cm}}^2}{I_s}\tau_{\text{load}} - 2r_{s,\text{cm}}\dot{r}_{s,\text{cm}}\,\omega_s\right).\end{aligned} \tag{11.63}$$

Hence,

$$\tau_c \leq \tau_{c,s,+}^{\text{stick}}, \tag{11.64}$$

with

Secondary Stick-Branch Upper Bound (11.65)

$$\tau_{c,s,+}^{\text{stick}} = \frac{\frac{r_{s,\text{eff}}}{R}\left[\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\,k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big) + 2\mu\tan\beta\,k_{s,x}(x_{s,0}+s) + \rho_b A_b\phi_s\left(\frac{r_{s,\text{cm}}^2}{I_s}\tau_{\text{load}} - 2r_{s,\text{cm}}\dot{r}_{s,\text{cm}}\,\omega_s\right)\right]}{1 - r_{s,\text{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s} + r_{s,\text{eff}}\rho_b A_b\phi_s\frac{r_{s,\text{cm}}^2}{I_s}}.$$

Negative-direction sticking bound $\tau_c \leq 0$ For $\tau_c \leq 0$, $|\tau_c| = -\tau_c$, so Equation (11.61) becomes

$$\begin{aligned}-\tau_c \leq \frac{r_{s,\text{eff}}}{R}\Bigg[&\mu \tan\beta\,\big(R\tau_c + k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big)\big)\,\frac{\mathrm{d}\theta_s}{\mathrm{d}s} \\ &+ 2\mu\tan\beta\,k_{s,x}(x_{s,0}+s) - \rho_b A_b \phi_s\left(\frac{r_{s,\text{cm}}^2}{I_s}(R\tau_c - \tau_{\text{load}}) + 2r_{s,\text{cm}}\dot{r}_{s,\text{cm}}\,\omega_s\right)\Bigg].\end{aligned} \tag{11.66}$$

Collecting the τ_c terms on the left gives

$$\left(1 + r_{s,\text{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s} - r_{s,\text{eff}}\rho_b A_b \phi_s \frac{r_{s,\text{cm}}^2}{I_s}\right)\tau_c \;\geq\; -\frac{r_{s,\text{eff}}}{R}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) - \frac{r_{s,\text{eff}}}{R} 2\mu\tan\beta\, k_{s,x}(x_{s,0} + s) - \frac{r_{s,\text{eff}}}{R}\rho_b A_b \phi_s \left(\frac{r_{s,\text{cm}}^2}{I_s}\tau_{\text{load}} - 2 r_{s,\text{cm}}\dot{r}_{s,\text{cm}}\,\omega_s\right). \tag{11.67}$$

Hence,

$$\tau_c \geq \tau_{c,s,-}^{\text{stick}}, \tag{11.68}$$

with

Secondary Stick-Branch Lower Bound **(11.69)**

$$\tau_{c,s,-}^{\text{stick}} = -\frac{\frac{r_{s,\text{eff}}}{R}\left[\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\, k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big) + 2\mu\tan\beta\, k_{s,x}(x_{s,0} + s) + \rho_b A_b \phi_s \left(\frac{r_{s,\text{cm}}^2}{I_s}\tau_{\text{load}} - 2 r_{s,\text{cm}}\dot{r}_{s,\text{cm}}\,\omega_s\right)\right]}{1 + r_{s,\text{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s} - r_{s,\text{eff}}\rho_b A_b \phi_s \frac{r_{s,\text{cm}}^2}{I_s}}.$$

Therefore, in the sticking regime the secondary pulley contributes the interval

$$\boxed{\tau_{c,s,-}^{\text{stick}} \leq \tau_c \leq \tau_{c,s,+}^{\text{stick}}.} \tag{11.70}$$

Slip regime In the slipping regime, the belt-speed derivative is given directly by the regularized closure Equation (9.28):

$$\dot{\hat{v}}_b = \frac{v_b^* - \hat{v}_b}{T_b}.$$

Applying Equation (11.34) with $j = s$, and again using $\tau_s = R\tau_c$ from Equation (11.55), gives

$$|R\tau_c| \leq r_{s,\text{eff}}\left[2\mu F_{\text{ax},s}\tan\beta - \rho_b A_b \phi_s \left(r_{s,\text{cm}}\frac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{s,\text{cm}}\,\hat{v}_b\right)\right]. \tag{11.71}$$

Dividing by $R > 0$ gives

$$|\tau_c| \leq \frac{r_{s,\text{eff}}}{R}\left[2\mu F_{\text{ax},s}\tan\beta - \rho_b A_b \phi_s \left(r_{s,\text{cm}}\frac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{s,\text{cm}}\,\hat{v}_b\right)\right]. \tag{11.72}$$

Using the secondary axial-force model Equation (8.16), with $\tau_s = R\tau_c$ substituted through Equation (11.55), gives

$$F_{\text{ax},s}(s, R\tau_c) = \frac{R\tau_c + k_{s,\theta}\big(\theta_{s,0} + \theta_s(s)\big)}{2}\frac{\mathrm{d}\theta_s}{\mathrm{d}s} + k_{s,x}(x_{s,0} + s). \tag{11.73}$$

Substituting this into Equation (11.72) yields

$$|\tau_c| \le \frac{r_{s,\mathrm{eff}}}{R}\Bigg[\mu\tan\beta\left(R\tau_c + k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big)\right)\frac{\mathrm{d}\theta_s}{\mathrm{d}s} + 2\mu\tan\beta\, k_{s,x}(x_{s,0}+s) - \rho_b A_b \phi_s\left(r_{s,\mathrm{cm}}\frac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{s,\mathrm{cm}}\,\hat{v}_b\right)\Bigg]. \tag{11.74}$$

In contrast to the sticking case, the inertial correction no longer depends on τ_c. However, the secondary bound remains implicit because the torque-reactive clamp force still depends on $R\tau_c$.

Positive-direction slipping bound $\tau_c \ge 0$ For $\tau_c \ge 0$, $|\tau_c| = \tau_c$. Collecting the τ_c terms on the left gives

$$\begin{aligned}\left(1 - r_{s,\mathrm{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\right)\tau_c \;\le\; &\frac{r_{s,\mathrm{eff}}}{R}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\,k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big)\\ &+\frac{r_{s,\mathrm{eff}}}{R}2\mu\tan\beta\,k_{s,x}(x_{s,0}+s)\\ &-\frac{r_{s,\mathrm{eff}}}{R}\rho_b A_b\phi_s\left(r_{s,\mathrm{cm}}\frac{v_b^*-\hat{v}_b}{T_b}+\dot{r}_{s,\mathrm{cm}}\,\hat{v}_b\right).\end{aligned} \tag{11.75}$$

Hence,

$$\tau_c \le \tau_{c,s,+}^{\mathrm{slip}}, \tag{11.76}$$

with

Secondary Slip-Branch Upper Bound **(11.77)**

$$\tau_{c,s,+}^{\mathrm{slip}} = \frac{\frac{r_{s,\mathrm{eff}}}{R}\left[\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\,k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big) + 2\mu\tan\beta\,k_{s,x}(x_{s,0}+s) - \rho_b A_b\phi_s\left(r_{s,\mathrm{cm}}\frac{v_b^*-\hat{v}_b}{T_b}+\dot{r}_{s,\mathrm{cm}}\,\hat{v}_b\right)\right]}{1 - r_{s,\mathrm{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}}.$$

Negative-direction slipping bound $\tau_c \le 0$ For $\tau_c \le 0$, $|\tau_c| = -\tau_c$. Collecting the τ_c terms on the left gives

$$\begin{aligned}\left(1 + r_{s,\mathrm{eff}}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\right)\tau_c \;\ge\; &-\frac{r_{s,\mathrm{eff}}}{R}\mu\tan\beta\frac{\mathrm{d}\theta_s}{\mathrm{d}s}\,k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big)\\ &-\frac{r_{s,\mathrm{eff}}}{R}2\mu\tan\beta\,k_{s,x}(x_{s,0}+s)\\ &+\frac{r_{s,\mathrm{eff}}}{R}\rho_b A_b\phi_s\left(r_{s,\mathrm{cm}}\frac{v_b^*-\hat{v}_b}{T_b}+\dot{r}_{s,\mathrm{cm}}\,\hat{v}_b\right).\end{aligned} \tag{11.78}$$

Hence,

$$\tau_c \ge \tau_{c,s,-}^{\mathrm{slip}}, \tag{11.79}$$

with

Secondary Slip-Branch Lower Bound **(11.80)**

$$\tau_{c,s,-}^{\text{slip}} = -\frac{\dfrac{r_{s,\text{eff}}}{R}\left[\mu\tan\beta\,\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}\,k_{s,\theta}\big(\theta_{s,0}+\theta_s(s)\big) + 2\mu\tan\beta\,k_{s,x}(x_{s,0}+s) - \rho_b A_b \phi_s\left(r_{s,\text{cm}}\dfrac{v_b^* - \hat{v}_b}{T_b} + \dot{r}_{s,\text{cm}}\,\hat{v}_b\right)\right]}{1 + r_{s,\text{eff}}\,\mu\tan\beta\,\dfrac{\mathrm{d}\theta_s}{\mathrm{d}s}}.$$

Therefore, in the slipping regime the secondary pulley contributes the interval

$$\boxed{\tau_{c,s,-}^{\text{slip}} \le \tau_c \le \tau_{c,s,+}^{\text{slip}}.} \tag{11.81}$$

11.9 Section Conclusion and Transition

This section derived the traction-capacity model of the CVT from the local belt–pulley force balance and converted it into explicit torque bounds for the primary and secondary pulleys.

With these traction limits now established, the torque-transfer model is fully closed. The transmitted coupling torque is no longer just the value required by the no-slip dynamics; it is now constrained by the admissible belt traction, which determines whether the transmission remains in stick or saturates onto a slipping branch.

12 Complete CVT Dynamic Model

This section collects the results of the preceding derivations into one closed nonlinear dynamical model. No new physics is introduced here. The purpose is simply to summarize, in one place,

1. the state variables,
2. the geometry and belt-kinematic quantities computed from those states,
3. the torque and regime closures,
4. and the final governing equations of motion.

12.1 State Definition

The state vector is

$$\mathbf{x}(t) = \big(\omega_p,\ \omega_s,\ s,\ \dot{s},\ \hat{v}_b\big), \tag{12.1}$$

where

- ω_p is the primary pulley angular velocity,
- ω_s is the secondary pulley angular velocity,
- s is the axial shift coordinate,
- $\dot{s}$ is the axial shift velocity,
- $\hat{v}_b$ is the modeled belt transport speed.

12.2 Geometry and Belt Kinematics

At each instant, the shift coordinate s determines the pulley geometry through the belt-length constraint and the conical kinematic relations derived earlier. In particular,

$$r_{p,\text{eff}}, \quad r_{s,\text{eff}}, \quad r_{p,\text{cm}}, \quad r_{s,\text{cm}}, \quad \phi_p, \quad \phi_s$$

are all functions of s, and the geometric CVT ratio is

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \tag{12.2}$$

Its time derivative is

$$\dot{R} = \dot{R}(s, \dot{s}), \tag{12.3}$$

as given by Equation (7.57).

The pulley-imposed belt-line speeds from Section 9.2 are

$$v_{p,\text{cm}} = r_{p,\text{cm}}(s)\,\omega_p, \qquad v_{s,\text{cm}} = r_{s,\text{cm}}(s)\,\omega_s. \tag{12.4}$$

The regularized target belt speed is

$$v_b^* = \frac{1}{2}\left(v_{p,\text{cm}} + v_{s,\text{cm}}\right). \tag{12.5}$$

In the sticking regime, the belt speed is imposed exactly by the adhered kinematics:

$$\hat{v}_b = v_{p,\text{cm}} = v_{s,\text{cm}}. \tag{12.6}$$

The corresponding sticking derivative is obtained by differentiating this compatibility relation, for example from the primary side,

$$\dot{\hat{v}}_{b,\text{stick}} = \dot{r}_{p,\text{cm}}\,\omega_p + r_{p,\text{cm}}\,\dot{\omega}_p, \tag{12.7}$$

with an equivalent expression obtained from the secondary side under sticking compatibility.

In the slipping regime, the belt-speed derivative is given by the regularized law

$$\dot{\hat{v}}_{b,\text{slip}} = \frac{v_b^* - \hat{v}_b}{T_b}. \tag{12.8}$$

12.3 Torque and Regime Closure

The internal torque transmitted through the CVT is the coupling torque τ_c.

No-slip torque demand. If sticking is maintained, the required coupling torque is the no-slip torque derived in Equation (10.23):

$$\tau_{\text{ns}} = \frac{\tau_{\text{eng}} + \dfrac{I_p}{I_s} R\,\tau_{\text{load}} - I_p\,\omega_s\,\dot{R}}{1 + \dfrac{I_p}{I_s} R^2}. \tag{12.9}$$

Stick and slip traction intervals. The branchwise primary and secondary traction bounds derived in Section 11.7 and Section 11.8 combine into a sticking admissible interval

$$\tau_-^{\text{stick}} \le \tau_c \le \tau_+^{\text{stick}}, \tag{12.10}$$

with

$$\tau_-^{\text{stick}} = \max\Big(\tau_{c,p,-}^{\text{stick}}, \tau_{c,s,-}^{\text{stick}}\Big)\,, \qquad \tau_+^{\text{stick}} = \min\Big(\tau_{c,p,+}^{\text{stick}}, \tau_{c,s,+}^{\text{stick}}\Big)\,, \tag{12.11}$$

and a slipping admissible interval

$$\tau_-^{\text{slip}} \le \tau_c \le \tau_+^{\text{slip}}, \tag{12.12}$$

with

$$\tau_-^{\text{slip}} = \max\Big(\tau_{c,p,-}^{\text{slip}}, \tau_{c,s,-}^{\text{slip}}\Big)\,, \qquad \tau_+^{\text{slip}} = \min\Big(\tau_{c,p,+}^{\text{slip}}, \tau_{c,s,+}^{\text{slip}}\Big)\,. \tag{12.13}$$

Slip metric. The slip metric from Equation (9.20) is

$$v_\Delta = r_{p,\text{eff}}\,\omega_p - r_{s,\text{eff}}\,\omega_s. \tag{12.14}$$

Regime selection. Using τ_{ns}, the sticking interval, the slipping interval, and v_Δ, the active transmission regime selects both the actual coupling torque and the appropriate belt-speed derivative.

The coupling torque is

$$\tau_c = \begin{cases} \tau_{\text{ns}}, & v_\Delta = 0 \text{ and } \tau_-^{\text{stick}} \le \tau_{\text{ns}} \le \tau_+^{\text{stick}}, \\ \tau_+^{\text{slip}}, & v_\Delta > 0 \text{ or } \big(v_\Delta = 0 \text{ and } \tau_{\text{ns}} > \tau_+^{\text{stick}}\big), \\ \tau_-^{\text{slip}}, & v_\Delta < 0 \text{ or } \big(v_\Delta = 0 \text{ and } \tau_{\text{ns}} < \tau_-^{\text{stick}}\big). \end{cases} \tag{12.15}$$

The belt-speed derivative is

$$\dot{v}_b = \begin{cases} \dot{v}_{b,\text{stick}}, & v_\Delta = 0 \text{ and } \tau_-^{\text{stick}} \le \tau_{\text{ns}} \le \tau_+^{\text{stick}}, \\ \dot{v}_{b,\text{slip}}, & \text{otherwise.} \end{cases} \tag{12.16}$$

Equation (12.15) and Equation (12.16) define the ideal piecewise analytical regime structure under Assumption 4 (Idealized Belt–Pulley Friction Law) and Assumption 14 (Single Active Slip Interface). For numerical simulation, this branch logic is realized using the regularized/blended transition law in Appendix C.3, which smooths switching near stick–slip transitions while preserving the same limiting branch behavior.

12.4 Governing ODEs

With the coupling torque and belt-speed derivative now closed, the rotational dynamics are

$$\dot{\omega}_p = \frac{1}{I_p}\left(\tau_{\text{eng}}(\omega_p) - \tau_c\right), \tag{12.17}$$

$$\dot{\omega}_s = \frac{1}{I_s}\left(R(s)\,\tau_c - \tau_{\text{load}}(\omega_s)\right), \tag{12.18}$$

from Equation (10.1) and Equation (10.2).

The axial shift dynamics are

$$\dot{s} = \dot{s}, \tag{12.19}$$

$$\ddot{s} = \frac{\Big(F_{p,\mathrm{ax}}(s,\omega_p) + F_{c,p}(s,\hat{v}_b)\Big) - \Big(F_{s,\mathrm{ax}}(s,R(s)\tau_c) + F_{c,s}(s,\hat{v}_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}, \tag{12.20}$$

from Equation (8.35).

Finally, the belt-speed state evolves according to

$$\dot{\hat{v}}_b = \text{the value selected by Equation (12.16).} \tag{12.21}$$

12.5 Model Evaluation Sequence

Given the current state

$$\mathbf{x}(t) = \big(\omega_p, \omega_s, s, \dot{s}, \hat{v}_b\big),$$

the model is evaluated in the following order:

1. Compute the pulley geometry from s, including the effective radii, centroid radii, wrap angles, and the ratio $R(s)$.
2. Compute $\dot{R}(s, \dot{s})$ from Equation (12.3).
3. Compute the pulley-imposed belt-line speeds $v_{p,\mathrm{cm}}$ and $v_{s,\mathrm{cm}}$ from Equation (12.4), then form v_b^* from Equation (12.5).
4. Evaluate the no-slip torque demand τ_{ns} from Equation (12.9).
5. Evaluate the branchwise traction bounds $\tau_{c,p,\pm}^{\mathrm{stick}}$, $\tau_{c,p,\pm}^{\mathrm{slip}}$, $\tau_{c,s,\pm}^{\mathrm{stick}}$, and $\tau_{c,s,\pm}^{\mathrm{slip}}$, then combine them into $\tau_{\pm}^{\mathrm{stick}}$ and $\tau_{\pm}^{\mathrm{slip}}$ using Equation (12.11) and Equation (12.13).
6. Evaluate the slip metric v_Δ from Equation (12.14), then select both τ_c and $\dot{\hat{v}}_b$ from Equation (12.15) and Equation (12.16).
7. Evaluate the ODE right-hand sides Equation (12.17), Equation (12.18), Equation (12.19), Equation (12.20), and Equation (12.21).

12.6 Complete Nonlinear ODE System

The complete closed CVT model is therefore

$$\boxed{\begin{aligned}
\dot{\omega}_p &= \frac{1}{I_p}\left(\tau_{\text{eng}}(\omega_p) - \tau_c\right), \\
\dot{\omega}_s &= \frac{1}{I_s}\left(R(s)\,\tau_c - \tau_{\text{load}}(\omega_s)\right), \\
\frac{\mathrm{d}}{\mathrm{d}t}s &= \dot{s}, \\
\frac{\mathrm{d}}{\mathrm{d}t}\dot{s} &= \frac{\Big(F_{p,\text{ax}}(s,\omega_p) + F_{c,p}(s,\hat{v}_b)\Big) - \Big(F_{s,\text{ax}}(s,R(s)\tau_c) + F_{c,s}(s,\hat{v}_b)\Big)}{m_{p,m} + m_{s,m} + \dfrac{\rho_b A_b L_b}{4}}, \\
\frac{\mathrm{d}}{\mathrm{d}t}\hat{v}_b &= \begin{cases} \dot{\hat{v}}_{b,\text{stick}}, & v_\Delta = 0 \text{ and } \tau_-^{\text{stick}} \le \tau_{\text{ns}} \le \tau_+^{\text{stick}}, \\ \dot{\hat{v}}_{b,\text{slip}}, & \text{otherwise}, \end{cases}
\end{aligned}} \tag{12.22}$$

with the algebraic closures

$$R = R(s), \qquad \dot{R} = \dot{R}(s,\dot{s}), \qquad v_b^* = \frac{1}{2}\big(r_{p,\text{cm}}(s)\omega_p + r_{s,\text{cm}}(s)\omega_s\big),$$

$$\dot{\hat{v}}_{b,\text{stick}} = \dot{r}_{p,\text{cm}}\,\omega_p + r_{p,\text{cm}}\,\dot{\omega}_p, \qquad \dot{\hat{v}}_{b,\text{slip}} = \frac{v_b^* - \hat{v}_b}{T_b}, \qquad \tau_c \text{ from Equation (12.15)}.$$

In summary, the shift state determines the pulley geometry, the geometry determines both the no-slip torque demand and the belt kinematics, the regime law selects the transmitted coupling torque and belt-speed evolution, and those quantities feed back into both the rotational and axial dynamics.

13 Simulation Results

This section compares four launch simulations: a nominal baseline case, a lighter-flyweight case, a steeper-helix case with reduced secondary torque reactivity, and a combined case with both a shallower initial primary ramp and a steeper helix. The purpose of these results is not to match experimental vehicle data directly, but to show that the model produces mechanically interpretable launch behaviour and responds to tuning changes in physically sensible ways.

The results are organized around the launch shift curves of these four cases. These curves first provide an overall picture of the launch, including engagement, low-ratio operation, active upshift, and high-ratio behaviour. The ratio terminology follows Section 7.9: low ratio means larger R (greater speed reduction), and high ratio means smaller R. following subsections then explain those trends by examining the axial force balance across the CVT and the resulting torque-transfer and slip behaviour. In particular, the combined case is included to show a clearly slip-dominated launch, in which the model predicts loss of the low-ratio hold and sustained slip through the early portion of the launch.

Taken together, the four cases are intended to show three things: first, that the baseline tuning produces a coherent and interpretable launch trajectory; second, that changes to primary and secondary tuning shift the launch behaviour in the expected direction; and third, that when the tuning is made sufficiently unfavorable, the model captures the transition to prolonged

slipping rather than continuing to predict an unrealistic no-slip response. The implementation used to generate these simulations, along with repository, release, and data-access details, is documented in Section 14.

13.1 Simulation Cases and Phase Convention

The results section compares four full-throttle launch simulations from rest. One case is treated as the nominal baseline, while the remaining three are targeted tuning variations chosen to isolate specific changes in CVT behaviour. All cases use the same vehicle, gearbox, tire, and engine model parameters; they differ only in the clutch tuning parameters noted below.

Table 6: Launch cases considered in the results section. All parameters not listed remain equal to the nominal baseline.

Case	Description	Purpose in results section
D	Default tuning	Baseline reference case
F	Lighter flyweights	Shows the effect of reduced primary centrifugal clamping
H	Steeper helix	Shows the effect of reduced secondary torque reactivity
C	Shallower initial primary ramp + steeper helix	Produces a strongly slip-dominated launch

To keep the discussion consistent across the different plots, the launch is described in terms of a small number of recurring operating phases. These phases are not introduced as separate events for each case, but as common regions of behaviour used to interpret the launch trajectories and their underlying force and torque balances.

Throughout the following subsections, these phases will be indicated directly on the plots using low-opacity background shading. The same color convention is used wherever possible so that the reader can track the evolution of each case across the shift, clamping-force, and torque-transfer results.

The phase colors are

- **engagement** — light blue,
- **low-ratio operation** — light green,
- **active upshift** — light yellow, and
- **high-ratio operation** — light red.

Not every case is expected to exhibit all four phases equally clearly. The phase shading is therefore used as a visual guide to the dominant operating regime, rather than as a claim that every boundary is perfectly sharp.

13.2 Launch Shift Curves

The overall launch behaviour of the four tuning cases is first compared in Figure 31. Each trajectory is plotted in engine-speed–vehicle-speed coordinates, so that engagement, ratio change, and the final high-ratio condition can all be viewed on the same axes. All four simulations begin from the same launch initial conditions,

$$\omega_p(0) = 1800\,\text{rpm}, \qquad \omega_s(0) = 0,$$

where ω_p is the primary angular speed and ω_s is the driven angular speed corresponding to zero vehicle speed. From this state, the system is released into a full-throttle forward simulation using the complete coupled model developed in the preceding sections.

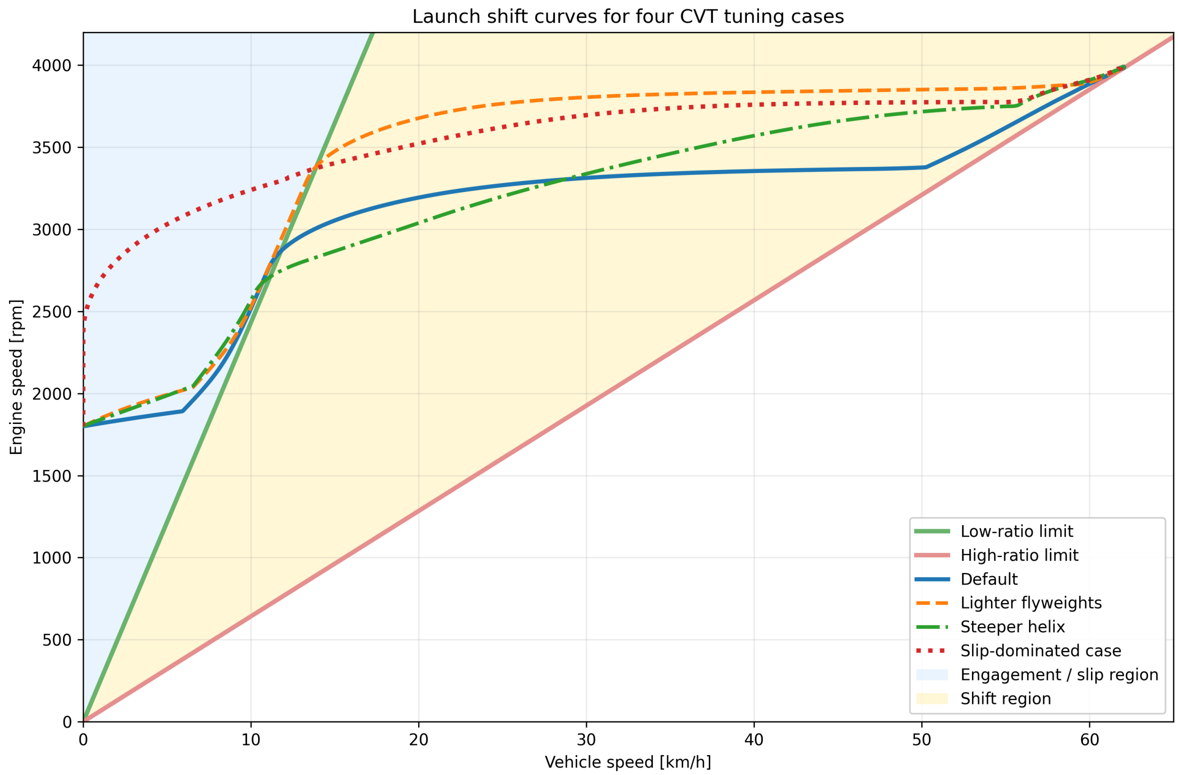


Figure 31: Engine-speed–vehicle-speed launch trajectories for the four tuning cases. The green line is the low-ratio limit, corresponding to a fully unshifted CVT. The red line is the high-ratio limit, corresponding to a fully shifted CVT. The blue shaded region to the left of the low-ratio line corresponds to launch engagement, where slip must be present. The yellow region between the two limit lines is the no-slip shift region.

The two reference lines are the kinematic no-slip limits of the CVT. The **low-ratio line** represents the fully unshifted transmission, where the CVT is in its torque-multiplying launch condition. The **high-ratio line** represents the fully shifted transmission, where the CVT has reached its final speed-increasing condition. Between these two limits lies the admissible shift region. A trajectory cannot move to the right of the high-ratio line, and any portion of a trajectory lying to the left of the low-ratio line must involve slip, since even the fully unshifted CVT cannot support such a low vehicle speed for the corresponding engine speed.

Read from left to right, the figure therefore shows the full launch sequence. The launch begins in the blue engagement region, where the vehicle is still too slow for no-slip operation. After crossing the low-ratio boundary, the trajectory enters the shift region. In a well-behaved launch, the CVT then tends to move approximately horizontally across this region, increasing vehicle speed while holding engine speed near a preferred operating value. This nearly horizontal portion will be referred to here as the *flat-shift region*. As the shift completes, the trajectory approaches the high-ratio line, which forms the final no-slip boundary of the launch.

With that interpretation in place, the differences between the four tuning cases are already visible. The **default** case shows the clearest baseline behaviour: a short engagement region, a brief low-ratio hold, and then a relatively flat shift through the middle of the plot before approaching the high-ratio boundary. This is the most balanced of the four trajectories, since it neither leaves low ratio too early nor rises excessively in engine speed during the main shift.

The **lighter-flyweight** case retains low ratio for longer before crossing into the shift region. Its flat-shift portion is also noticeably higher in engine speed than the default case, so the launch spends more of the main shift at elevated primary speed. The overall shape is still recognizable as a conventional launch trajectory, but the shift is delayed and carried out at a higher engine-speed level.

The **steeper-helix** case behaves differently. It leaves the low-ratio boundary much earlier, so the launch shows little sustained low-ratio hold. Its shift path is also less flat: instead of moving nearly horizontally across the shift region, it climbs upward more strongly as vehicle speed increases. This indicates that the launch does not settle into the same nearly constant engine-speed shift seen in the default case.

The **combined slip-dominated** case is the most extreme. It remains in the engagement region for much longer, showing no clear low-ratio hold before entering the shift region. Once it does cross into the no-slip range, its shift still occurs at a comparatively high engine speed, and the early launch is much more dominated by slip than by clean low-ratio operation.

At this stage, Figure 31 is intended to establish the overall shape of the four launches and the main qualitative differences between them. The key observations are the amount of low-ratio retention, the height and flatness of the main shift segment, and the extent of the initial slip-dominated portion of the launch. The following subsections explain these differences in terms of the underlying axial force balance and torque-transfer behaviour of the CVT.

13.3 Axial Force Balance Through the Launch

The shift-curve trajectories of Figure 31 indicate when each case remains near low ratio, when it begins to upshift, and how quickly it reaches high ratio. To explain those differences mechanically, Figure 32 compares the total axial force acting on the primary and secondary pulleys for each tuning case, with the same launch phases shaded in the background.

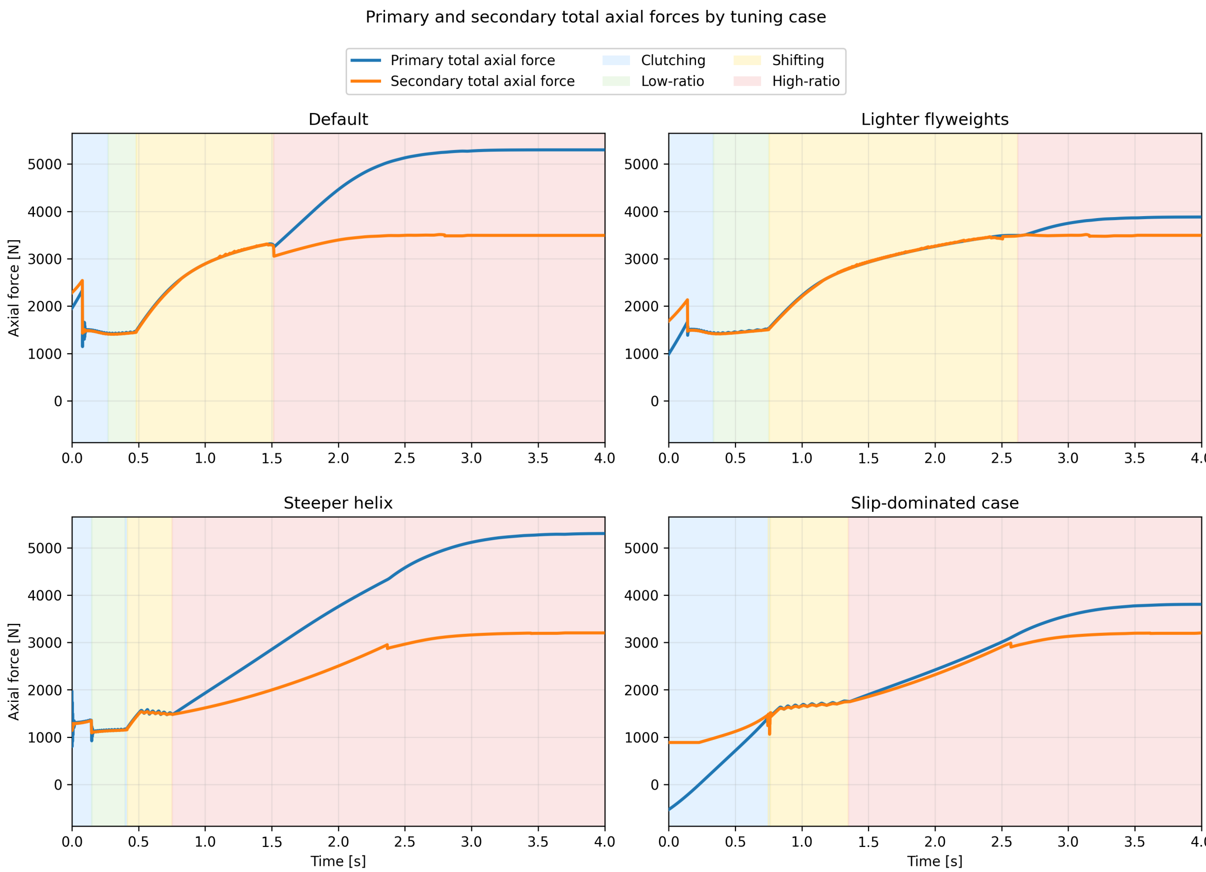


Figure 32: Primary and secondary total axial forces during launch for the four tuning cases. Blue curves show total primary axial force, orange curves show total secondary axial force. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as force-balance diagrams for the shift. While the secondary axial force exceeds the primary axial force, the transmission is held toward low ratio. Once the primary force overtakes the secondary, the pulley pair is driven into upshift. The timing of that crossover therefore determines how long the launch remains near low ratio and how early the transmission begins moving through its ratio range.

The **default** case shows the most balanced behaviour. During the initial clutching phase, the secondary force begins noticeably above the primary force, so the transmission is held near its low-ratio condition rather than shifting immediately. This produces the short but clear low-ratio hold seen previously in Figure 31. As the launch progresses, the primary axial force rises and eventually overtakes the secondary, initiating the main upshift. Importantly, the secondary force does not remain fixed while this happens. It also rises throughout the shift, so the system continues to resist ratio change even after upshift begins. This is mechanically important: if the secondary force stayed nearly constant while the primary force rose rapidly, the transmission would sweep through the shift range much more abruptly. Instead, both sides grow together, giving the default case its smoother and more controlled shift.

The **lighter-flyweight** case makes this point especially clear. Relative to the default case, the primary force is reduced in the early launch, while the secondary remains comparatively strong.

The initial force gap is therefore larger, meaning the low-ratio condition is held more firmly and for longer. This matches the longer low-ratio segment seen in the shift curves. Later in the launch, the primary still rises enough to drive upshift, but it does so later and at a higher engine-speed level than in the default case. In other words, this case does not fundamentally change the character of the shift; it mainly delays it by weakening the primary closing tendency during engagement and early launch.

The **steeper-helix** case behaves differently. Here, the initial difference between secondary and primary force is very small, and the primary overtakes the secondary much earlier. This explains why the corresponding shift curve showed little sustained low-ratio hold and entered the shifting region so quickly. We also see a differently shaped force evolution throughout the shift itself. In the default and lighter-flyweight cases, both forces continue rising together in a gradual way through the main shift region. In the steeper-helix case, both forces rise sharply early on and then level off much sooner. The two forces still remain relatively close to one another, but the overall shape they trace is different. This helps explain why the earlier shift curve did not display the same flat-shift behaviour as the default case. Rather than maintaining a long, nearly constant-speed shift, the transmission shows a more rising shift trajectory, consistent with this less gradual force evolution.

The **slip-dominated** case is the most extreme. Its early launch is characterized by a prolonged clutching phase and no meaningful low-ratio hold. This is consistent with the earlier shift-curve figure, where the trajectory spent much longer in the slip-dominated region before entering the no-slip shift region. Importantly, the beginning of geometric shift does not imply that the system has already stopped slipping. It only indicates that the shift coordinate has begun to move away from its initial value. The more important feature in this case is the low magnitude of both axial forces in the early launch. Compared with the other cases, neither pulley develops a strong axial loading during the initial portion of the event. This weak overall clamping is a key reason why this case behaves poorly, and it will be seen more directly in the following torque-transfer plots where the associated slip remains elevated for much longer.

One additional feature is visible near the beginning of each case. The early spike in secondary axial force is the torque-reactive contribution of the secondary. It appears most strongly during the initial launch when transmitted torque rises rapidly, and it provides an immediate resisting effect against premature upshift. This feature is especially useful because it shows that the secondary is responding dynamically to transmitted torque rather than acting as a static spring-only restraint. The influence of this torque reactivity will be seen more directly in the following torque-transfer results, but its signature is already visible here in the early force histories.

Taken together, Figure 32 provides the mechanical explanation for the differences observed in Figure 31. A larger initial gap between secondary and primary force corresponds to stronger low-ratio retention, as seen in the lighter-flyweight case. A very small gap corresponds to premature upshift, as seen in the steeper-helix case. The default case lies between these extremes and therefore produces the most balanced launch. Perhaps most importantly, the figure shows that the shift is not driven by the primary alone. The secondary rises with it throughout the launch, and the actual shift behaviour emerges from the evolving balance between the two rather than from either force in isolation. At the same time, this figure only explains the shift tendency and overall clamping levels; it has not yet quantified the resulting slip behaviour, which is addressed next.

13.4 Torque Bounds, No-Slip Demand, and Coupling Torque

The force-balance results of Figure 32 explain when each case tends to remain at low ratio and when it begins to shift. However, force balance alone does not determine whether the belt is actually able to transmit the torque required for sticking. To examine that question, Figure 33 compares four quantities during the launch: the primary and secondary traction bounds, the no-slip coupling torque τ_{ns}, and the actual transmitted coupling torque τ_c.

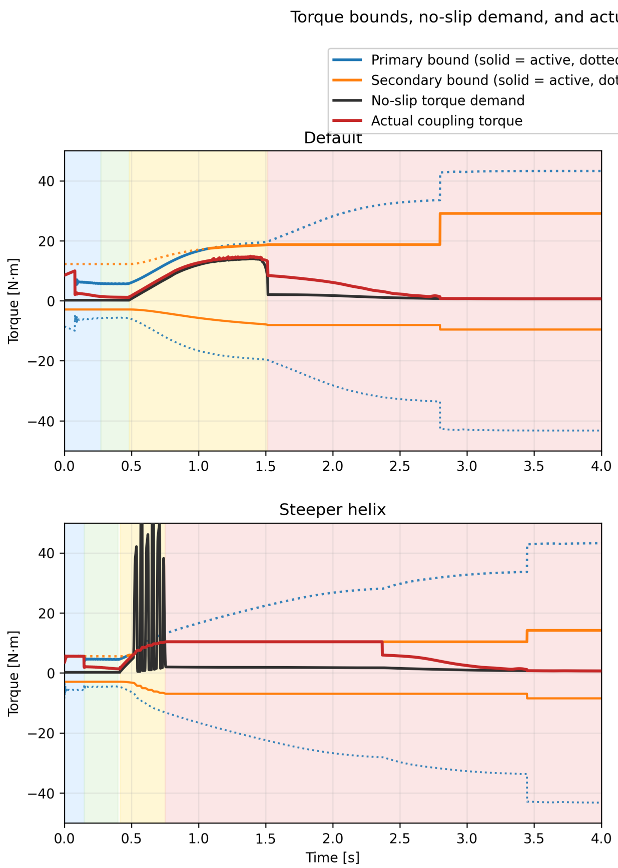


Figure 33: Torque bounds, no-slip demand, and actual coupling torque for the four tuning cases. Blue and orange curves show the primary and secondary bounds, with the active bound drawn solid and the inactive bound dotted. The black curve shows the no-slip torque demand τ_{ns}, and the red curve shows the actual transmitted coupling torque τ_c. Background shading indicates the clutching, low-ratio, shifting, and high-ratio phases defined earlier.

These plots should be read as traction-capacity diagrams. At any instant, the belt can only remain in stick if the required no-slip torque lies inside the admissible interval defined by the active upper and lower bounds. When τ_{ns} lies inside that interval, the actual coupling torque can follow the no-slip demand closely. When τ_{ns} exceeds the available bounds, sticking is no longer admissible and the transmitted torque must saturate at the corresponding active limit.

A first broad trend is immediately visible. The cases that settle out of slip early in the launch are precisely the cases with larger admissible positive torque capacity during the initial clutching and low-ratio phases. In the **default** and **lighter-flyweight** cases, the upper torque bound rises to a relatively large value early in the launch. This gives the system enough admissible coupling torque to reduce the belt-speed mismatch and recover toward sticking. By contrast, the **slip-dominated** case begins with essentially no admissible positive torque and only develops usable capacity more gradually. That lower early torque capacity directly explains why the slip-dominated case spends much longer in the clutching region.

This same distinction appears in the evolution of τ_c. In the cases that recover stick early, the actual coupling torque relaxes toward and then tracks the no-slip demand through the end of clutching and into low-ratio operation. Once low-ratio launch is established, the red and black curves remain close, indicating that the required torque stays admissible and sustained slip is not reintroduced immediately.

A second important pattern appears once the transmission begins to shift. In all four cases, the no-slip torque rises during the shift phase. This is consistent with the form of τ_{ns}, in which the ratio-rate term becomes active once the CVT begins moving away from its initial ratio. In the **default** and **lighter-flyweight** cases, this rise remains below the admissible upper torque bound, so the actual coupling torque continues to follow the no-slip demand and large renewed slip does not occur. In the **steeper-helix** and **slip-dominated** cases, however, τ_{ns} rises above the admissible upper bound during the shift. At that point, sticking is no longer feasible, the actual coupling torque saturates at the active maximum, and renewed slip appears until the state returns closer to the sticking condition.

The **steeper-helix** and **slip-dominated** cases also show visibly oscillatory behaviour in the no-slip torque during the shift onset. This is consistent with the oscillations already seen in the preceding axial-force plots. The most direct source is the ratio-rate contribution to τ_{ns}: if the shift dynamics are not smooth, then the implied no-slip torque need not evolve smoothly either. In these two poorer tuning cases, the shift motion appears more weakly damped, so the no-slip torque exhibits stronger excursions during the transition into active shifting. A plausible reason is that the present model neglects parasitic losses and other non-critical frictional effects, which in a real system would provide additional damping and suppress some of this oscillatory behaviour. This is therefore a useful limitation to note: the model captures the overall stick–slip transitions and the active limiting mechanism, but the transient oscillations in the most aggressive cases are likely amplified by the intentionally idealized loss assumptions.

Another common feature is the gradual relaxation of the actual coupling torque toward the no-slip demand rather than an instantaneous snap to it. This comes from the regularized transition law used in the simulation, discussed in Appendix C.3, which replaces a discontinuous switch with a smooth tanh-based approximation. Its purpose is numerical: it reduces chattering and improves stability near the stick–slip transition. The effect is visible here as a slight lag in the recovery of perfect sticking, but it does not change the main qualitative behaviour of the four cases. The same limiting patterns are still present, and the cases that lose admissibility remain clearly distinguishable from those that do not.

Finally, each case shows a noticeable jump in the admissible torque bound once full sticking is re-established near the end of the launch. This reflects the change in limiting behaviour used by the model after slip is removed, together with the corresponding increase in effective friction behaviour assumed in the sticking regime. The important point for the present discussion is that this jump occurs only after the system has already relaxed back toward the no-slip state; it is therefore a consequence of stick recovery rather than its cause.

Taken together, Figure 33 closes the interpretation begun in the previous two subsections. The shift curves showed the overall launch behaviour, and the axial-force plots explained the tendency to hold or leave low ratio. The present figure then shows whether the required coupling torque is actually admissible throughout those phases. The default and lighter-flyweight cases maintain sufficient positive torque capacity to recover stick early and remain near the no-slip demand through the main shift. The steeper-helix and slip-dominated cases do not: in both, the no-slip torque exceeds the available upper bound during the shift, forcing the actual coupling torque to saturate and reintroducing slip. In this way, the torque plots both confirm the solid patterns seen earlier and reveal one of the model's clearest shortcomings: the most weakly damped cases display exaggerated oscillatory transients due to the idealized loss assumptions.

13.5 Launch Performance Comparison

The preceding subsections established the mechanical differences between the four tuning cases. The shift-curve plots showed the overall launch trajectory, the axial-force plots explained the tendency to hold or leave low ratio, and the torque-bound plots showed whether the required coupling torque remained admissible throughout the launch. The final step is to compare how those differences affect overall launch performance.

Two simple metrics are used here: the time required to travel 50 m, and the time required to reach 60 km/h. These are useful because they summarize the launch in two complementary ways. The 50 m time reflects the integrated quality of the launch as a whole, while the 60 km/h time emphasizes how quickly the drivetrain can build vehicle speed through the main acceleration phase.

Table 7: Launch performance metrics for the four tuning cases.

Case	**Time to** 50 m [s]	**Time to** 60 km/h [s]
Default	3.93	2.27
Lighter flyweights	4.26	2.78
Steeper helix	4.19	2.82
Slip-dominated case	4.41	2.98

The default case performs best on both metrics, which is consistent with the earlier mechanism-level results. It develops enough admissible coupling torque to reduce slip early, retains low ratio long enough to launch effectively, and then transitions into a controlled shift without reintroducing large sustained slip. This is the most balanced use of the CVT: the transmission first allows a strong initial torque transfer into the driveline, and then continues accelerating the vehicle while keeping the shift progression smooth and admissible.

This point is worth emphasizing. The coupling torque plays two roles at once. It is the torque that loads the engine through the transmission, limiting how freely the primary (and therefore the engine) can accelerate, and it is also the torque delivered through the CVT that accelerates the vehicle from rest. A strong early coupling torque is therefore beneficial when it is admissible, because it quickly pulls the belt-speed mismatch down and simultaneously applies useful drive torque to the vehicle. The default case benefits from exactly this behaviour: it uses the initial slip region to establish a relatively large transmitted torque early in the launch, which helps "kick-start" vehicle acceleration before the main shift begins.

The lighter-flyweight case performs worse on both metrics, even though its shift remains comparatively well behaved. This matches the earlier interpretation that its main drawback is delayed primary loading during the early launch. By holding low ratio longer but developing less aggressive early coupling, it sacrifices initial acceleration and does not recover that loss later.

The steeper-helix case shows a different trade-off. It leaves low ratio too early and re-enters slip during the shift, which degrades the quality of the launch. However, because it still develops appreciable coupling torque early in the event, its 50 m time remains slightly better than the lighter-flyweight case. Its slower time to 60 km/h, on the other hand, is consistent with the less stable shift behaviour and the renewed slip seen in the torque-bound plots.

The slip-dominated case performs worst overall. This is the expected result from the earlier figures: its early clamping forces are weakest, its admissible torque rises too slowly, and its launch remains slip-dominated for too long. As a result, it spends too much of the start of the event establishing torque transfer rather than converting that torque cleanly into forward acceleration.

Overall, these performance metrics support the mechanism-level interpretation developed throughout the section. The best launch is not produced by simply holding low ratio as long as possible, nor by shifting as early as possible. Instead, good performance requires a balance: sufficient early admissible coupling torque to accelerate the vehicle strongly from rest, enough low-ratio retention to avoid premature upshift, and a smooth enough shift that the required no-slip torque does not repeatedly exceed the available traction limits.

14 Implementation and Reproducibility

The source code for the CVT simulator used in this work is available at `https://github.com/gr812b/CVT-Simulator`. This implementation includes the mathematical model developed in the main derivation sections along with the practical numerical regularization strategy described in Appendix C.3.

To support reproducibility, the exact version used to generate the figures and tables in this manuscript will be archived as a tagged release:

- repository release (to be finalized): `https://github.com/gr812b/CVT-Simulator/releases/tag/vX.Y.Z`,
- archival DOI landing page (placeholder): `https://doi.org/XX.XXXX/zenodo.XXXXXXX`.

Unless otherwise stated, simulation cases in Section 13 correspond to that archived release. The repository README and release notes describe the execution workflow, configuration files, and plotting scripts used to produce the reported results.

An interactive web deployment of the simulator is also available for demonstration purposes: `https://cvt.kaiarseneau.dev`. This live endpoint is provided for accessibility and does not replace the archived release as the canonical reproducibility artifact.

15 Conclusion

This work developed a unified first-principles dynamic model for a mechanically actuated dry rubber V-belt CVT and its coupling to the vehicle drivetrain. Starting from pulley geometry, belt-length closure, actuator mechanics, and rotational dynamics, the derivation produced a closed nonlinear state-space formulation in which ratio evolution, torque transmission, slip admissibility, and vehicle response are solved consistently in time.

The completed formulation brings together several mechanisms that strongly govern mechanically actuated CVT behaviour: primary flyweight actuation, secondary torque-reactive helix clamping, belt geometric closure, axial shift dynamics, traction-limited torque transfer, and longitudinal drivetrain coupling. The main outcome is a single dynamic framework in which these effects interact directly, while remaining mechanically interpretable in terms of the underlying pulley forces, ratio change, and torque limits.

The simulation results showed coherent launch behaviour across the tuning cases considered. The shift-curve results captured distinct differences in engagement, low-ratio retention, main-shift shape, and approach to high ratio. The axial-force results showed that these differences could be traced back to the evolving balance between primary and secondary clamping. The torque-bound results then showed when the no-slip coupling demand remained admissible and when renewed slip was forced by the available traction limits. Together, these results formed a consistent mechanism chain from tuning changes, to force balance, to torque admissibility, to overall launch behaviour.

An important outcome of the study is that the model preserved physical intuition across multiple levels. Changes in tuning did not simply produce different output curves; they produced changes that remained understandable in mechanical terms. Stronger or weaker low-ratio retention appeared through the initial clamping-force gap, earlier or later upshift appeared through the timing of force crossover, and reintroduced slip appeared when the no-slip torque rose beyond the admissible coupling bounds. This consistency gives confidence that the formulation is capturing the dominant interactions it was intended to represent.

The results also highlighted useful limitations of the present model. In the more aggressive cases, the no-slip torque demand exhibited visibly underdamped oscillatory transients during shift onset. These were consistent with irregular shift-rate evolution in the model and suggest that the idealized loss assumptions, particularly the neglect of parasitic losses and other non-critical frictional effects, leave the system with less damping than would be expected in real hardware. The formulation also remains limited to one-dimensional vehicle motion and does not yet include thermal effects, wear progression, or more detailed tribological contact evolution.

Overall, the present work provides a physically transparent dynamic foundation for understanding mechanically actuated CVT behaviour during transient launch events. Its main value is not only that it closes the governing mechanisms into one consistent dynamical system, but also that the resulting behaviour remains traceable back to clear mechanical causes. With further parameter identification, experimental validation, and extension to include more realistic damping and loss mechanisms, the framework developed here can serve as a useful basis for both engineering interpretation and future design-oriented CVT simulation.

A Belt Length Small-Angle Approximation

This appendix quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models. Using parameter values representative of the physical system considered in this work, we demonstrate that the approximation introduces substantial error in the computed center-to-center distance, then further in the ratio, which is critical for accurate dynamic modeling.

A.1 Standard Approximation

Many simplified treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this approximation, the belt-length constraint Equation (7.11) reduces to

$$L_b \approx \pi(r_p + r_s) + 2\sqrt{C^2 - (r_s - r_p)^2}. \tag{A.1}$$

A.2 Derivation of the Approximate Center Distance

To determine the center-to-center distance at the initial configuration, we substitute $r_p = r_{p,\text{outer,min}}$ and $r_s = r_{s,\text{outer,max}}$ into Equation (A.1) and solve for C.

Step 1: Isolate the square root. Rearranging Equation (A.1):

$$L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) = 2\sqrt{C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}. \tag{A.2}$$

Step 2: Square both sides.

$$\big[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\big]^2 = 4\big[C^2 - (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2\big]. \tag{A.3}$$

Step 3: Solve for C^2.

$$C^2 = \frac{\big[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\big]^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2. \tag{A.4}$$

Step 4: Take the square root. Expanding the squared term in the numerator:

$$\big[L_b - \pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})\big]^2 = L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2,$$

the approximate center distance becomes

$$\boxed{C_{\text{approx}} = \sqrt{\frac{L_b^2 - 2L_b\pi(r_{p,\text{outer,min}} + r_{s,\text{outer,max}}) + \pi^2(r_{p,\text{outer,min}} + r_{s,\text{outer,max}})^2}{4} + (r_{s,\text{outer,max}} - r_{p,\text{outer,min}})^2}.} \tag{A.5}$$

This closed-form expression avoids the need for numerical root finding but neglects the $\sin^{-1}$ term in the exact constraint.

A.3 Error Analysis

We now evaluate the approximation error using parameter values from the physical CVT system considered in this work.

System Parameters

Parameter	Value	SI Value
Belt length L_b	37.538 in	0.9535 m
Belt height h_b	0.613 in	0.015 57 m
Groove half-angle β	11.5°	0.2007 rad
Inner primary radius	0.8125 in	0.020 637 5 m
Primary outer radius $r_{p,\text{outer,min}}$	1.4255 in	0.036 21 m
Secondary outer radius $r_{s,\text{outer,max}}$	4.0 in	0.1016 m

The primary outer radius is computed as the inner radius plus belt height: $r_{p,\text{outer,min}} = 0.8125 + 0.613 = 1.4255\,\text{in}$.

Computed Center Distances Solving the exact belt-length constraint Equation (7.13) numerically yields

$$C_{\text{exact}} = 0.251\,74\,\text{m} \approx 9.911\,\text{in}.$$

Evaluating the approximate formula Equation (A.5) yields

$$C_{\text{approx}} = 0.268\,37\,\text{m} \approx 10.566\,\text{in}.$$

Error Metrics The absolute and relative errors are:

$$\Delta C = C_{\text{approx}} - C_{\text{exact}} = 0.016\,63\,\text{m} = 0.655\,\text{in}, \tag{A.6}$$

$$\varepsilon_{\text{rel}} = \frac{\Delta C}{C_{\text{exact}}} = \frac{0.01663}{0.25174} = 6.61\%. \tag{A.7}$$

Assessment The small-angle approximation introduces a relative error of nearly 7% in the center-to-center distance for this system. This error arises because the ratio

$$\frac{r_{s,\text{outer,max}} - r_{p,\text{outer,min}}}{C} = \frac{0.1016 - 0.03621}{0.25174} = 0.260$$

is not small. The assumption $\lambda \approx 0$ is therefore not justified for this CVT geometry.

Since the center distance C propagates into all subsequent geometric calculations—wrap angles, tangent span lengths, and derived quantities such as the transmission ratio—a 7% error in C can compound into larger errors in predicted system behavior.

A.4 Propagation to Secondary Radius and Transmission Ratio

To demonstrate how the center distance error propagates into operationally relevant quantities, we now examine the secondary radius and CVT ratio at two shift positions: zero shift and maximum shift.

A.4.1 Derivation of the Approximate Secondary Radius

Given a known primary outer radius $r_{p,\text{outer}}$ and center distance C, the secondary outer radius can be obtained by solving the belt-length constraint. Under the small-angle approximation Equation (A.1), this yields a closed-form expression.

Step 1: Isolate the square root. From Equation (A.1):

$$L_b - \pi(r_{p,\text{outer}} + r_{s,\text{outer}}) = 2\sqrt{C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2}. \tag{A.8}$$

Step 2: Square both sides.

$$\left[L_b - \pi r_{p,\text{outer}} - \pi r_{s,\text{outer}}\right]^2 = 4\left[C^2 - (r_{s,\text{outer}} - r_{p,\text{outer}})^2\right]. \tag{A.9}$$

Step 3: Expand and collect terms. Expanding the left side:

$$(L_b - \pi r_{p,\text{outer}})^2 - 2\pi(L_b - \pi r_{p,\text{outer}})r_{s,\text{outer}} + \pi^2 r_{s,\text{outer}}^2.$$

Expanding the right side:

$$4C^2 - 4r_{s,\text{outer}}^2 + 8r_{p,\text{outer}}r_{s,\text{outer}} - 4r_{p,\text{outer}}^2.$$

Collecting powers of $r_{s,\text{outer}}$ yields a quadratic equation:

$$(\pi^2+4)r_{s,\text{outer}}^2 - 2\left[\pi L_b - \pi^2 r_{p,\text{outer}} + 4r_{p,\text{outer}}\right]r_{s,\text{outer}} + \left[(L_b - \pi r_{p,\text{outer}})^2 + 4r_{p,\text{outer}}^2 - 4C^2\right] = 0. \tag{A.10}$$

Step 4: Apply the quadratic formula. The physically meaningful root (smaller secondary radius for overdrive configuration) is

$$\boxed{r_{s,\text{approx}} = \frac{\pi L_b + (4-\pi^2)r_{p,\text{outer}} - 2\sqrt{(\pi^2+4)C^2 - L_b^2 + 4\pi L_b r_{p,\text{outer}} - 4\pi^2 r_{p,\text{outer}}^2}}{\pi^2+4}.} \tag{A.11}$$

This closed-form expression enables direct computation of the approximate secondary radius without numerical root finding.

A.4.2 Zero Shift Configuration

At zero shift, the CVT is in its lowest ratio (maximum reduction) configuration. This is also the geometry used to compute the approximate center distance C_{approx}, so much of that initial center-distance error cancels when the same configuration is used again to recover the secondary radius.

Primary Radius at Zero Shift At $s = 0$, the primary is within the deadzone, so from Equation (7.3):

$$r_{p,\text{outer}}(0) = r_{p,\text{outer,min}} = 0.036\,21\,\text{m}.$$

Secondary Radius Comparison Solving the exact belt-length constraint with C_{exact} yields

$$r_{s,\text{exact}} = 0.101\,60\,\text{m}.$$

Evaluating the approximate formula Equation (A.11) with $C_{\text{approx}} = 0.268\,37\,\text{m}$ yields

$$r_{s,\text{approx}} = 0.101\,60\,\text{m}.$$

Transmission Ratio Error Computing effective radii with $h_b/2 = 0.007\,79\,\mathrm{m}$:

Quantity	Exact	Approximate
$r_{p,\mathrm{eff}}$	$0.0284\,\mathrm{m}$	$0.0284\,\mathrm{m}$
$r_{s,\mathrm{eff}}$	$0.0938\,\mathrm{m}$	$0.0938\,\mathrm{m}$
$R = r_{s,\mathrm{eff}}/r_{p,\mathrm{eff}}$	3.300	3.300

The ratio error at zero shift is

$$\varepsilon_R = \frac{|3.300 - 3.300|}{3.300} \approx 0.000\%.$$

Thus, although C_{approx} is already 6.61% too large, the induced error has little visible effect at minimum shift because the radius calculation is being evaluated at the same operating point used to construct that approximate center distance in the first place.

Figure 34: Scaled diagram of the CVT at zero shift ($s = 0$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The two solutions nearly overlap at this operating point, indicating negligible ratio error in this case.

A.4.3 Maximum Shift Configuration

At maximum shift, the primary and secondary pulleys are closest in size, representing a near-crossover configuration.

Primary Radius at Maximum Shift Using Equation (7.3) with $s = s_{\max} = 0.75\,\mathrm{in} = 0.019\,05\,\mathrm{m}$ and deadzone $s_{\mathrm{dz}} = 0.002\,489\,2\,\mathrm{m}$:

$$\begin{aligned} r_{p,\mathrm{outer}}(s_{\max}) &= r_{p,\mathrm{outer,min}} + \frac{s_{\max} - s_{\mathrm{dz}}}{2\tan\beta} \\ &= 0.03621 + \frac{0.01905 - 0.0024892}{2 \times \tan(11.5^\circ)} \\ &= 0.03621 + \frac{0.01656}{0.4070} \\ &= 0.0769\,\mathrm{m}. \end{aligned} \tag{A.12}$$

Secondary Radius Comparison Solving the exact belt-length constraint Equation (7.13) numerically with $C_{\mathrm{exact}} = 0.251\,74\,\mathrm{m}$ yields

$$r_{s,\mathrm{exact}} = 0.066\,73\,\mathrm{m}.$$

Evaluating the approximate formula Equation (A.11) with $C_{\mathrm{approx}} = 0.268\,37\,\mathrm{m}$ yields

$$r_{s,\mathrm{approx}} = 0.056\,26\,\mathrm{m}.$$

Transmission Ratio Error The CVT ratio is defined using effective radii, which account for belt seating depth (Equation (7.23)):

$$r_{\text{eff}} = r_{\text{outer}} - \frac{h_b}{2}.$$

With $h_b/2 = 0.007\,79\,\text{m}$:

Quantity	**Exact**	**Approximate**
$r_{p,\text{eff}}$	0.0691 m	0.0691 m
$r_{s,\text{eff}}$	0.0589 m	0.0485 m
$R = r_{s,\text{eff}}/r_{p,\text{eff}}$	0.853	0.701

Both models predict overdrive ($R < 1$), but the approximation substantially underestimates the ratio. The ratio error is

$$\varepsilon_R = \frac{|0.701 - 0.853|}{0.853} = 17.8\%.$$

Unlike the minimum-shift case, this operating point does not benefit from the same cancellation, so the center-distance error propagates into a much larger error in secondary radius and transmission ratio.

Figure 35: Scaled diagram of the CVT at maximum shift ($s = s_{\max}$). The primary pulley (left) and exact secondary pulley (right, solid line) are shown alongside the approximate secondary pulley (dotted line). The approximation underestimates the secondary radius, leading to a lower predicted ratio than the exact model.

A.5 Summary of Approximation Errors

Configuration	R_{exact}	R_{approx}	**Ratio Error**
Zero shift ($s = 0$)	3.300	3.300	0.000%
Maximum shift ($s = s_{\max}$)	0.853	0.701	17.8%

The small-angle approximation error is strongly operating-point dependent. At zero shift, most of the center-distance error cancels because the approximate center distance was itself constructed from that same geometry. At maximum shift, that cancellation is lost and the ratio error becomes substantial.

These errors arise from the initial 6.61% error in center distance, which propagates nonlinearly through the belt-length geometry in a state-dependent way. Since the transmission ratio directly enters the rotational equations of motion (Section 12), this approximation can still produce materially inaccurate dynamics near maximum shift.

This analysis justifies the use of exact numerical solutions for all geometric computations in this work.

B Common Ramp Profile Designs

This appendix presents common ramp profile parameterizations for CVT primary pulley actuation. From Equation (8.8), the centrifugal axial force is

$$F_{p,\mathrm{cent}}(s,\omega_p) = m_f\omega_p^2\big(r_{f,0} + r_f(s)\big)\frac{dr_f}{ds}, \tag{B.1}$$

where m_f is the flyweight mass, $r_{f,0}$ is the initial flyweight radius, and $r_f(s)$ is the radial displacement along the ramp as a function of axial shift s.

The profile $r_f(s)$ must satisfy the admissibility conditions from Section 8: continuously differentiable, non-negative slope, and finite slope throughout $[0, s_{\max}]$.

B.1 Linear Ramp

A linear ramp with slope k_r (radial displacement per unit axial shift):

$$r_f(s) = k_r \cdot s, \qquad \frac{dr_f}{ds} = k_r = \mathrm{const.} \tag{B.2}$$

Substituting into Equation (B.1):

$$F_{p,\mathrm{cent}}(s,\omega_p) = m_f\omega_p^2 k_r(r_{f,0} + k_r s) = m_f\omega_p^2 k_r r_{f,0} + m_f\omega_p^2 k_r^2 s. \tag{B.3}$$

At constant ω_p, axial force increases linearly with s because the growing radius is multiplied by a constant slope. Simple to manufacture, but may cause aggressive shifting at high ratio.

B.2 Circular Arc Ramp

A circular arc ramp is designed to reduce force variation across the shift range by achieving *partial* cancellation. The key insight is to parameterize the arc by desired slope angles α_1 and α_2 at the start and end of the segment, where $|dr_f/ds| = \tan\alpha$ specifies the force multiplier. This provides direct control over actuation behavior at engagement (high slope) versus overdrive (low slope).

For a circle of radius R_c centered at (s_c, r_c), the ramp profile satisfies:

$$(s - s_c)^2 + (r_f - r_c)^2 = R_c^2. \tag{B.4}$$

Solving for $r_f(s)$ on the upper arc:

$$r_f(s) = r_c + \sqrt{R_c^2 - (s - s_c)^2}. \tag{B.5}$$

Implicit differentiation yields:

$$\frac{dr_f}{ds} = \frac{-(s - s_c)}{\sqrt{R_c^2 - (s - s_c)^2}} = \frac{-(s - s_c)}{r_f - r_c}. \tag{B.6}$$

Parameterization from slope angles. At any point on the arc where the local slope angle is α (i.e. $dr_f/ds = \tan\alpha$), the circle constraint combined with Equation (B.6) gives the parametric form:

$$s = s_c - R_c \sin\alpha, \qquad\qquad r_f = r_c + R_c \cos\alpha. \tag{B.7}$$

The slope angle α decreases monotonically from α_1 (entry) to $\alpha_2 < \alpha_1$ (exit) as s increases along the ramp. Applying Equation (B.7) at both boundary shifts s_1 and s_2 and subtracting:

$$s_2 - s_1 = R_c(\sin\alpha_1 - \sin\alpha_2). \tag{B.8}$$

With $\Delta s = s_2 - s_1$ as the total axial span of the ramp segment, the arc radius and center are determined entirely by the two slope angles:

$$R_c = \frac{\Delta s}{\sin\alpha_1 - \sin\alpha_2}, \qquad s_c = s_1 + R_c \sin\alpha_1. \tag{B.9}$$

The boundary condition $r_f(s_1) = 0$ (flyweights at their initial radius at ramp entry) then fixes the radial center:

$$r_c = -R_c \cos\alpha_1. \tag{B.10}$$

Substituting into Equation (B.1):

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 (r_{f,0} + r_f) \cdot \frac{-(s - s_c)}{r_f - r_c}. \tag{B.11}$$

To see the cancellation explicitly, split $r_{f,0} + r_f = (r_{f,0} + r_c) + (r_f - r_c)$:

$$\begin{aligned} F_{p,\text{cent}} &= m_f \omega_p^2 \left[\underbrace{(r_f - r_c)}_{\text{cancels}} \cdot \frac{-(s - s_c)}{r_f - r_c} + (r_{f,0} + r_c) \cdot \frac{-(s - s_c)}{r_f - r_c} \right] \\ &= -m_f \omega_p^2 \,(s - s_c) \left[1 + \frac{r_{f,0} + r_c}{r_f - r_c} \right]. \end{aligned} \tag{B.12}$$

This form reveals the structure clearly. The dominant factor is $-(s - s_c)$, which is linear in shift: the $(r_f - r_c)$ terms cancel exactly in the first part of the split. The bracket $\left[1 + \frac{r_{f,0} + r_c}{r_f - r_c}\right]$ is a nonlinear correction multiplier. Using the parametric form Equation (B.7) with Equation (B.10), the bracket evaluates to

$$1 + \frac{r_{f,0} + r_c}{r_f - r_c} = 1 + \frac{r_{f,0} - R_c \cos\alpha_1}{R_c \cos\alpha}, \tag{B.13}$$

where α is the local slope angle at shift s, sweeping from α_1 at entry to α_2 at exit. This makes the effective proportionality between force and $(s - s_c)$ position-dependent.

The numerator $r_{f,0} - R_c \cos\alpha_1$ is the key residual: it is controlled entirely by the choice of α_1, α_2, and Δs through $R_c = \Delta s/(\sin\alpha_1 - \sin\alpha_2)$. Increasing either α_1 or Δs raises R_c, making $R_c \cos\alpha_1$ larger and the residual smaller (or negative). The entry force scale, the exit force scale, and the degree of position-dependence between them are therefore all coupled to the same two angle choices. There is no freedom to set the overall force scale independently of the nonlinearity — changing α_1 and α_2 adjusts both simultaneously.

The condition $r_{f,0} = R_c \cos\alpha_1$ makes the residual zero, so the bracket equals 1 everywhere and force is exactly proportional to shift. This is the angle configuration at which the circular arc best approximates the hyperbolic ideal — it gives a practical design rule for tuning.

The practical consequence is important: at a fixed engine speed ω_p, the centrifugal force is approximately proportional to $(s - s_c)$, growing linearly with shift. This directly matches the linear restoring force of the primary spring (force $\propto k_s \cdot s$). A spring-loaded primary can therefore be tuned to balance the centrifugal flyweights at a target shift, and the linear character of the circular arc ensures the equilibrium is stable across the shift range rather than concentrated at a single point.

Circular arcs are widely used in practice due to ease of manufacture and sufficient compensation. The nonlinear residual is small when the ramp center offset $(r_{f,0} + r_c)$ is small relative to the arc radius R_c, but it cannot be eliminated entirely. The following section shows how to *exactly* achieve the linear target by choosing a different ramp geometry.

B.3 Hyperbolic Ramp (Perfect Cancellation)

From Equation (B.13), the circular arc force-to-shift proportionality constant is:

$$\frac{F_{p,\mathrm{cent}}}{-m_f\omega_p^2(s - s_c)} = 1 + \frac{r_{f,0} - R_c\cos\alpha_1}{R_c\cos\alpha}. \tag{B.14}$$

This varies with position because α changes along the arc. The natural question is: what ramp profile makes this ratio a *true constant*? That constant is what we call K, and the profile that achieves it satisfies:

$$(r_{f,0} + r_f(s))\frac{dr_f}{ds} = Ks, \tag{B.15}$$

where $K > 0$. This produces centrifugal force exactly proportional to shift:

$$F_{p,\mathrm{cent}}(s, \omega_p) = m_f\omega_p^2 Ks. \tag{B.16}$$

K is not a free parameter any more than R_c is for the circular arc: just as R_c is uniquely determined by α_1, α_2, and Δs (Equation (B.9)), K is uniquely determined by α_1, α_2, s_1, and s_2 (derived in the parameterization below). What differs is not the design inputs, but that the hyperbolic profile makes K exactly constant along the entire ramp, whereas the circular arc's equivalent ratio changes with position.

Deriving the profile. Equation (B.15) is separable. Substituting $u = r_{f,0} + r_f$, so $du/ds = dr_f/ds$:

$$u\,du = Ks\,ds. \tag{B.17}$$

Integrating both sides and writing the constant of integration as $\frac{1}{2}Ks_0^2$:

$$\tfrac{1}{2}u^2 = \tfrac{1}{2}K(s^2 + s_0^2). \tag{B.18}$$

Substituting back $u = r_{f,0} + r_f$:

$$\boxed{(r_{f,0} + r_f(s))^2 = K(s^2 + s_0^2),} \tag{B.19}$$

giving the explicit profile:

$$r_f(s) = \sqrt{K(s^2 + s_0^2)} - r_{f,0}. \tag{B.20}$$

In the (s, r_f) plane, the locus $(r_{f,0} + r_f)^2 = Ks^2$ (taking $s_0 = 0$) is a branch of a *hyperbola* — hence the name. Unlike the circular arc, where the residual multiplier in Equation (B.12) could not be removed, this profile forces the product $(r_{f,0} + r_f) \cdot dr_f/ds = Ks$ exactly for all s.

Local slope and its variation. To see how K relates to actual ramp angles, differentiate Equation (B.19) implicitly:

$$\frac{dr_f}{ds} = \frac{Ks}{r_{f,0} + r_f} = \frac{\sqrt{K}\, s}{\sqrt{s^2 + s_0^2}}. \tag{B.21}$$

Since $dr_f/ds = \tan\alpha(s)$ by definition of the ramp slope angle, the local angle satisfies:

$$\tan\alpha(s) = \frac{\sqrt{K}\, s}{\sqrt{s^2 + s_0^2}}. \tag{B.22}$$

Two observations follow. First, the slope is zero at $s = 0$ and grows monotonically toward the asymptote $\sqrt{K}$ as $s \to \infty$. This is the opposite trend from a circular arc, where the slope decreases from entry to exit. The reason is that the growing lever arm $(r_{f,0} + r_f)$ at large shift would otherwise amplify force super-linearly; a rising slope exactly compensates to keep the product Ks linear. Second, the asymptotic slope $\lim_{s\to\infty} \tan\alpha(s) = \sqrt{K}$ gives a simple interpretation: $K = \tan^2\alpha_\infty$ where α_∞ is the slope the profile approaches at large shift.

Parameterization by entry and exit angles. Following the same convention as Appendix B.2, specify desired slope angles α_1 and α_2 at entry shift s_1 and exit shift s_2. From Equation (B.22):

$$\tan^2\alpha_1 = \frac{Ks_1^2}{s_1^2 + s_0^2}, \qquad \tan^2\alpha_2 = \frac{Ks_2^2}{s_2^2 + s_0^2}. \tag{B.23}$$

Dividing the two equations eliminates K:

$$\frac{\tan^2\alpha_1}{\tan^2\alpha_2} = \frac{s_1^2\,(s_2^2 + s_0^2)}{s_2^2\,(s_1^2 + s_0^2)}. \tag{B.24}$$

Solving for s_0^2:

$$s_0^2 = \frac{s_1^2 s_2^2\,(\tan^2\alpha_2 - \tan^2\alpha_1)}{\tan^2\alpha_1\, s_2^2 - \tan^2\alpha_2\, s_1^2}. \tag{B.25}$$

Then K follows from either condition in Equation (B.23):

$$K = \frac{\tan^2\alpha_1\,(s_1^2 + s_0^2)}{s_1^2}. \tag{B.26}$$

For $s_0^2 > 0$ a consistent solution requires the numerator and denominator of Equation (B.25) to have the same sign. Since the hyperbolic slope increases with s (Eq. (B.21)), the physically consistent case is $\alpha_2 > \alpha_1$ (exit angle larger than entry), with $\tan^2\alpha_1 s_2^2 > \tan^2\alpha_2 s_1^2$ ensuring a positive denominator. No solution exists when the two angle conditions give a degenerate system ($\tan^2\alpha_1 s_2^2 = \tan^2\alpha_2 s_1^2$).

Practical impact. In most cases the difference between circular and hyperbolic profiles is negligible. For the specific ramps implemented in this work, the integrated force deviation across the full shift range is less than 1%. The hyperbolic derivation is nonetheless useful: it establishes the theoretical ideal and shows that the circular arc's partial cancellation is not a fundamental limitation but rather a small correction from the constant offset $(r_{f,0} + r_c)$.

Limitations. Both profiles address only the centrifugal force term. The secondary pulley torque feedback ($F_{s,\text{ax}}(s, \tau_s)$ in Equation (12.20)) introduces load-dependent nonlinear terms that no ramp geometry alone can cancel. Circular arcs therefore remain the practical standard—simpler to manufacture, well-understood, and sufficient given these irreducible secondary-side effects.

B.4 Piecewise Ramps and Smoothing

Any of the profiles above can be combined in a piecewise fashion: multiple linear segments, multiple circular arcs with different angle pairs, or a mix of profile types joined at transition shifts $s_1, s_2, \ldots, s_{n-1}$. This allows region-specific tuning — steeper slopes at engagement, moderate for cruise, gentler near overdrive — without committing to a single global geometry.

Admissibility at junctions. The admissibility conditions from Section 8 require $r_f(s)$ to be continuously differentiable (C^1). A piecewise ramp satisfies C^0 by construction (the profile value is continuous if segments are joined at a common point), but slope continuity may be violated: if the left-segment slope at s_k differs from the right-segment slope, Eq. (B.1) gives a finite force jump at that point. No kinematic singularity occurs, but the force has a step discontinuity.

Numerical stiffness considerations. A sharp slope discontinuity is not the same as a smooth rapid transition: many real implementations use a finite but steep blend region, which the ODE solver sees as an extremely large second derivative $d^2 r_f/ds^2$ near the junction. This can create a locally stiff region and force an adaptive integrator to take very small steps near the knot, even if the discontinuity is physically mild. If simulation performance is a concern, slope transitions should be blended — a short circular arc segment tangent to both adjoining profiles is sufficient, as it restores C^1 continuity and bounds $d^2 r_f/ds^2$ throughout the ramp.

C Numerical Methods Summary

This section documents the numerical algorithms used for belt-geometry root finding and dynamic simulation, including the specific tolerances, bracketing strategies, and solver configurations employed in the implementation.

C.1 Root Finding for Belt-Length Constraint

Given a known primary outer radius r_1 and fixed center-to-center distance C, the secondary outer radius r_2 is the unique root of the belt-length constraint. This scalar root problem is obtained by rearranging the exact belt length closure from Equation (7.13) into the residual form:

$$g(r_2;\, r_1) = \pi(r_1 + r_2) + 2(r_2 - r_1)\arcsin\left(\frac{r_2 - r_1}{C}\right) + 2\sqrt{C^2 - (r_2 - r_1)^2} - L_b = 0, \qquad \text{(C.1)}$$

where L_b is the fixed belt length. A separate invocation solves for the center-to-center distance C itself at initialization, using the same equation evaluated at the boundary radii ($r_{1,\min}$, $r_{2,\max}$) with

C as the unknown; this is exactly the initialization step implied by Equation (7.13) before the ratio law Equation (12.2) can be evaluated over the shift range.

Bracketing strategy. The arcsin term in Equation (C.1) is real only when $|r_2 - r_1| \leq C$, so the natural domain for r_2 is the open interval $(r_1 - C,\, r_1 + C)$. Within this interval a sign change is guaranteed: at the lower boundary $r_2 \to r_1 - C$, the arcsin saturates to $-\pi/2$ and the square root vanishes, giving $g \to 2\pi r_1 - 2\pi C - L_b < 0$ for all physically valid parameters. At the upper boundary $r_2 \to r_1 + C$, by symmetry $g \to 2\pi r_1 + 2\pi C - L_b > 0$, satisfied whenever the belt length is less than the circumference subtended by the wider pulley pair — a requirement any real CVT must satisfy. A small safety margin $\varepsilon = 10^{-9}$ m is applied at both ends to keep arcsin away from its domain boundary where the function becomes numerically steep.

For center-to-center distance solving, the bracket is instead the interval $(\Delta r + \varepsilon,\ L_b/2)$, where $\Delta r = |r_{2,\text{eff}} - r_{1,\text{eff}}|$ is the effective-radius difference at the boundary configuration. The lower bound ensures arcsin is defined; the upper bound is a conservative geometric limit (a straight two-span belt cannot exceed L_b across a gap of $L_b/2$). Both brackets are validated by an explicit sign-change check before invoking the solver, so any failure to bracket immediately identifies an infeasible geometry rather than producing a silent wrong answer.

Brent's method. Both roots are found using `scipy.optimize.brentq`, an implementation of Brent's algorithm Brent (1973). The method combines three strategies: bisection (guaranteed bracket-halving), secant interpolation (superlinear convergence near smooth roots), and inverse quadratic interpolation (order-≈ 1.84 convergence when three previous iterates are well-conditioned). Whenever the faster interpolation steps would place the next iterate outside the current bracket, the algorithm falls back to bisection, preserving the worst-case $O(\log_2(1/\varepsilon))$ guarantee of pure bisection. Because the sign-change brackets described above are guaranteed for all valid CVT geometries, `brentq` is certain to converge. Both root solves use an absolute tolerance of 10^{-9} m, well below any physically meaningful length precision.

C.2 ODE Integration for Dynamic Simulation

The full CVT state vector $\mathbf{x} = (\omega_p, \omega_s, s, \dot{s}, \hat{v}_b)$ is integrated using `scipy.integrate.solve_ivp`. The governing system is written as a first-order ODE system in Equation (12.22), with rotational sub-equations Equation (12.17) and Equation (12.18), axial dynamics Equation (12.20), belt-speed dynamics for $\hat{v}_b$, and algebraic geometry closures Equation (12.2) and Equation (12.3). Although Equation (12.20) is second-order in s, introducing $(s, \dot{s})$ as separate states converts the complete model to first-order form for time integration. The time integrator is used with the `RK45` method — the Dormand–Prince explicit embedded Runge–Kutta pair Dormand and Prince (1980). The method propagates the state with a 5th-order formula and simultaneously computes a 4th-order companion estimate for local error control, using the same six function evaluations. The tableau is FSAL (first-same-as-last): the final stage of each accepted step is reused as the first stage of the next, so only five new evaluations are needed per step in steady operation.

Adaptive step size control. At each candidate step the per-component local error estimate $\hat{e}_i$ is compared to a mixed tolerance threshold. A step is accepted when

$$\max_i \frac{|\hat{e}_i|}{\text{atol} + \text{rtol} \cdot |y_i|} \leq 1, \tag{C.2}$$

and the next step size is scaled proportionally to $(\mathrm{tol}/\hat{e})^{1/5}$. The tolerances used in this work are atol $= 10^{-6}$ and rtol $= 10^{-4}$ (Table 8). The absolute floor prevents spurious step rejection when a state variable passes through zero (e.g. $\dot{s} \approx 0$ at shift equilibrium or $\omega_p \approx 0$ at engine start), where a purely relative criterion would demand unreachable accuracy.

Parameter	**Value**	**Interpretation**
`atol`	10^{-6}	absolute floor: 1 µm on s, 1 µrad/s on ω
`rtol`	10^{-4}	relative tolerance: 0.01% of current state magnitude
`t_eval`	10 000 pts	uniform output grid over 30 s

Table 8: ODE solver tolerance parameters.

Event detection. Phase transitions (shift engagement, boundary contact, back-shift onset) are located using `solve_ivp`'s event mechanism. These events correspond to qualitative changes in the dynamics around the bounded shift coordinate used in Equation (12.20) and the full state evolution in Equation (12.22). After each accepted step the solver evaluates the event functions on a degree-4 dense-output interpolant and applies a root-finding pass to detect any sign change within the step. When a crossing is found it is bracketed at sub-tolerance precision before being reported, so transitions are located with the same accuracy as the integration tolerance rather than being snapped to the coarse output grid.

Phase splitting at hard constraints. The shift bounds $s \in [0, s_{\max}]$ with $\dot{s} = 0$ at both endpoints are handled by partitioning the trajectory into separate `solve_ivp` calls rather than imposing inequality constraints inside the ODE. Each phase — normal shifting, clamped-at-maximum, and back-shifting — presents a smooth right-hand side to the integrator, allowing RK45 to operate at full 5th-order accuracy throughout. The terminal event of each phase supplies the initial condition of the next. This is numerically cleaner than directly clamping Equation (12.20) inside a single solve because it preserves a smooth right-hand side within each phase. Up to ten alternating full-shift / back-shift cycles are supported before the simulation terminates, preventing unbounded phase loops for unusual parameter regimes.

C.3 Slip-Law Regularization and Blend Parameter Selection

The traction law in Equation (12.15) is physically clear but discontinuous at $v_{\mathrm{rel}} = 0$, which can induce step-size chatter near stick–slip transitions. A practical regularized formulation uses three ideas: a smooth saturation law, a clamped no-slip torque request, and a blending rule between those two limits.

First define a smooth saturation torque

$$\tau_{\mathrm{sat}} = \tau_{\max} \tanh\left(\frac{v_{\mathrm{rel}}}{v_{\mathrm{smooth}}}\right), \tag{C.3}$$

where $v_{\mathrm{smooth}} > 0$ sets the transition width in slip-speed units. For $|v_{\mathrm{rel}}| \gg v_{\mathrm{smooth}}$, this approaches the usual friction-limited value $\pm\tau_{\max}$, while near zero it removes the sharp sign discontinuity.

Next, clamp the no-slip torque request:

$$\tau_{\mathrm{req}} = \mathrm{clip}(\tau_{\mathrm{ns}},\, -\tau_{\max},\, \tau_{\max})\,. \tag{C.4}$$

This is the largest admissible torque consistent with the no-slip request, without allowing the requested value to exceed traction capacity.

Then define the blend factor

$$\alpha = \mathrm{clip}\left(\frac{|v_{\mathrm{rel}}|}{v_{\mathrm{blend}}},\, 0,\, 1\right), \tag{C.5}$$

with $v_{\mathrm{blend}} > 0$ defining the width of the blending region, and use the transmitted torque

$$\tau_c = (1 - \alpha)\,\tau_{\mathrm{req}} + \alpha\,\tau_{\mathrm{sat}}. \tag{C.6}$$

This has a useful interpretation. For very small mismatch, $\alpha \approx 0$, so the model stays close to the clamped no-slip demand. For larger mismatch, $\alpha \to 1$, so the law transitions to the smooth Coulomb limit. In other words, the regularized model behaves like a requested no-slip torque near stick and like a friction-limited sliding law once the mismatch becomes appreciable.

How to choose v_{smooth}. Choose the smoothing scale small enough that $\tanh(v_{\mathrm{rel}}/v_{\mathrm{smooth}})$ is close to a sign law through most of the slip regime, but not so small that the solver again sees an effectively discontinuous transition. A practical workflow is:

1. Start with v_{smooth} around 0.5–2% of a representative relative-speed magnitude seen during actual slip.
2. Decrease it until predicted macro outputs (ratio trajectory, launch time, peak speeds) stop changing materially.
3. Increase it slightly if event localization or adaptive step-size behavior becomes noisy near $v_{\mathrm{rel}} \approx 0$.

The blend scale v_{blend} should generally be chosen greater than or equal to v_{smooth}, so that the handoff from clamped demand to the smooth Coulomb limit is not sharper than the smoothing law itself.

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